

Basis risk

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Basis risk**

In theory the futures market provides a fixed and stable outcome when hedging currency or interest rate risk, but in practice futures contracts are exposed to basis risk.

Basis is the difference between the futures and spot prices and, for the purposes of recommending a hedging strategy, it is often assumed to diminish at a constant rate. Basis risk arises when the price of a futures contract does not have a predictable relationship with the spot price of the instrument being hedged. When basis risk is introduced to a scenario, it may mean an alternative hedging method would provide a better result.

In order to illustrate this point, we will use the information provided in a sample question from the September/December 2017 exam sessions, Wardegul Co, and investigate the impact of basis risk on the hedging recommendation. The company's treasury department would like to hedge the following transaction.

Transaction to be hedged

Today's date is 1 October 2017. The treasury department plans to hedge a receipt, in Eurian Dinar (D), of D27m. The receipt is expected on 31 January 2018 and will need to be invested until 30 June 2018.

The central bank base rate in Euria is currently 4.2% and the treasury team believes that it can invest funds in Euria at the central bank base rate less 30 basis points. However, treasury staff have seen predictions that the central bank base rate could increase by up to 1.1% or fall by up to 0.6% between now and 31 January 2018.

For the purposes of this example we will consider the following possibilities to hedge the receipt:

- Interest rate futures
- Exchange-traded options on interest rate futures

Three month D futures, D500,000 contract size

Prices are quoted in basis points at 100 – annual % yield:

December 2017: 94.84

March 2018: 94.78

June 2018: 94.66

Exchange-traded options on three month D futures, D500,000 contract size, option premiums are in annual %

Calls			Strike price	Puts		
December	March	June		December	March	June
0.417	0.545	0.678		0.071	0.094	0.155
0.078	0.098	0.160		0.393	0.529	0.664

Assume futures and options contracts are settled at the end of each month.

In the first instance we will follow the approach used in the past exam question and assume there is no basis risk and that basis diminishes at a constant rate, based on monthly time intervals. We will then introduce basis risk and consider the impact on our recommended hedging strategy.

(1) Ignore basis risk (as per sample question)

Futures

Wardegul Co's treasury department would buy March futures as the hedge is against a fall in interest rates and the investment will be made on 31 January.

Number of contracts = $D27,000,000 / D500,000 \times 5 \text{ months} / 3 \text{ months} = 90 \text{ contracts}$

Unexpired basis

Spot price (1 October) – futures price = basis

$(100 - 4.20) - 94.78 = 1.02$

Unexpired basis on 31 January = $2/6 \times 1.02 = 0.34$

If central bank base rate increases to 5.3%

D

Investment return $5.0\% \times 5/12 \times D27,000,000$	562,500
--	---------

Expected futures price: $100 - 5.3 - 0.34 = 94.36$

Loss on the futures market: $(0.9436 - 0.9478) \times D500,000 \times 3/12 \times 90$	(47,250)
---	----------

Net receipt	<u>515,250</u>
-------------	----------------

Effective annual interest rate $515,250 / 27,000,000 \times 12/5$	4.58%
---	-------

If central bank base rate falls to 3.6%

D

Investment return $3.3\% \times 5/12 \times D27,000,000$	371,250
--	---------

Expected futures price: $100 - 3.6 - 0.34 = 96.06$

Profit on the futures market: $(0.9606 - 0.9478) \times \$500,000 \times 3/12 \times 90$ 144,000

Net receipt 515,250

Effective annual interest rate $515,250 / 27,000,000 \times 12/5$ 4.58%

Options on interest rate futures

The treasury department would buy March call options to hedge against a fall in interest rates. As above, 90 contracts are required.

If central bank base rate increases to 5.3%

Exercise price	94.25	95.25
Expected futures price, as above	94.36	94.36
Exercise?	Yes	No
Gain in basis points	11	0
	D	D
Investment return as above	562,500	562,500
Profit on option		0
$0.0011 \times \$500,000 \times 3/12 \times 90$	12,375	

Premium	-	-
$0.00545 \times D500,000 \times 3/12 \times 90$	(<u>61,313</u>)	-
$0.00098 \times D500,000 \times 3/12 \times 90$	-	(11,025)
Net receipt	<u>513,562</u>	<u>551,475</u>
Effective annual interest rate		
$513,562 / 27,000,000 \times 12/5$	4.56%	
$551,475 / 27,000,000 \times 12/5$		4.90%
If central bank base rate falls to 3.6%		
Exercise price	94.25	95.25
Expected futures price, as above	96.06	96.06
Exercise?	Yes	Yes
Gain in basis points	181	81
	D	D
Investment return as above	371,250	371,250
Profit on option		0
$0.0181 \times \$500,000 \times 3/12 \times 90$	203,625	

$0.0081 \times \$500,000 \times 3/12 \times 90$		91,125
Premium as above	(61,313)	(11,025)
Net receipt	<u>513,562</u>	<u>451,350</u>
Effective annual interest rate		
$513,562 / 27,000,000 \times 12/5$	4.56%	
$451,350 / 27,000,000 \times 12/5$		4.01%

Comments

As expected, the futures market provides a fixed return of 4.58% whether the central bank base rate increases to 5.3% or reduces to 3.6%. The 95.25 option provides a better outcome as long as interest rates rise but is significantly lower if interest rates fall. If the board is at all risk averse the futures outcome would be preferable. The 94.25 option is marginally lower than the futures outcome under both scenarios but may be preferable if the base rate rises higher than 5.41%, the point at which the option would not be exercised.

(2) Impact of basis risk

Today's date is 31 January. The prediction that the central bank base rate might increase by 1.1% to 5.3% turns out to be exactly correct. Based on our previous basis calculation we would have expected today's price for the March futures contracts to fall to 94.36.

However, futures contracts are exposed to basis risk, which means the closing March futures price on 31 January could be more or less than predicted. For the purposes of this example, we will assume the closing futures price on 31 January is 94.16, only marginally less than our prediction from before.

Futures

Central bank rate increases to 5.3%

Investment return as above	562,500
Loss on the futures market: $(0.9416 - 0.9478) \times \$500,000 \times 3/12 \times 90$	<u>(69,750)</u>
Net receipt	<u>492,750</u>
Effective annual interest rate $492,750 / 27,000,000 \times 12/5$	4.38%

Options on interest rate futures

If central bank base rate increases to 5.3%

Exercise price	94.25	95.25
Futures price, as above	94.16	94.16
Exercise?	No	No
	D	D
Investment return as above	562,500	562,500
Premium	<u>(61,313)</u>	<u>(11,025)</u>
Net receipt	501,187	551,475
Effective annual interest rate		
$501,187 / 27,000,000 \times 12/5$	4.45%	
$551,475 / 27,000,000 \times 12/5$		4.90%

Implications for hedging strategy

The futures market provides a lower return than both option contracts although we have already determined that a risk averse treasury manager would have ignored the 95.25 option. However, the 94.25 option now looks more attractive than the futures outcome even though the central bank base rate increased exactly as predicted.

If the relationship between the futures and spot price is not predictable, there is no guarantee that the closing futures price on 31 January will match the price predicted by our basis calculation. On 1 October the futures price prediction for 31 January is 94.36, assuming the base rate increases to 5.3%, but it could be either more or less even when the base rate increases exactly as predicted.

This exposure to basis risk means the actual futures price fell to 94.16 on 31 January instead of 94.36. An unexpected change in basis, even marginally, reduces the return on the futures hedge from 4.58% to 4.38% as opposed to a return of 4.45% using the 94.25 option. With the benefit of hindsight the 94.25 option would have provided a better outcome. However, the treasury department have limited visibility of the future when choosing an optimal strategy at the time the hedge is set up on 1 October. The treasury department's best estimate of the closing futures price is based on a simplifying assumption that basis diminishes at a constant rate, which may not hold true in practice. Basis risk, therefore, introduces an element of unpredictability which the treasury staff need to be aware of at the outset. Scenario analysis would be useful in determining the final strategy in accordance with the company's risk preferences.

Written by a member of the AFM examining team



Related Links

- [Student Accountant hub page](#)



Think Ahead

How to answer an interest rate risk management question

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **How to answer an interest rate risk management question**

Questions on risk management feature regularly in the *Advanced Financial Management* exam. Performance information from recent exams suggests students tend to do less well on interest rate risk management questions than questions about foreign exchange risk management. This article will therefore explain the significance of the information you'll be given in interest rate risk management questions and show you what you'll be asked to do.

The scenario is adapted from Wardegul Co, Question 4 in the September/December 2017 sample questions which ACCA has published.

Scenario

Assume Wardegul Co has a newly-acquired subsidiary in Euria, where the local currency is the dinar (D). The subsidiary expects to receive D27,000,000 and wants to invest this D27,000,000. Assume it is now 1 October 2017 and the subsidiary expects to receive the money on 31 January 2018. It wishes the money to be invested for five months until 30 June 2018.

Currently the central bank base rate in Euria is 4.2%, but Wardegul Co's treasury team has seen predictions that the central bank base rate could increase by up to 1.1% or fall by up to 0.6% between now and 31 January 2018. The treasury team believes that Wardegul Co can invest funds at the central bank base rate less 30 basis points.

The treasury team normally hedge interest rate exposure by using whichever of the following products is most appropriate:

- Forward rate agreements (FRAs)
- Interest rate futures
- Options on interest rate futures

Treasury function guidelines emphasise the importance of mitigating the impact of adverse movements in interest rates. However, they also allow staff to take into consideration upside risks associated with interest rate exposure when deciding which instrument to use.

A local bank in Euria, with which Wardegul Co has not dealt before, has offered the following FRA rates:

- 4–9: 5.02%
- 5–10: 5.10%

The treasury team has also obtained the following information about exchange traded Dinar futures and options:

Three-month D futures, D500,000 contract size

Prices are quoted in basis points at 100 – annual % yield

December 2017	94.84
March 2018	94.78
June 2018	94.66

Options on three-month D futures, D500,000 contract size, option premiums are in annual %

Call

Put

December	March	June		December	March	June
0.417	0.545	0.678	94.25	0.071	0.094	0.155
0.078	0.098	0.160	95.25	0.393	0.529	0.664

It can be assumed that futures and options contracts are settled at the end of each month. Basis can be assumed to diminish to zero at contract maturity at a constant rate, based on monthly time intervals. It can also be assumed that there is no basis risk and there are no margin requirements.

Requirements

Recommend a hedging strategy for the D27,000,000 investment, based on the hedging choices which treasury staff are considering, if interest rates increase by 1·1% or decrease by 0·6%. Support your answer with appropriate calculations and discussion. (18 marks)

Approaching the question

Read the requirements carefully

You should read the requirements first before reading the scenario in detail. Knowing what your answer has to cover, and therefore what the key data will be, will help you analyse the scenario.

Breaking down the requirements for Wardegul Co:

Recommend a hedging strategy

You'll have to make a clear recommendation based on your calculations. Anyone reading the recommendation should be able to see:

- How much would be received under each instrument
- The effective annual interest rate for each instrument

so that they can compare the results of the hedging choices with the interest rate currently available.

Based on the
hedging choices
which treasury staff
are considering

You need to consider all the hedging
instruments for which data is given, including
both the options.

If interest rates
increase by 1·1% or
decrease by 0·6%

You should assess, for all the hedging
instruments, what will happen if interest rates
rise or fall.

Support your answer
with appropriate
calculations and
discussion

You should make some comment on any
calculation you carry out in the *Advanced
Financial Management* exam. However,
mentioning discussion in the question
requirements here indicates that a number of
marks will be available for comments (four
marks maximum per the marking scheme).
Therefore, a single sentence comment won't
be enough.

Identify the important data in the scenario

For interest rate hedging questions, you need to identify the information that will affect the calculations for each instrument. Let's have another look at the scenario, with the important data highlighted and referenced to explanations below.

Assume Wardegul Co has a newly-acquired subsidiary in Euria, where the local currency is the dinar (D). The subsidiary expects to receive D27,000,000 and **wants to invest this D27,000,000**.

Forward rate
agreements

Futures

Options

	Forward rate agreements	Futures	Options
Wants to invest this D27,000,000	Possibilities are: • Pay money to bank if base rate exceeds FRA rate • Receive money from bank if FRA rate is greater than base rate	Buy now (go long), sell later	Buy call option

Assume it is now 1 October 2017 and the subsidiary expects to receive the money on 31 January 2018.

	Forward rate agreements	Futures	Options
<i>It is now 1 Oct 2017 and the subsidiary expects to receive the money on 31 Jan 2018</i>	A period of four months, so look for a 4-x agreement	Choose futures dated after January – March is closest date	Choose options dated after January – March is closest date

It wishes the money to be invested for five months until 30 June 2018.

Forward rate agreements	Futures	Options
-------------------------	---------	---------

	Forward rate agreements	Futures	Options
<u>The money to be invested for five months until 30 June 2018</u>	Four months to start of investment + five months to end of investment = nine months, so select 4– 9 agreement	Contracts are for three months, so adjust contracts calculation, so that five month period is covered	Contracts are for three months, so adjust contracts calculation, so that five month period is covered
	Calculate investment return for five months	Calculate investment return for five months	Calculate investment return for five months
	Calculate transaction with bank for five months		
	Adjust effective annual interest rate calculation for interest being received for five months	Adjust effective annual interest rate calculation for interest being received for five months	Adjust effective annual interest rate calculation for interest being received for five months

Currently the central bank base rate in Euribor is 4.2%,

Forward rate agreements	Futures	Options
----------------------------	---------	---------

	Forward rate agreements	Futures	Options
Currently the central bank base rate in Euribor is currently 4.2%	Affects calculation of: • Future interest rates	Affects calculations of: • Future interest rates • Basis	Affects calculations of: • Future interest rates • Basis

but Wardegul Co's treasury team has seen predictions that **the central bank base rate could increase by up to 1.1% or fall by up to 0.6%** between now and 31 January 2018.

	Forward rate agreements	Futures	Options
The central bank base rate could increase by up to 1.1% or fall by up to 0.6%	Affects future interest rates and hence: • Actual investment return • Transaction with bank	Affects future interest rates and hence: • Actual investment return • Calculation of expected futures price and hence result on futures market	Affects future interest rates and hence: • Actual investment return • Calculation of expected futures price and hence whether options are exercised or not • Calculation of gain if options are exercised

The treasury team believes that Wardegul Co can invest funds at the central bank base rate less 30 basis points.

	Forward rate agreements	Futures	Options
<u>Wardegul Co can invest funds at the central bank base rate less 30 basis points</u>	Affects actual investment return: • If rate rises to 5.3%, investment return will be 5.0% • If rate falls to 3.6%, investment return will be 3.3%	Affects actual investment return: • If rate rises to 5.3%, investment return will be 5.0% • If rate falls to 3.6%, investment return will be 3.3%	Affects actual investment return: • If rate rises to 5.3%, investment return will be 5.0% • If rate falls to 3.6%, investment return will be 3.3%

The treasury team normally hedges interest rate exposure by using whichever of the following products is most appropriate:

- Forward rate agreements (FRAs)
- Interest rate futures
- Options on interest rate futures

Treasury function guidelines emphasise the importance of mitigating the impact of adverse movements in interest rates.

However, they also allow staff to take into consideration upside risks associated with interest rate exposure when deciding which instrument to use.

A local bank in Euria, with which Wardegul Co has not dealt before, has offered the following FRA rates:

- 4–9: 5.02%
- 5–10: 5.10%

The treasury team has also obtained the following information about exchange traded Dinar futures and options:

Three-month D futures, D500,000 contract size

Forward rate
agreements

Futures

Options

Three-month
D500,000 futures

Affects calculations of:

- Number of futures contracts
- Result on futures contracts

Prices are quoted in basis points at 100 – annual % yield

December 2017 94.84

March 2018 94.78

June 2018 94.66

Options on three-month futures, D500,000 contract size, option premiums are in annual %

Forward rate
agreements

Futures

Options

Forward rate agreements	Futures	Options
<u>Options on three-month futures,</u> <u>D500,000 contract size</u>		Affects calculation is of:
		• Number of options contracts • Gain if options are exercised • Option premium

Call				Put		
December	March	June		December	March	June
0.417	0.545	0.678	94.25	0.071	0.094	0.155
0.078	0.098	0.160	95.25	0.393	0.529	0.664

It can be assumed that futures and options contracts are settled at the end of each month. **Basis can be assumed to diminish to zero at contract maturity at a constant rate, based on monthly time intervals.** It can also be assumed that there is no basis risk and there are no margin requirements.

Forward rate agreements	Futures	Options
-------------------------	---------	---------

Forward rate
agreements

Futures

Options

**Basis can be
assumed to
diminish to
zero at contract
maturity at a
constant rate,
based on
monthly time
intervals**

Use in basis
calculation:

- Period between investment date (31 January) and contract maturity date (31 March) (two months)

- Period between today's date (1 October) and contract date (31 March) (six months)

Use in basis
calculation:

- Period between investment date (31 January) and contract maturity date (31 March) (two months)

- Period between today's date (1 October) and contract date (31 March) (six months)

Let's now review the answer:

Forward rate agreement

FRA 5.02% (4 – 9) since the investment will take place in *four months' time* for a period of five months.

If interest rates **increase by 1.1% to 5.3%**

D

Actual investment return **5.0%** × 5/12 × D27,000,000 562,500

Payment to bank (5.3% – 5.02%) × 5/12 × D27,000,000 (31,500)

Net receipt 531,000

Effective annual interest rate $531,000/27,000,000 \times 12/\underline{5}$ 4.72%

D

Actual investment return $\underline{3.3\%} \times \underline{5}/12 \times D27,000,000$ 371,250

Receipt from bank $(5.02\% - \underline{3.6\%}) \times \underline{5}/12 \times D27,000,000$ 159,750

Net receipt 531,000

Effective annual interest rate as above 4.72%
 $531,000/27,000,000 \times 12/\underline{5}$

Comment

The two calculations should give the same effective annual interest rate.

Futures

Buy futures now (go long in the futures market), as the hedge is against a fall in interest rates.

Use March contracts, as investment will be made on 31 January.

Number of contracts = $D27,000,000 \div \underline{D500,000} \times \underline{5 \text{ months}} \div \underline{3 \text{ months}} = 90 \text{ contracts}$

Basis

Current price (1 October) – futures price = basis

$(100 - \underline{4.20}) - 94.78 = 1.02$

Unexpired basis on 31 January = $\underline{2/6} \times 1.02 = 0.34$

If interest rates **increase by 1.1% to 5.3%**

D

Actual investment return $\underline{5.0\%} \times \underline{5/12} \times \text{D}27,000,000$ 562,500

Expected futures price: $100 - \mathbf{5.3} - \underline{0.34} = 94.36$

Loss on the futures market: $(0.9436 - 0.9478) \times$ (47,250)
 $\underline{\text{D}500,000} \times \underline{3/12} \times 90$

Net return 515,250

Effective annual interest rate $515,250 / 27,000,000 \times 12 / \underline{5}$ 4.58%

If interest rates **fall by 0.6% to 3.6%**

D

Actual investment return $\underline{3.3\%} \times \underline{5/12} \times \text{D}27,000,000$ 371,250

Expected futures price: $100 - \mathbf{3.6} - \underline{0.34} = 96.06$

Profit on the futures market: $(0.9606 - 0.9478) \times$ 144,000
 $\underline{\text{D}500,000} \times \underline{3/12} \times 90$

Net receipt 515,250

Effective annual interest rate $515,250 / 27,000,000 \times 12 / \underline{5}$ 4.58%

Comment

The two calculations should give the same effective annual interest rate.

As we are buying futures now, then selling futures later:

- we make a **PROFIT** if the expected futures price is **GREATER** than the current futures price
- we make a **LOSS** if the expected futures price is **LESS** than the current futures price

Options

Buy call options as need to hedge against a fall in interest rates.

Use *March contracts*, as investment will be made on 31 January.

Number of contracts = $D27,000,000 \div D500,000 \times \frac{5 \text{ months}}{3 \text{ months}} = 90 \text{ contracts}$

Basis

Current price (1 October) – futures price = basis

$(100 - 4.20) - 94.78 = 1.02$

Unexpired basis on 31 January = $\frac{2}{6} \times 1.02 = 0.34$

If interest rates **increase by 1.1% to 5.3%**

Exercise price	94.25	95.25
Expected futures price:	94.36	94.36
$100 - 5.3 - 0.34 =$		
94.36		
Exercise?	Yes	No
Gain in basis points	11	0

	D	D
Actual investment return <u>5.0%</u> × <u>5/12</u> × D27,000,000	562,500	562,500
Gain from options $0.0011 \times \underline{D500,000} \times$ <u>3/12</u> × 90	12,375	0
Premium $0.00545 \times \underline{D500,000} \times$ <u>3/12</u> × 90	(61,313)	
$0.00098 \times \underline{D500,000} \times$ <u>3/12</u> × 90		(11,025)
Net return	<u>513,562</u>	<u>551,475</u>
Effective interest rate $513,562/27,000,000 \times$ <u>12/5</u>	4.56%	
$551,475/27,000,000 \times$ <u>12/5</u>		4.90%
Exercise price	94.25	95.25
Expected futures price: 100 – 3.6 – <u>0.34</u> = 96.06	96.06	96.06

Exercise?	Yes	Yes
Gain in basis points	181	81
Actual investment return $3.3\% \times \frac{5}{12} \times$ D27,000,000	371,250	371,250
Gain from options		
$0.0181 \times \underline{D500,000}$ $\times \frac{3}{12} \times 90$	203,625	
$0.0081 \times \underline{D500,000}$ $\times \frac{3}{12} \times 90$		91,125
Premium $0.00545 \times \underline{D500,000}$ $\times \frac{3}{12} \times 90$	(61,313)	
$0.00098 \times \underline{D500,000}$ $\times \frac{3}{12} \times 90$		(11,025)
Net return	<u>513,562</u>	<u>451,350</u>
Effective interest rate		
$513,562/27,000,000$ $\times 12/\underline{5}$	4.56%	
$451,350/27,000,000$ $\times 12/\underline{5}$		4.01%

Comment

If one of the options is exercised for both interest rates, as the 94.25 is here, the calculations should give the same result.

As these are **CALL** options, options to buy, choose the **LOWER** price and so:

- If the exercise price is **LOWER** than the expected futures price, **EXERCISE**
- If the exercise price is **HIGHER** than the expected futures price, **DO NOT EXERCISE**

Discussion

The forward rate agreement gives the highest guaranteed return. If Wardegul Co wishes to have a certain cash flow and is primarily concerned with protecting itself against a fall in interest rates it will most likely choose the forward rate agreement. The 95.25 option gives a better rate if interest rates rise, but a significantly lower rate if interest rates fall, so if Wardegul Co is at all risk averse it will choose the forward rate agreement.

This assumes that the bank with Wardegul Co deals with is reliable and there is no risk of default. If Wardegul Co believes that the current economic uncertainty may result in a risk that the bank will default, the choice will be between the futures and the options, as these are guaranteed by the exchange. Again the 95.25 option may be ruled out because it gives a much worse result if interest rates fall to 3.6%. The futures give a marginally better result than the 94.25 option in both scenarios but the difference is small. If Wardegul Co feels there is a possibility that interest rates will be higher than 5.41%, the point at which the 94.25 option would not be exercised, it may choose this option rather than the future.

Comment

Identifying which of the possible strategies gives the highest value is only the start of the discussion and you need to consider other factors that may influence the decision to obtain four marks:

- The level of risk aversion that Wardegul Co has. The treasury team appears to be weighing limiting downside against the possibility of taking advantage of upside.
- Other risk considerations are also important. There may be counterparty risk, as FRAs are over-the-counter instruments.
- The decision may depend upon what is believed about future interest rates. Here, as rates are volatile, you should consider whether the decision would change depending on what interest rates are expected.

The discussion should be in full sentences and use information relevant to the scenario. A bullet point list or generic statements relating to hedging are unlikely to be awarded many marks.

Conclusion

This article has demonstrated how to use the data given in the question to calculate the impact of interest rate hedging. Hopefully it will help you tackle interest rate risk management questions in a structured way, which should mean that you score well.

Written by a member of the examining team for *Advanced Financial Management*



Related Links

- [Student Accountant hub](#)



Think Ahead

Our sites

[myACCA](#)

[ACCA Mail](#)

[ACCA Careers](#)

[ACCA Learning Community](#)

[Your Future](#)

Useful links

[Make a payment](#)

[ACCA-X online courses](#)

[Find an accountant](#)

[ACCA Rulebook](#)

[News](#)

Most popular

[Professional insights](#)

[ACCA Qualification](#)

[Member events and CPD](#)

[Covid-19 hub](#)

[Past exam papers](#)



Call Us

[+44 \(0\)141 582 2000](tel:+441415822000)



ACCA Mail

[Login to ACCA mail](#)

[Accessibility](#)

[Legal policies](#)

[Data protection & cookies](#)

[Advertising](#)

[Site map](#)

[Contact us](#)

Reverse takeovers

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Reverse takeovers**

Reverse takeovers is a topic that is examinable in *Advanced Financial Management*. This article aims to provide an explanation of reverse takeovers and to discuss the potential benefits and drawbacks associated with reverse takeovers (RTO).

Reverse takeovers – an explanation

An RTO involves a smaller quoted company taking over a larger unquoted company by a share-for-share exchange. In order to acquire the larger unquoted company, a large number of shares in the quoted company will have to be issued to the shareholders of the larger unquoted company. Hence, after the takeover the current shareholders in the larger unquoted company will hold the majority of the shares in the quoted company and will therefore have control of the quoted company.

On completion of an RTO, it is usual for the quoted company to be managed by the senior management team from the previously unquoted company and to take the name of the previously unquoted company.

Through the RTO, the previously unquoted company has effectively achieved a listing on the stock exchange. Eddie Stobart, a road haulage company based in the UK, achieved a listing in this way in 2007 by combining with Westbury, a property and

logistics company.

It is worth noting that in the USA, the term 'reverse merger' is often used as opposed to the term reverse takeover.

As ever, there are many variations on the basic idea. For instance, an RTO may involve a quoted company, which is actively trading, or a shell company, which is not actively trading.

RTOs have often been deemed to be the poor man's initial public offering (IPO) perhaps due to US studies showing that companies achieving a listing through a reverse merger generally have lower survival rates and underperform compared to companies who have achieved their listing through a traditional IPO.

However, studies in the UK have shown that this is not necessarily the case. Indeed, during the period 1995 to 2012, RTOs seem to have survival rates similar to those for IPOs. The best results seem to arise with RTOs, which involve a quoted company that is actively trading, as the takeover is then able to benefit from synergy gains. Equally, small RTOs seem to perform better than larger ones.

Reverse takeovers – the potential benefits

As previously stated, an RTO is effectively a way that a currently unquoted company can achieve a listing. Hence, just as with an IPO, the company obtains the benefits of the public trading of its securities. These benefits include:

Easier access to capital markets

As a listed company, more finance is likely to be available and the cost of that finance is likely to be lower than if the company was still unquoted.

Higher company valuation

As the shares in the company will be listed, potential investors will deem the shares to be less risky as the company will have to abide by the relevant rules and regulations. Additionally, they will know that the shares are liquid and that whenever they wish to sell there will be a willing buyer. As a result of this, investors are likely to attribute a higher value to the shares.

Enhanced ability to carry out further takeovers

Once the shares in a company are listed, the company is able to acquire other companies through further share-for-share exchanges.

Enhanced ability to use share based incentive plans

Once the shares of a company are listed, share based incentive plans can be used as a key tool to attract and retain good quality employees.

In addition to the above, an RTO has a number of other potential benefits when compared to a normal IPO. These include the following:

Speed

An IPO can often take between one and two years to complete whereas an RTO can be completed in as little as 30 days. Furthermore, the work required to complete an IPO can mean that the managers of a company have less time to run the company, which may prove detrimental to the growth prospects of the company. The variability of market conditions can also make the speed of an RTO attractive, as in the time taken to prepare for an IPO, the market may deteriorate such that the IPO is not finally worth completing. Furthermore, particular circumstances in a market may make RTOs attractive. For instance, in China the IPO process is notoriously slow and there is usually a significant queue of companies waiting to carry out an IPO. An RTO allows a company to jump this queue.

Cost

Just as an IPO is a time-consuming process, it is also an expensive one due to the volume of work required by investment banks, sponsors, accountants and other advisers. An RTO will usually, but not always, cost less.

Availability

In a market downturn it is not easy to convince investors to support an IPO, whereas this does not seem to be the case with RTOs. Studies have shown that the volume of RTO transactions is far more resilient to market downturns. During the market correction that followed the bursting of the dotcom bubble, the number of RTOs actually increased while the number of IPOs fell very significantly.

Similarly, the fall in the number of RTOs was less than the fall in the number of IPOs following the more recent financial crisis. This is probably because, with an RTO, the deal is fundamentally between the shareholders of the quoted and unquoted companies involved and, hence, market sentiment has much less import.

Furthermore, while an RTO is often accompanied by a concurrent secondary offering to raise new finance, the amount of new finance being raised in both \$ and % terms is usually less than that which is raised during an IPO. Hence, even in a downturn, investors are often more willing to support an RTO rather than an IPO.

Existing analyst coverage

A listed company subject to an RTO is likely to have existing analyst coverage and, after the RTO, this analyst coverage usually continues. However, companies that use an IPO may struggle to get significant analyst coverage especially if they are smaller. Without reasonable analyst coverage, potential investors may not have much awareness of the company and, hence, are unlikely to want to invest in the company.

Reverse takeovers – the potential drawbacks

RTOs do, however, have a number of potential drawbacks when compared to an IPO and any company considering an RTO should be aware of these.

Lack of expertise

A company achieving a listing through an RTO may find that it does not have the expertise to understand and deal with all the regulations and procedures that listed companies must comply with. The long process of listing through an IPO can be viewed as a valuable training period and any company that has been through the process is in a better position to deal with the

requirements of the exchange than a company catapulted onto the market through an RTO. Hence, any company considering an RTO must consider the need to hire and/or retain staff from the existing listed company who are able to keep the company compliant with all the relevant regulations.

Reputation

As previously discussed, an RTO has often been viewed as a poor man's IPO. Hence, companies that achieve their listing in this way may be viewed less favourably by investors than companies that have completed an IPO. To some extent, the reasons for this lie in the recent past.

In 2011 and 2012, there were a number of accounting scandals involving Chinese firms that gained access to the US markets through RTOs. Indeed, over 100 companies were suspended or delisted as a result. A public bulletin issued by the US Securities and Exchange Commission in June 2011 warned investors by stating that 'many companies either fail or struggle to remain viable following a reverse merger' and that there have been 'instances of fraud and other abuses involving reverse merger companies'.

Since that time, the regulations and standards that apply to reverse mergers / RTOs on the US stock exchanges and other exchanges around the world have been tightened up in order to prevent similar problems arising in the future.

Risk

As a result of the lower level of scrutiny that is applied to an RTO compared to an IPO, investors must be aware of the higher level of risk that is attached to companies achieving a listing in this way. In particular, the unquoted company carrying out an RTO must ensure that there is a thorough investigation of the listed company which they are taking over so that all potential problems and liabilities are revealed.

Regulation

Although RTOs can generally be completed more quickly than an IPO as there is less regulation and scrutiny involved, it must be recognised that there are still a significant amount of regulatory hurdles to overcome. It should be understood that RTOs are, to some extent, combinations of acquisitions and IPOs and, as such, are potentially complex and difficult deals to manage. By way of example, two regulatory issues that may arise are now discussed:

- **Suspension**

The Financial Conduct Authority's (FCA) standard view is that when an RTO is announced or leaked, there will generally be insufficient information publicly available on the proposed transaction. In particular, information on the unquoted company contemplating the takeover could well be limited compared to the information that is available on listed companies. As a result of this, the listed company will not be able to accurately assess its financial position and inform the market. Hence, the FCA will often consider that a suspension of trading in the shares is appropriate. This standard view can be rebutted, but there is significant work required to achieve this. However, this work is essential as the listed company will not want to contemplate a scenario where its listing is suspended and is quite likely to walk away from the proposed transaction were this to occur.

- **Mandatory offer**

If, individually or with their closely connected persons or friends, any shareholder in the unquoted company carrying out an RTO will on completion of the transaction hold shares that carry 30% or more of the voting rights of the listed company, then that shareholder will be required to make a general cash offer for the remaining shares in the listed company under the mandatory bid rule. This would obviously undermine the reason for doing the RTO in the first place. While the takeover panel will usually consent to a waiver of this requirement as long as certain conditions are satisfied, it is another regulatory obstacle which must be navigated around carefully.

Share price decrease

Many listed companies which could make potential RTO targets are in that position because of past problems. Hence, they may have shareholders who are keen to exit from the company as soon as a suitable opportunity arises and, hence, they may 'dump' their shares shortly after the RTO has completed. To safeguard against the risk of a 'dump' occurring, the shareholders may need to guarantee that they will not sell their shares until a certain period of time has elapsed since the deal is completed. This is called a lock-up and/or a lock-up period.

Cost

While a reverse takeover is usually cheaper than an IPO, there are still significant direct and indirect costs involved and, hence, the total cost can easily be far more than was originally anticipated. A number of these costs are now considered:

- **Regulatory costs**

As mentioned previously, an RTO is a complex transaction and to ensure that the regulatory hurdles are successfully overcome will incur significant cost.

- **Acquisition cost**

As a result of an RTO being seen as an easier and quicker option than an IPO, especially in the Chinese market, the value of potential listed company targets are often at a significant premium to their true value. Furthermore, the pressure to find a target has resulted in some unusual combinations such as a mobile computer game developer getting listed through the acquisition of a shoe company! It is hard to imagine there were any synergy gains available here and, indeed, resolving cultural and other issues that may well have arisen would have further added to the indirect cost of achieving the listing.

- **Investor relations**

Although an RTO may benefit from existing analyst coverage, RTO transactions only really introduce liquidity to a previously private company if there is real investor interest in the company. In many cases, in order to generate this interest, a comprehensive investor relations and investor marketing programme will be required. This is another potential indirect cost of an RTO.

Conclusion

As with anything that seems too good to be true, it must be recognised that an RTO is not without significant complication and cost. Just as there is no such thing as a free lunch, there is also no easy way to achieve a listing.

William Parrott, freelance tutor and senior FM tutor, MAT Uganda

References:

- BPP ACCA Approved Study Text
- Reverse takeovers: No longer the poor man's IPO, M&A Research Centre at Cass Business School
- Reverse Takeovers, Stepping through the looking glass, Karen Davies, Dominic Ross and James Fletcher of Ashurst LLP
- Leapfrogging the IPO gridlock: Chinese companies get a taste for reverse takeovers, [Elzio Barreto](#)

Conditional probability

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Conditional probability**

Management decision making in both financial and performance management involves making choices based upon what may happen in the future. Such future events can be predicted but decision makers can rarely be 100% certain that these future events will actually occur or will occur in the way in which they are forecast.

There are a variety of ways of dealing with such uncertainty in forecast outcomes. In this article, we shall look at how to consider all of the possible outcomes that may arise, together with their associated chances of occurring. This is referred to as a situation where we are evaluating risk, since there is past information or experience which can provide statistical evidence in order to assist in determining the possibility of each event occurring.

Expected values, decision trees and combined probabilities

A commonly used way of evaluating decisions is via the use of expected values.

An expected value summarises all the different possible outcomes by weighting the possible outcomes by their probabilities and then summing the result.

Problems where one or more decisions have to be taken can become more complex and may require the use of a decision tree, with expected values being used to evaluate each of the decisions.

A decision tree is a diagrammatic representation of a problem, where the decision maker needs to consider the logical sequence of events.

Since one event may depend upon another, we may get situations where event one has a certain probability of occurring and event two, which depends on event one occurring, has another probability of occurring. In such circumstances, we have a situation of combined probabilities

For example, if event one has a 0.6 chance of occurring and subsequent event two a 0.75 chance of occurring, then overall the probability of both events occurring is:

$$0.6 \times 0.75 = 0.45$$

ie a 45% chance of occurring.

We shall look at such concepts in the following example, which demonstrates how techniques acquired from the Applied Skills exams can be used in the Strategic Professional exams.

Scenario

Brisport Master Motor Co (Brisport) designs, manufactures and sells a range of components for the motor car industry. The design team has recently designed a new component for inclusion into hybrid cars. The component greatly enhances the battery 'road time' and therefore reduces the frequency with which the battery has to be recharged.

The company can either sell the design now, for its initial market value of \$400,000, or attempt to develop the design into a marketable product, which can be supplied to the motor industry. This development would have an initial outlay of \$300,000 now and the component would take one year to be developed. In such a fast moving market, the component is likely to have a market life as a saleable product of just five years after development.

If the company decides to develop the component, the chances of succeeding in developing the design into the marketable product are 80%. If the attempt to develop fails, the design can only be sold, in one year's time, for half of its earlier market value.

If the attempt to develop the design succeeds the company has a choice of either selling both the design and the rights to sell the developed component, or marketing the component themselves.

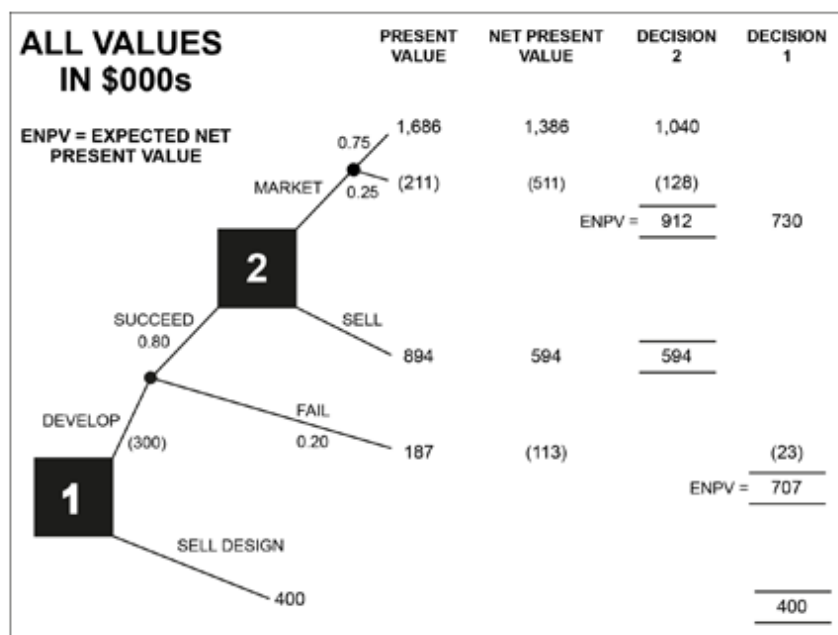
Selling the design would yield \$300,000 in one year's time and \$160,000 in royalty payments for each of the five years thereafter (years 2 to 6).

If the component is marketed by Brisport then there is a 75% probability that the product will be popular and will generate cash inflows of \$440,000 per annum but there is a 25% probability that it will be unpopular and it will generate cash outflows of \$55,000 per annum. Both cash flow figures are also for each of years 2 to 6.

Brisport uses a weighted average cost of capital of 7% to discount its future cash flows.

The management of Brisport Master Motor Co seeks your advice as to their best course of action.

Solution



There are two decisions which need to be taken by the company:

1. sell the design or develop it
2. if it is developed, then whether to sell the design and the rights to sell the developed component, or market it themselves

Initial workings relating to present value of future cash flows using discount factors at 7% (all values rounded to the nearest \$000s):

Discount factor for $T_1 = 0.935$

Annuity factor ($T_2 - T_6$) 3.832 (Annuity factor ($T_1 - T_6$) $- 0.935 = (4.767 - 0.935)$)

'Development fails'

Selling the design in one year's time for half of its initial market value:

$200 \times 0.935 = \mathbf{187}$

'Development succeeds'

Selling the design in one year's time plus royalties ($T_2 - T_6$)

$(300 \times 0.935) + (160 \times 3.832) = 281 + 613 = \mathbf{894}$

Successful marketing of the component ($T_2 - T_6$)

$440 \times 3.832 = \mathbf{1,686}$

Unsuccessful marketing of the component ($T_2 - T_6$)

$-55 \times 3.832 = \mathbf{-211}$

In order to evaluate decision 1, decision 2 needs to be evaluated first. In other words, the values we use in decision 1 need to be determined by the decision we take in decision 2.

Decision 2

The net present value (\$000s) on the 75% path is

$$1,686 - 300 = 1,386.$$

Taken together with the net present value (\$000s) on the 25% path of

$$(211) - 300 = (511)$$

there is an expected net present value of choosing to market the component of (\$000s):

$$\begin{aligned} & [0.75 \times 1,386] + [0.25 \times (511)] \\ &= 1,040 - 128 \\ &= 912 \end{aligned}$$

This is a higher value than the option of selling the design and the rights to sell the developed component for a net present value of \$594,000 (\$894,000 – \$300,000). Therefore, if the development goes ahead, it will be more beneficial to market the product.

Of course, we still need to evaluate decision 1, whether to develop at all. The 'success' of the expected present value of \$912,000 in decision 2 has an 80% chance of arising, but there is a 20% chance of the development not succeeding and recouping just half of the initial market value, that being \$187,000 in present value terms, resulting in the company being worse off by \$113,000 in present value terms after taking the development costs into account.

Hence, the expected net present value of the development option of decision 1 can be calculated (\$000s):

$$\begin{aligned}
 &= [0.80 \times 912] + [0.20 \times (113)] \\
 &= 730 - 23 \\
 &= 707
 \end{aligned}$$

Since this is higher than the option to sell the design at time 0, \$400,000, on an expected value basis, the component should be developed and marketed.

Attitude to risk

The expected value approach assumes risk neutrality, but not all management decision makers are risk neutral. A risk averse management would, in this scenario, be concerned with the 20% probability of being \$113,000 worse off in present value terms should the development decision go on to fail.

Furthermore, having taken the decision (at node 2) that marketing the component is preferred to selling both the design and developed component there is a further risk of losses, since there is a 25% chance of the component being unpopular leaving the company worse off by \$511,000 in present value terms.

Combined with the 80% probability of the development being successful, there is an overall 20% chance of this \$511,000 loss. This 20% is known as a conditional probability since it depends upon the 80% (0.80) success rate firstly and then depends on the 25% (0.25) unpopularity chance.

Hence,

$$0.80 \times 0.25 = 0.20 \text{ ie } 20\%$$

For completeness, there is of course a 75% chance of the component being popular if marketed, and hence the overall combined probability of a successful development together with a marketing campaign which results in popularity is:

$$\begin{aligned}
 0.80 \times 0.75 &= 0.60 \\
 &\text{ie } 60\%
 \end{aligned}$$

Summary

Outcome	Probability	Net present value (\$000s)
---------	-------------	-------------------------------

Outcome	Probability	Net present value (\$000s)
Development succeeds and component is popular	60%	1,386
Development succeeds but component is unpopular	20%	(511)
Development fails	20%	(113)

Therefore, be aware that expected values can lead to a false sense of security. The expected NPV of \$707,000 is an average. In other words, it is the average NPV if the decision is repeated over and over again. But is that useful in this situation? This is a one-off development of a product and therefore only one of the outcomes listed in the table above will actually occur. (This is analogous to tossing a coin once. We know that the outcome will either be a head or a tail, not the expected value of 'half a head' or 'half a tail'). As can be seen above, there is a 40% chance that the NPV will be negative, and that is maybe a risk that the company is not prepared to take.

Furthermore, be aware that the analysis largely depends upon the values of the probabilities prescribed. Often these are subjective estimates made by the decision makers and it would only take relatively small changes in these to alter one of the decisions.

For example, in decision 2, if the probability of successful marketing falls to 55%, then the expected NPV of 'marketing' falls to:

$$\begin{aligned}
 & [0.55 \times 1,386] + [0.45 \times (511)] \\
 & = 762 - 230 \\
 & = 532
 \end{aligned}$$

This is now a lower value than the option of selling the design and the rights to sell the developed component for a net present value of \$594,000.

Such sensitivity analysis can be performed on other variables within the model.

Of course, decision models such as this are only as good as the information used. In reality there would probably be a much wider range of possible outcomes than the discrete outcomes described above. In other words, the problems in examination questions are a simpler version of what is usually found in reality, but nonetheless are very useful techniques of which you should be aware.

Currency swaps

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Currency swaps**

A currency swap is an agreement in which two parties exchange the principal amount of a loan and the interest in one currency for the principal and interest in another currency.

At the inception of the swap, the equivalent principal amounts are exchanged at the spot rate.

During the length of the swap each party pays the interest on the swapped principal loan amount.

At the end of the swap the principal amounts are swapped back at either the prevailing spot rate, or at a pre-agreed rate such as the rate of the original exchange of principals. Using the original rate would remove transaction risk on the swap.

Currency swaps are used to obtain foreign currency loans at a better interest rate than a company could obtain by borrowing directly in a foreign market or as a method of hedging transaction risk on foreign currency loans which it has already taken out.

We will consider how a fixed for fixed currency swap works by looking at an example.

An American company may be able to borrow in the United States at a rate of 6%, but requires a loan in rand for an investment in South Africa, where the relevant borrowing rate is 9%. At the same time, a South African company wishes to finance a project in the United States, where its direct borrowing rate is 11%, compared to a borrowing rate of 8% in South Africa.

Each party can benefit from the other's interest rate through a fixed-for-fixed currency swap. In this case, the American company can borrow U.S. dollars for 6%, and then it can lend the funds to the South African company at 6%. The South African company can borrow South African rand at 8%, then lend the funds to the U.S. company for the same amount.

Currency swaps can also involve exchanging two variable rate loans, or fixed rate borrowing for variable rate borrowing. Let's consider a case where a company exchanges fixed rate borrowing for variable rate borrowing.

Barrow Co, a company based in the USA, wants to borrow €500m over five years to finance an investment in the Eurozone.

Today's spot exchange rate between the Euro and US \$ is $\text{€}1 \cdot 1200 = \$1$.

Barrow Co's bank can arrange a currency swap with Greening Co. The swap would be for the principal amount of €500m, with a swap of principal immediately and in five years' time, with both these exchanges being at today's spot rate.

Barrow Co's bank would charge an annual fee of 0.4% in € for arranging the swap.

The benefit of the swap will be split equally between the two parties.

The relevant borrowing rates for each party are as follows:

	Barrow Co	Greening Co
USA	3.6%	4.5%
Eurozone	EURIBOR + 1.5%	EURIBOR + 0.8%

We will see what the gain on the swap for each party will be.

	Barrow Co	Greening Co	Benefit
USA	3.6%	4.5%	0.9%
Eurozone	EURIBOR + 1.5%	EURIBOR + 0.8%	0.7%
Gain on swap	0.8%	0.8%	1.6%
Bank fee	(0.2%)	(0.2%)	(0.4%)
Final gain	0.6%	0.6%	1.2%

Using this gain to work out the overall result for each company, we can provide an illustration of how the swap could work as follows:

	Barrow Co	Greening Co
Barrow Co borrows	3.6%	
Greening Co borrows		EURIBOR + 0.8%
Swap		
Greening Co receives		(EURIBOR)
Barrow Co pays	EURIBOR	
Barrow Co receives	(2.9%)	

	Barrow Co	Greening Co
Greening Co pays		2.9%
Net result	EURIBOR + 0.7%	3.7%
Bank fee	0.2%	0.2%
Overall result	EURIBOR + 0.9%	3.9%

The overall result show each party paying 0.6% less than they would have paid in they had borrowed directly in the foreign markets.

Barrow Co's original principal amount of €500m would be exchanged at the inception of the swap for \$446,428,517. The principal would be swapped back five years later, at the end of the agreement, at the original spot rate.

Written by a member of the examining team for *Advanced Financial Management*



Related Links

- [Student Accountant hub](#)



Think Ahead

Patterns of behaviour

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Patterns of behaviour**

Are all financial decisions rational? The assumption that they are underpins theories of economic behaviour and stock market models, such as the efficient market hypothesis.

Why then do stock market booms and busts occur if investors are acting rationally? Rational behaviour surely implies no shocks, with stock markets showing steady movements in share prices, but not sudden spurts. However, unexpected and significant news could still result in sudden shocks.

Also, why are some mergers and acquisitions considered to be poor deals? If a listed company is being acquired, surely the acquisition price should be based on the market value of its shares, if the markets are valuing it fairly. Why then is there uncertainty about the true value of many acquired companies? Why also do many acquisitions run into difficulties?

If proper due diligence has been done and decisions are made rationally, surely the directors of the acquiring company will only go ahead if the combination stands a very good chance of success.

Behavioural finance attempts to explain how decision makers take financial decisions in real life, and why their decisions might not appear to be rational every time and, hence, have unpredictable consequences. Behavioural finance has been described as

'the influence of psychology on the behaviour of financial practitioners' (Sewell, 2005). Behavioural finance seeks to examine the following assumptions of rational decision making by investors and financial managers:

1. Financial decision makers seek to maximise their utility and do so by trying to maximise portfolio or company value.
2. They take financial decisions based on analysis of relevant information.
3. The analysis of financial information that they undertake is rational, objective and risk-neutral.

Let's look at how behavioural factors may influence decision making and, therefore, stock markets' and companies' financial strategies.

Investors

Maximisation of utility

Rational decision making by investors implies that their decisions about their investment portfolios will aim to maximise their long-term wealth and, hence, their utility. However, behavioural factors may influence investors to take decisions that are not the best ones for achieving maximum value from their portfolios. Investors may have preferences for particular stocks on non-financial grounds – for example, companies that they consider are acting with social responsibility. They may also avoid 'sin stocks' – companies operating in sectors that they regard as unethical.

Investor utility may also be linked to the process of decision making. Some investors hold on to shares with prices that have fallen over time and are unlikely to recover. They may do this because it will cause them psychological hurt to admit, even only to themselves, that their decision to invest was wrong. This is known as *cognitive dissonance*.

Analysis of relevant information

Behavioural finance next looks at the basis that investors use to take decisions. It suggests that decisions may not be based on an assessment of relevant financial information, but on other grounds. Investors may use information that is not relevant but is readily available, possibly to simplify the decision making process (known as *anchoring*). For example, investors may buy shares that in the past have had high values, on the grounds that these represent their true potential values, even though rational analysis suggests that the prices of these shares will remain low in the future

Investors may also believe that the probability of a future outcome will be influenced by how often the same outcome has occurred in the past. A non-financial example of this idea would be the situation when a coin is flipped eight times, comes up as tails every time and it is said that heads is more likely the ninth time as, by the 'law of averages', heads must come up soon.

If the value of a company's shares has risen for some time, investors will be using similar logic to the coin example if they sell those shares on the grounds that the shares have gained in value for 'long enough' and their price must therefore soon start to fall, even if rational analysis suggest that the rise in price will continue. This is known as the *gambler's fallacy*.

Another deviation from rational analysis is the *herd* instinct, where investors buy or sell shares in a company or sector because many other investors have already done so. Explanations for investors following a herd instinct include social conformity, the desire not to act differently from others. Following a herd instinct may also be due to individual investors lacking the confidence to

make their own judgements, believing that a large group of other investors cannot be wrong. If many investors follow a herd instinct to buy shares in a certain sector, for example the IT sector, this can result in significant price rises for shares in that sector and lead to a stock market bubble.

Investors may not therefore base their decisions on rational analysis, but there is also evidence to suggest that stock market 'professionals' often don't do so either. Studies have shown that there are traders in stock markets who do not base their decisions on fundamental analysis of company performance and prospects. They are known as *noise traders*.

Characteristics associated with noise traders include making poorly timed decisions and following trends. Chartism, using analysis of past share prices as a basis for predicting the future, is an example of noise trading.

Fund managers may also be subject to behavioural influences. Fund managers who wish to give the impression that they are actively managing their investment portfolios, may periodically reposition their portfolios into new sectors, even though the old sectors continue to have good prospects. Some fund managers also ignore companies with low market capitalisation, with the result that their shares are not purchased and their value remains low (known as *small capitalisation discount*).

Rational, objective and risk-neutral analysis

Investors may base their decision on an analysis of available information, but behavioural finance has highlighted that this analysis can be subjective. One aspect is *confirmation bias*, taking an approach or paying attention to evidence that confirms investors' current beliefs about their investments and ignoring evidence that casts doubt on their beliefs. In the dotcom boom, some investors used a variety of methods to value high-tech companies at a large premium, but ignored models such as cash flow valuation models that indicated the worth of those companies was much lower.

Another aspect of investor bias is attitudes towards risks. Rational theory suggests that risk-neutral investors will adopt a long-term approach based on expected values. However, behavioural finance has highlighted various attitudes towards the risks of making profits or losses. Some investors may be attracted by a company that offers the possibility of making very high returns, even if the possibility is not very great (again, the dotcom boom provides evidence of this).

Other investors may have *regret aversion*, avoiding investments that have the risk of making losses, even though expected value analysis suggests that, in the long-term, they will make significant capital gains. Investors with regret aversion may also prefer to invest in companies that look likely to make stable, but low, profits, rather than companies that may make higher profits in some years but possibly losses in others.

There is also evidence that many investors pay most attention to the last set of financial results and other recent information about a company, and take less notice of data that has been available for a while. Explanations for this have included recent information being more readily accessible and more immediate in investors' minds than older information. A consequence of this may be over-reaction when companies release information, with share prices rising or falling quickly after information is released and then going back in the opposite direction to an equilibrium value over time.

Behavioural finance also suggests that there may be a *momentum* effect in stock markets. A period of rising share prices may result in a general feeling of optimism that price rises will continue and an increased willingness to invest in companies that show prospects for growth. If a momentum effect exists, then it is likely to lengthen periods of stock market boom or bust.

Finance managers

Behavioural finance studies have also looked at decision making by managers of companies. They have identified factors that affect investment decisions of all types, but particularly focused on mergers and acquisitions, since many do not appear to fulfil the expectations of the acquiring company.

Maximisation of utility

Companies can be regarded as maximising their utility by long-term maximisation of their shareholders' wealth. However, it is not just behavioural finance that casts doubt on whether company managers are seeking this objective for their shareholders. Agency theory also highlights that managers may have different objectives from shareholders, such as maximising their own short-term rewards and expanding the company by acquisition or other means in order to enhance their own reputation.

However, behavioural finance has highlighted that managers' objectives may not be explainable rationally. Studies have looked at contested takeovers, where different companies bidding against each other has forced the acquisition price up to a level that was significantly greater than many outside the companies involved thought was reasonable. One theory for this is that once managers enter into competition, it makes acquiring a company that others have sought to buy as well, a source of satisfaction in itself. The acquirer's managers are unwilling to let someone else have what they have been trying to acquire (known as *loss aversion bias*).

Analysis of relevant information

There is also evidence that when managers choose to bid for another company, the factor is sometimes not a rational assessment of the target's potential, but their belief in their own abilities. Some managers of acquiring companies seem to believe that, however poor the outlook for the target seems, their own considerable management skills will improve its prospects after the merger takes place. A symptom of this belief could be managers arguing that the target should be valued not using its own price-earnings ratio, but using the (higher) price-earnings ratio of the acquirer.

Once an acquisition or any other strategy has been implemented, what influences managers may be the need to show that they have made the right decisions. Managers may feel that a failing strategy would damage their reputation, and possibly their future prospects. Therefore, they may decide to commit more funds trying to ensure that the strategy is successful, rather than admitting defeat and taking steps to mitigate losses (known as *entrapment*).

Rational, objective and risk-neutral analysis

Managers may also be subjective when they analyse information. Also, they may have confirmation bias, paying attention to information that suggests that an acquisition will enhance value and ignoring evidence that indicates that the target will not be a good buy. They may also seek information that provides a simple yardstick for their own decision making, however flawed that information may be. The value put on the target company by its own directors may be subject to considerable bias, but the acquirer's directors may regard it as a good indication of what the target's fair value is.

Limitations of behavioural finance

Critics of the behavioural finance approach have argued that even if individuals make irrational decisions when left by themselves, participating in finance markets helps discipline them to act rationally by giving them opportunities to learn from their experiences. The consequences of irrational decisions are short-term anomalies. In the longer-term general theories, such as the efficient market hypothesis, will apply.

Conclusion

Behavioural finance has identified a number of factors that may take individuals away from a process of taking decisions to maximise economic utility on the basis of rational analysis of all the information supplied. If these factors apply in practice, they can lead to movements from what would be considered a fair price for an individual company's shares, and the market as a whole to a period where share prices are collectively very high or low. For an acquisition, it can lead to a purchase price that differs significantly from what appears to be a rational valuation.

If you're asked in the *Advanced Financial Management* exam to consider behavioural factors that may influence the decisions of investors or managers, you'll need to read the question scenario very carefully. Look out for information about how investors or managers may be taking decisions, or factors in the situation that may trigger biases that the decision makers have.

You may not be able to come to a firm conclusion about what decision makers will do and why, but you should be looking to discuss various possibilities. Bringing real life into your answer has to mean questioning the assumption that all financial decisions are taken rationally, and at least admitting that behavioural factors may influence decision makers.

Written by a member of the examining team for *Advanced Financial Management*



Related Links

- [**Student Accountant hub**](#)



Think Ahead

Using real options when making financial strategy decisions

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Using real options when making financial strategy decisions**

While the use of real options in investment appraisal is increasingly accepted, the practicalities of using option pricing techniques and ideas in making actual financial strategy decisions is less well understood.

This article will initially consider how an individual traditional project could be reassessed using option valuation. It will then consider how option valuation could assist when assessing a portfolio of projects and will conclude with a brief discussion regarding inter-dependent projects.

A traditional project and option valuation

Let us imagine that a company is proposing a major expansion project. The operational management team have forecast the cash flows relevant to the project in line with the standard assumptions and policies set down by the financial management. The financial management have then discounted these cash flows at a cost of capital of 11% and this has resulted in an NPV of just \$5m. For the sake of simplicity tax, inflation and other real world complications have been ignored.

NPV calculation – overall:

\$'m	T ₀	T ₁	T ₂	T ₃₋₁₀
Relevant cash flows	(600)	110	(190)	158
11% discount factors	1	0.901	0.812	0.812
11% annuity factor				5.146
PV	(600)	99.1	(154.3)	660.2
NPV	<u>5.0</u>			

Those executives who are keen proponents of this expansion are disappointed by the low NPV and fear that the project is unlikely to win approval when competing for funds against other projects. They consider that the project is being undervalued and, hence, a meeting with the financial management team is arranged. At this meeting the financial management team query the large net cash outflow that is forecast to occur at the end of year two. As a result of this, it becomes apparent that the project comprises an initial investment of \$600m which will produce net cash inflows of \$110m for the following 10 years, followed by a further investment of \$300m after two years which will increase the net cash inflows by \$48m to \$158m per year for the remaining eight years. Further discussion reveals that the additional investment after two years is discretionary and does not necessarily need to be made.

Hence, the project could be viewed as an initial expansion costing \$600m – phase 1 – followed by an option to expand further after two years – phase 2.

If separate NPV calculations are carried out for each of these phases. The results below are obtained.

NPV calculation – phase 1:

\$'m	T ₀	T ₁₋₁₀
Relevant cash flows	(600)	110
11% discount factors	1	
11% annuity factor		5.889
PV	(600)	647.8
NPV	<u>47.8</u>	

NPV calculation – phase 2:

\$'m	T ₂	T ₃₋₁₀
Relevant cash flows	(300)	48
11% discount factors	0.812	0.812
11% annuity factor		5.146
PV	(243.6)	200.6
NPV	(43)	

The total NPV is fundamentally unchanged as $\$47.8\text{m} - \$43\text{m} = \$4.8\text{m}$, which is about the same as the $\$5\text{m}$ calculated initially. To the extent there is a difference, this can be attributed to rounding. This analysis alone provides some insight as, given that there is no obligation for the company to carry out phase two, the overall NPV must be at least $\$47.8\text{m}$ which far exceeds the initial NPV calculated of $\$5\text{m}$.

It is worth noting that the discretionary spend at the end of year two has also been discounted at the 11% cost of capital. Although this is the approach commonly taken it could be more accurate to discount such discretionary expenditure at the risk free rate. This is because discretionary expenditure has much less operational risk than the net cash inflows that it is hoped will arise from such expenditure. If a risk free rate of 5% is used the present value of the $\$300\text{m}$ expenditure at the end of year two would be $\$272.1\text{m}$. This is $\$28.5\text{m}$ ($272.1 - 243.6$) more than the present value if the cost of capital is used. Hence, the original overall NPV and the NPV of phase two should perhaps be reduced by this $\$28.5\text{m}$. This approach would be consistent with the treatment of the exercise price in the Black Scholes Option Pricing model.

Although phase two does not currently seem worthwhile the option to carry out this phase can only add value as an option can never have a negative value. If the cash flows expected from phase two were to become favourable, the company will have the ability to carry out phase two and reap the benefit. Hence, the overall NPV will be $\$47.8\text{m}$ plus the value of the option to carry out phase two.

In order to value the option to carry out phase two we must first attribute figures to the inputs required for the Black Scholes option pricing (BSOP) formula:

- P_e = the investment required after two years to carry out phase two = $\$300\text{m}$
- P_a = the PV of the net cash inflows currently forecast to arise from phase two = $\$200.6\text{m}$ (this must exclude the P_e)
- t = the time until phase two will begin = 2 years
- s = the volatility – assumed to be 0.4 (standard deviation)
- r = the risk free rate – assumed to be 5%

The P_e and P_a figures can be seen in the calculation of the NPV for phase two and, as we know, the company has the option to expand into phase two after two years. The s and r will both be given within any exam question and, hence, suitable figures have been assumed.

Using these inputs the option value can now be calculated.

Option value:

$$d_1 = (\ln(200.6/300) + (0.05 + 0.5 \times 0.4^2) \times 2) / (0.4 \times \sqrt{2}) = (-0.402 + 0.260) / 0.566 = -0.251 \approx -0.25$$

$$d_2 = -0.251 - 0.566 = -0.817 \approx -0.82$$

$$N(d_1) = 0.5 - 0.0987 = 0.4013$$

$$N(d_2) = 0.5 - 0.2939 = 0.2061$$

$$\text{Call option value} = (200.6 \times 0.4013) - (300 \times 0.2061 \times e^{(-0.05 \times 2)}) = 80.5 - 55.9 = \underline{\underline{\$24.6m}}$$

Hence, the total NPV for the project with the option to expand = \$47.8m + \$24.6m = **\$72.4m**. As a result of the financial management taking the time to better understand the project and the real options within it a project which seemed fairly marginal, has been shown to be attractive and is far more likely to win approval.

The attractiveness of the project arises because phase one of the project is itself attractive and the company can potentially benefit if the phase two expansion finally becomes worthwhile.

As discussed in the previous real options article, there are significant problems associated with using BSOP to value real options and, hence, the option value of \$24.6m calculated should be treated as indicative only and should be used with care.

A portfolio of projects and option valuation

As soon as executives launch a strategy conditions change in the environment within which they are operating and indeed their knowledge of that environment is updated. Hence, managers must actively manage and respond accordingly. Traditional NPV analysis is probably too static a tool to reflect this active management. This is because it tends to assume a company will follow a previously agreed plan and does not account so well for how events may unfold. Instead managers should perhaps view the projects they could undertake to achieve their goals as a portfolio of real options which they could potentially exercise over time.

Imagine a company is faced with six independent projects as follows:

\$'m	Project:	U	V	W	X	Y	Z
PV of expected inflows		50	50	50	50	50	50
Initial investment required		45	47	53	54	56	56
NPV		5	3	-3	-4	-6	-6

Given this basic NPV analysis projects U and V would be accepted and a total NPV of \$8m would be generated for the company.

Let us now imagine that projects U and X have to be carried out immediately or not at all. In other words there is no option to delay these projects. Hence, project U would be accepted to generate an NPV of \$5m and project X would be rejected. A useful analogy here is that of fruit growing in a garden. Projects U and X represent fruits which have to be picked now. Project U represents the perfectly ripe fruit which can be sold or eaten, while project X represents the rotten fruit which must be picked but then discarded.

The remaining projects can be delayed and represent fruits which do not have to be picked immediately and which have the potential to develop into perfectly ripe fruit. Given the volatility of the cash flows from these projects and the period by which they could be delayed option values can be calculated for these projects. This data and the resulting option values are presented below:

Project:	V	W	Y	Z
Volatility (standard deviation)	0.2	0.1	0.3	0.2
Delay period (years)	3	1	3	2
Option value (\$m)	12.2	1.8	11.0	5.5

These option values have been calculated using an assumed risk free rate of 5%.

These option values total to a value of \$30.5m. If the NPV of project U, which has already been accepted, is added to this a total value of \$35.5m is created. This value is significantly higher than the original NPV of \$8m which could have been generated by accepting projects U and V. However, careful management is necessary if as much value as possible is to be generated.

We will now consider each of projects V, W, Y and Z separately:

Project V:

This project looks most promising. It has a positive NPV if exercised now but its value as an option is significantly higher. Hence, exercising now would appear sub-optimal as with further nurturing a higher value could be generated. To continue the fruit analogy this project represents a fruit which could be picked now and eaten or sold but which, with careful cultivation, could become a larger and better fruit. However, just as ripening fruit can be eaten by pests, there is potential for project value to be lost – through the actions of competitors for instance. Hence, the company may decide to exercise the option early to realise the existing positive NPV.

Project W:

This project is not at all promising. It has a negative NPV if exercised now and, as it has a low volatility and there is a relatively short time until a decision has to be made regarding this project, it has a low option value. Hence, this project will probably never be worth exercising. This project could be thought of as the small, late developing fruit which is unlikely to ripen before the season ends.

Project Y:

Despite currently having a negative NPV this project has a high value as an option. This is due to the fact that it will not expire for

three years and has a relatively high volatility when compared to the other projects. Hence, this project will probably be worth exercising at some later date. This project represents the unripe fruit which cannot be picked now, but which is expected to mature into a perfectly ripe fruit in the future.

Project Z:

This project is similar to project Y but is much less promising. It currently has the same negative NPV as project Y but as it has a lower volatility and a shorter time until it expires it has a lower value as an option when compared to project Y. This project seems unlikely to be worth exercising but there is still a reasonable chance that it could move 'into the money' and, hence, may become worth exercising in the future. As with project Y this project is also an unripe fruit which cannot be picked now. However, its chances of maturing into a perfectly ripe fruit seem much less.

How does the use of option valuation analysis help when compared to a traditional NPV analysis?

As previously stated traditional NPV analysis would lead to the acceptance of projects U and V, resulting in an NPV of \$8m, and the rejection of the other four projects. However, option valuation analysis results in the acceptance of project U generating an NPV of \$5m, the rejection of project X and options to carry out the four other projects in the future the total value of which has been estimated as \$30.5m.

Further the analysis encourages active management:

Project V has been identified as worthy of careful management to ensure it is carried out at the optimal time to maximise the value created but at the same time ensuring that the existing positive NPV is not destroyed.

Of the remaining projects Project W has been identified as the project deserving least attention while with careful nurturing project Y and even project Z could finally create value for the company.

It is crucial to recognise that projects, just like fruit, do require nurturing. This is because as the time approaches when the project must be carried out or abandoned, the option value will always tend to decline assuming all other variables remain constant. This is because the time to maturity falls and the present value of the exercise price, the investment to be made to carry out the project, is rising.

Hence, without nurturing all the option values will move to zero over time. For this not to happen there must be good luck or active management. Good luck could add value to a project because, for instance, sudden growth in the economy could mean the returns from a project become higher than originally forecast. Active management could involve taking action to reduce the costs or increase the revenues associated with a project. Equally action could be taken to reduce the initial investment required in the project.

Using the option valuation approach has helped identify those projects most likely to benefit from such nurturing and, hence, is useful as a company follows and develops its financial strategy.

Inter-dependent projects

So far this article has looked at how our understanding of a single project and a portfolio of independent projects could be enhanced by the use of option valuation. Let us now consider briefly interdependent projects.

Just as you and I are faced with a myriad of options when buying a new car the car manufacturer also has many options.

For instance a car manufacturer may have a project to launch a new saloon model. From this saloon model, an estate model could then be launched, and from this a 4x4 version of the estate model could be launched. This is a strategy followed by Skoda with their Octavia and Superb ranges. In effect the company has the initial project (the launch of the saloon), followed by a call option on the launch of the estate, which itself has a call option on the launch of the 4x4 estate. Hence, the company has a call on a call! This is known as a nest of options or nested options.

Another car manufacturer may also have a project to launch a new saloon model. From this saloon, an estate model could be launched and a sports utility vehicle (SUV) model could also be launched. This is a strategy followed by BMW with their 3 series and 5 series ranges. In effect the company has the initial project (the launch of the saloon), followed by a call option on the launch of the estate and a call option on the launch of the SUV model (the BMW X3 and X5). While these options are not nested they are obviously still very much related.

Understanding how these options inter-relate is obviously useful to a company as they develop their strategy. Once again good luck and active management play their part. For instance a few bad winters is likely to enhance future sales of SUV's and 4x4 estates. This in turn enhances the value of the call options to manufacture these models and, hence, the value of the original product launch.

Active management could involve the development of a new more fuel efficient 4x4 system. Any such increase in fuel efficiency is also likely to enhance the sales of 4x4 equipped SUV's and estates, and enhance the value of the call options to manufacture these models and, hence, the value of the original product launch. Conversely, extra sales of 4x4 equipped models may reduce sales of other models.

Obviously the initial product launch should be marketed in such a way as to maximise its success. Active management should also ensure that this initial product launch should ensure consumers' perception of the new model is developed in such a way that the chances of success of the follow on models is optimised.

Our knowledge of the determinants of option values can also be useful. For instance if sales of SUV's are high in a more volatile market such as China this adds value to the option to develop the SUV variant of the model in the future, and, hence, enhances the value of the initial project to launch the saloon.

Conclusion

This article has demonstrated how option valuation techniques can help understand the potential value of projects and how financial strategy decisions can be made using this knowledge in order to maximise the results arising from projects and, hence,

maximise company value.

William Parrott, freelance tutor and senior FM tutor, MAT Uganda

References

- Investment Opportunities as Real Options: Getting Started on the Numbers
- Timothy A. Luehrman, Harvard Business Review, July/August 1998
- Strategy as a Portfolio of Real Options
- Timothy A. Luehrman, Harvard Business Review, September/October 1998



Related Links

- [**Student Accountant hub**](#)



Think Ahead

Our sites

myACCA

ACCA Mail

ACCA Careers

ACCA Learning Community

Your Future

Investment appraisal and real options

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Investment appraisal and real options**

Real options' valuation methodology adds to the conventional net present value (NPV) estimations by taking account of real life flexibility and choice. This is the first of two articles which considers how real options can be incorporated into investment appraisal decisions. This article discusses real options and then considers the types of real options calculations which may be encountered in *Advanced Financial Management*, through three examples. The article then considers the limitations of the application of real options in practice and how some of these may be mitigated.

The second article considers a more complex scenario and examines how the results produced from using real options with NPV valuations can be used by managers when making strategic decisions.

Net present value (NPV) and real options

The conventional NPV method assumes that a project commences immediately and proceeds until it finishes, as originally predicted. Therefore it assumes that a decision has to be made on a now or never basis, and once made, it cannot be changed. It does not recognise that most investment appraisal decisions are flexible and give managers a choice of what actions to undertake.

The real options method estimates a value for this flexibility and choice, which is present when managers are making a decision on whether or not to undertake a project. Real options build on net present value in situations where uncertainty exists and, for example: (i) when the decision does not have to be made on a now or never basis, but can be delayed, (ii) when a decision can be changed once it has been made, or (iii) when there are opportunities to exploit in the future contingent on an initial project being undertaken. Therefore, where an organisation has some flexibility in the decision that has been, or is going to be made, an option exists for the organisation to alter its decision at a future date and this choice has a value.

With conventional NPV, risks and uncertainties related to the project are accounted for in the cost of capital, through attaching probabilities to discrete outcomes and/or conducting sensitivity analysis or stress tests. Options, on the other hand, view risks and uncertainties as opportunities, where upside outcomes can be exploited, but the organisation has the option to disregard any downside impact.

Real options methodology takes into account the time available before a decision has to be made and the risks and uncertainties attached to a project. It uses these factors to estimate an additional value that can be attributable to the project.

Estimating the value of real options

Although there are numerous types of real options, in *Advanced Financial Management*, candidates are only expected to explain and compute an estimate of the value attributable to three types of real options:

- (i) The option to delay a decision to a future date (which is a type of call option)
- (ii) The option to abandon a project once it has commenced if circumstances no longer justify the continuation of the project (which is a type of put option), and
- (iii) The option to exploit follow-on opportunities which may arise from taking on an initial project (which is a type of call option).

In addition to this, candidates are expected to be able to explain (but not compute the value of) redeployment or switching options, where assets used in projects can be switched to other projects and activities.

For the *Advanced Financial Management* exam purposes, it can be assumed that real options are European-style options, which can be exercised at a particular time in the future and their value will be estimated using the Black-Scholes Option Pricing (BSOP) model and the put-call parity to estimate the option values. However, assuming that the option is a European-style option and using the BSOP model may not provide the best estimate of the option's value (see the section on limitations and assumptions below).

Five variables are used in calculating the value of real options using the BSOP model as follows:

1. The underlying asset value (P_a), which is the present value of future cash flows arising from the project.
2. The exercise price (P_e), which is the amount paid when the call option is exercised or amount received if the put option is exercised.
3. The risk-free (r), which is normally given or taken from the return offered by a short-dated government bill. Although this is normally the discrete annualised rate and the BSOP model uses the continuously compounded rate, for *Advanced Financial Management* purposes the continuous and discrete rates can be assumed to be the same when estimating the value of real options.

4. The volatility (σ), which is the risk attached to the project or underlying asset, measured by the standard deviation.
5. The time (t), which is the time, in years, that is left before the opportunity to exercise ends.

The following three examples demonstrate how the BSOP model can be used to estimate the value of each of the three types of options.

Example 1: Delaying the decision to undertake a project

A company is considering bidding for the exclusive rights to undertake a project, which will initially cost \$35m.

The company has forecast the following end of year cash flows for the four-year project.

Year	1	2	3	4
Cash flows (\$m)	20	15	10	5

The relevant cost of capital for this project is 11% and the risk free rate is 4.5%. The likely volatility (standard deviation) of the cash flows is estimated to be 50%.

Solution:

NPV without any option to delay the decision

Year	Today	1	2	3	4
Cash flows (\$)	-35m	20m	15m	10m	5m
PV (11%) (\$)	-35m	18.0m	12.2m	7.3m	3.3m

NPV = \$5.8m

Supposing the company does not have to make the decision right now but can wait for two years before it needs to make the decision.

NPV with the option to delay the decision for two years

Year	3	4	5	6
Cash flows (\$)	20m	15m	10m	5m
PV (11%) (\$)	14.6m	9.9m	5.9m	2.7m

Variables to be used in the BSOP model

Asset value (P_a) = \$14.6m + \$9.9m + \$5.9m + \$2.7m = \$33.1m

Exercise price (P_e) = \$35m

Exercise date (t) = Two years

Risk free rate (r) = 4.5%

Volatility (s) = 50%

Using the BSOP model

d1	0.40
d2	-0.31
N(d1)	0.6554
N(d2)	0.3783
Call value	\$9.6m

Based on the facts that the company can delay its decision by two years and a high volatility, it can bid as much as \$9.6m instead of \$5.8m for the exclusive rights to undertake the project. The increase in value reflects the time before the decision has to be made and the volatility of the cash flows.

Example 2: Exploiting a follow-on project

A company is considering a project with a small positive NPV of \$3m but there is a possibility of further expansion using the technologies developed for the initial project. The expansion would involve undertaking a second project in four years' time.

Currently, the present values of the cash flows of the second project are estimated to be \$90m and its estimated cost in four years is expected to be \$140m. The standard deviation of the project's cash flows is likely to be 40% and the risk free rate of return is currently 5%.

Solution:

The variables to be used in the BSOP model for the second (follow-on) project are as follows:

Asset Value (P_a) = \$90m

Exercise price (P_e) = \$140m

Exercise date (t) = Four years

Risk free rate (r) = 5%

Volatility (s) = 40%

Using the BSOP model

d_1	0.10
d_2	-0.70
$N(d_1)$	0.5398
$N(d_2)$	0.242
Call value	\$20.8m

The overall value to the company is \$23.8m, when both the projects are considered together. At present the cost of \$140m seems substantial compared to the present value of the cash flows arising from the second project. Conventional NPV would probably return a negative NPV for the second project and therefore the company would most likely not undertake the first project either. However, there are four years to go before a decision on whether or not to undertake the second project needs to be made. A lot could happen to the cash flows given the high volatility rate, in that time. The company can use the value of \$23.8m to decide whether or not to invest in the first project or whether it should invest its funds in other activities. It could even consider the possibility that it may be able to sell the combined rights to both projects for \$23.8m.

Example 3: The option to abandon a project

Duck Co is considering a five-year project with an initial cost of \$37,500,000 and has estimated the present values of the project's cash flows as follows:

Year	1	2	3	4	5
Present values (\$ 000s)	1,496.9	4,938.8	9,946.5	7,064.2	13,602.9

Swan Co has approached Duck Co and offered to buy the entire project for \$28m at the start of year three. The risk free rate of return is 4%. Duck Co's finance director is of the opinion that there are many uncertainties surrounding the project and has assessed that the cash flows can vary by a standard deviation of as much as 35% because of these uncertainties.

Solution:

Swan Co's offer can be considered to be a real option for Duck Co. Since it is an offer to sell the project as an abandonment option, a put option value is calculated based on the finance director's assessment of the standard deviation and using the Black-Scholes option pricing (BSOP) model, together with the put-call parity formula.

Although Duck Co will not actually obtain any immediate cash flow from Swan Co's offer, the real option computation below, indicates that the project is worth pursuing because the volatility may result in increases in future cash flows.

Without the real option

Year	1	2	3	4	5
Present values (\$ 000s)	1,496.9	4,938.8	9,946.5	7,064.2	13,602.9

Present value of cash flows approx. = \$37,049,300

Cost of initial investment = \$37,500,000

NPV of project = \$37,049,300 – \$37,500,000 = \$(450,700)

With the real option

The asset value of the real option is the sum of the PV of cash flows foregone in years three, four and five, if the option is exercised (\$9.9m + \$7.1m + \$13.6m = \$30.6m)

Asset value (Pa)	\$30.6m
------------------	---------

Exercise Price (P_e)	\$28m
Risk-free rate (r)	4%
Time to exercise (t)	Two years
Volatility (s)	35%
d_1	0.59
d_2	0.09
$N(d_1)$	0.7224
$N(d_2)$	0.5359
Call Value	8.25
Put Value	\$3.50m

Net present value of the project with the put option is approximately \$3.05m (\$3.50m – \$0.45m).

If Swan Co's offer is not considered, then the project gives a marginal negative net present value, although the results of any sensitivity analysis need to be considered as well. It could be recommended that, if only these results are taken into consideration, the company should not proceed with the project. However, after taking account of Swan Co's offer and the finance director's assessment, the net present value of the project is positive. This would suggest that Duck Co should undertake the project.

Limitations and assumptions

Many of the limitations and assumptions discussed below stem from the fact that a model developed for financial products is used to assess flexibility and choice embedded within physical, long-term investments.

European-style options or American-style options

The BSOP model is a simplification of the binomial model and it assumes that the real option is a European-style option, which can only be exercised on the date that the option expires. An American-style option can be exercised at any time up to the expiry date. Most options, real or financial, would, in reality, be American-style options.

In many cases the value of a European-style option and an equivalent American-style option would be largely the same, because unless the underlying asset on which the option is based is due to receive some income before the option expires, there is no benefit in exercising the option early. An option prior to expiry will have a time-value attached to it and this means that the value of an option prior to expiry will be greater than any intrinsic value the option may have, if it were exercised.

However, if the underlying asset on which the option is based is due to receive some income before the option's expiry; say for example, a dividend payment for an equity share, then an early exercise for an option on that share may be beneficial. With real options, a similar situation may occur when the possible actions of competitors may make an exercise of an option before expiry the better decision. In these situations the American-style option will have a value greater than the equivalent European-style option.

Because of these reasons, the BSOP model will either underestimate the value of an option or give a value close to its true value. Nevertheless, estimating and adding the value of real options embedded within a project, to a net present value computation will give a more accurate assessment of the true value of the project and reduce the propensity of organisations to under-invest.

Estimating volatility

The BSOP model assumes that the volatility or risk of the underlying asset can be determined accurately and readily. Whereas for traded financial assets this would most probably be the case, as there is likely to be sufficient historical data available to assess the underlying asset's volatility, this is probably not going to be the case for real options. Real options would probably be available on large, one-off projects, for which there would be little or no historical data available.

Volatility in such situations would need to be estimated using simulations, such as the Monte-Carlo simulation model, with the need to ensure that the model is developed accurately and the data input used to generate the simulations reasonably reflects what is likely to happen in practice.

Other limitations of real options

The BSOP model requires further assumptions to be made involving the variables used in the model, the primary ones being:

- (a) The BSOP model assumes that the underlying project or asset is traded within a situation of perfect markets where information on the asset is available freely and is reflected in the asset value correctly. Further it assumes that a market exists to trade the underlying project or asset without restrictions (that is, that the market is frictionless)
- (b) The BSOP model assumes that interest rates and the underlying asset volatility remain constant until the expiry time ends. Further, it assumes that the time to expiry can be estimated accurately
- (c) The BSOP model assumes that the project and the asset's cash flows follow a lognormal distribution, similar to equity markets on which the model is based
- (d) The BSOP model does not take account of behavioural anomalies which may be displayed by managers when making decisions, such as over- or under-optimism
- (e) The BSOP model assumes that any contractual obligations involving future commitments made between parties, which are then used in constructing the option, will be binding and will be fulfilled. For example, in example three above, it is assumed that Swan Co will fulfil its commitment to purchase the project from Duck Co in two years' time for \$28m and there is therefore no risk of non-fulfilment of that commitment.

In any given situation, one or more of these assumptions may not apply. The BSOP model therefore does not provide a 'correct' value, but instead it provides an indicative value which can be attached to the flexibility of a choice of possible future actions that may be embedded within a project.

Conclusion

This article discussed how real options thinking can add to investment appraisal decisions and in particular NPV estimations by considering the value which can be attached to flexibility which may be embedded within a project because of the choice managers may have when making investment decisions. It then worked through computations of three real options situations, using the BSOP model. The article then considered the limitations of, and assumptions made when, applying the BSOP model to real options computations. The value computed can therefore be considered indicative rather than conclusive or correct.

The second article will consider how managers can use real options to make strategic investment appraisal decisions.

Written by a member of the *Advanced Financial Management* examining team



Related Links

Securitisation and tranching

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Securitisation and tranching**

Securitisation became a topical issue during the credit crunch, and still remains a relevant issue for financial management and *Advanced Financial Management* students

Please note:

This is an updated version of the 'Toxic Assets' technical article

Securitisation through Collateralised Debt Obligations (CDOs)

One common use of securitisation occurs when banks lend through mortgages, credit cards, car loans or other forms of credit, they invariably move to 'lay off' their risk by a process of securitisation. Such loans are an asset on the statement of financial position, representing cash flow to the bank in future years through interest payments and eventual repayment of the principal sum involved. By securitising the loans, the bank removes the risk attached to its future cash receipts and converts the loan back into cash, which it can lend again, and so on, in an expanding cycle of credit formation.

Securitisation is achieved by transferring the lending to specifically created companies called 'special purpose vehicles' (SPVs). In the case of conventional mortgages, the SPV effectively purchases a bank's mortgage book for cash, which is raised through the issue of bonds backed by the income stream flowing from the mortgage holder. In the case of sub-prime mortgages, the high levels of risk called for a different type of securitisation, achieved by the creation of derivative-style instruments known as 'collateralised debt obligations' or CDOs.

Securitisation may be also appropriate for an organisation which wants to enhance its credit rating by using low-risk cash flows, such as rental income from commercial property, which will be diverted into a "ring-fenced" SPV.

CDOs are a way of repackaging the risk of a large number of risky assets such as sub-prime mortgages. Unlike a bond issue, where the risk is spread thinly between all the bond holders, CDOs concentrate the risk into investment layers or 'tranches', so that some investors take proportionately more of the risk for a bigger return and others take little or no risk for a much lower return.

Each tranche of CDOs is securitised and 'priced' on issue to give the appropriate yield to the investors. The investment grade tranche of CDOs will be the most highly priced, giving a low yield but with low risk attached. At the other end, the 'equity' tranche carries the bulk of the risk – it will be very lowly priced but with a high potential, but very risky, yield. There is more detail on this in the next section.

CDOs are, therefore, a mechanism whereby losses are transferred to investors with the highest appetite for risk (such as hedge funds), leaving the bulk of CDOs' investors (mainly other banks) with a low risk source of cash flow.

The structure of CDOs

An example of a possible structure for a CDO is as follows. For a pool of mortgages taken over by the SPV, three tranches of CDOs are created:

- **Tranche 1** (highest risk) known as the 'equity' tranche and normally comprising about 10% of the value of the mortgages in the pool. Throughout the CDOs' life, the equity tranche will absorb any losses brought about by default on the part of mortgage holders, up to the point that the principal underpinning the tranche is exhausted. At this point the investment is worthless.
- **Tranche 2** (intermediate risk or 'mezzanine' tranche) consists of around 10% of the principal and will absorb any losses not absorbed by the equity tranche until the point at which its principal is also exhausted.
- **Tranche 3** (AAA or 'senior' tranche) consists of the balance of the pool value and will absorb any residual losses.

The proportion of the principal held in each tranche is known as the CDO 'structure', and if there is perceived to be little risk of default then the percentage of value in the mortgage pool forming the equity and mezzanine tranches will be quite small. However, if the risk is high then CDOs will be created with a greater proportion of the principal in the equity and mezzanine tranches and a relatively smaller proportion in the senior tranche.

When cash flows are received from borrowers in the form of interest payments and loan repayments, these payments are paid to tranche 3 first until their obligation is fulfilled, then tranche 2, and anything left over is paid to the equity tranche. Any defaults hit tranche 1 first, then tranche 2 and so on. The repayments represent a 'waterfall' of cash with the investors holding the tranches

like buckets. The senior tranches get filled first, the mezzanine holders get filled next and anything left falls into the equity pools at the bottom.

Example

A bank has made a number of mortgage loans to customers with a current total value of \$350 million. The mortgages have an average term to maturity of ten years. The net income from the loans is 7% per year. The bank will use 85% of the mortgage pool as collateral for a securitisation with the following structure:

- 75% of the collateral value to support a tranche of A-rated loan notes offering investors 6% per year.
- 15% of the collateral value to support a tranche of B-rated loan notes offering investors 11% per year.
- 10% of the collateral value to support a tranche of subordinated certificates which are unrated.

The estimated cash flows for this arrangement would be as follows:

Cash inflows

Inflows from mortgages $\$350\text{m} \times 7\% = \24.5m

Cash outflows

A-rated loan notes

$\$350\text{m} \times 85\% \times 75\% \times 6\% = \13.4m

B-rated loan notes

$\$350\text{m} \times 85\% \times 15\% \times 11\% = \4.9m

Total outflows = $\$13.4\text{m} + \$4.9\text{m} = \$18.3\text{m}$

The difference between the inflows and the outflows is returned to the high-risk unrated certificates.

Difference in cash flows = $\$24.5\text{m} - \$18.3\text{m} = \$6.2\text{m}$

The subordinated certificates have a value of $\$350\text{m} \times 85\% \times 10\% = \29.75m .

The return on this high-risk investment is $\$6.2\text{m}/\$29.75\text{m} = 20.8\%$

However this return is at risk should there be a reduction in the income from the mortgages resulting from customers defaulting on their mortgages. Because of this level of risk, the equity tranche may be unattractive to investors for some securitisation arrangements.

Note that all of the income from the mortgages is used to pay the tranche holders, not just 85% representing the securitised amount. The reason why the securitisation is performed is to get money in quickly. In order to sell the various tranches there needs to be an incentive; for this to be present for all tranches not all of the available pool of mortgages is securitised, but all of the income from the pool is distributed. The bank, in theory, loses out from this approach by distributing 100% of the income instead of keeping 15%, but has achieved the objective of getting the initial funds from the tranche holders as quickly as possible.

Written by a member of the examining team for *Advanced Financial Management*



Related Links

- [Student Accountant hub page](#)



Think Ahead

Our sites

[myACCA](#)

[ACCA Mail](#)

[ACCA Careers](#)

[ACCA Learning Community](#)

[Your Future](#)

Useful links

[Make a payment](#)

[ACCA-X online courses](#)

Exchange traded foreign exchange derivatives

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Exchange traded foreign exchange derivatives**

The aim of this article is to consider both foreign exchange futures and options using real market data. The basics, which have been well examined in the recent past, will be quickly revisited. The article will then consider areas which, in reality, are of significant importance but which, to date, have not been examined to any great extent.

Foreign exchange futures – the basics

Scenario

Imagine it is 10 July. A UK company has a US\$6.65m invoice to pay on 26 August. They are concerned that exchange rate fluctuations could increase the £ cost and, hence, seek to effectively fix the £ cost using exchange traded futures. The current spot rate is \$1.71110/£1.

Research shows that £/\$ futures, where the contract size is denominated in £, are available on the CME Europe exchange at the following prices:

September expiry – 1.71035

December expiry – 1.70865

The contract size is £100,000 and the futures are quoted in US\$ per £1.

Note:

CME Europe is a London based derivatives exchange. It is a wholly owned subsidiary of CME Group, which is one of the world's leading and most diverse derivatives marketplace, handling (on average) three billion contracts worth about \$1 quadrillion annually!

Setting up the hedge

1. Date? – September:

The first futures to mature after the expected payment date (transaction date) are chosen. As the expected transaction date is 26 August, the September futures which mature at the end of September will be chosen.

2. Buy/Sell? – Sell:

As the contract size is denominated in £ and the UK company will be selling £ to buy \$ they should sell the futures.

3. How many contracts? – 39

As the amount to be hedged is in \$ it needs to be converted into £ as the contract size is denominated in £. This conversion will be done using the chosen futures price. Hence, the number of contracts required is: $(\$6.65m \div 1.71035)/£100,000 \approx 39$.

Summary

The company will sell 39 September futures at \$1.71035/£1.

Outcome on 26 August:

On 26 August the following was true:

Spot rate – \$1.65770/£1

September futures price – \$1.65750/£1

Actual cost:

$\$6.65m / 1.65770 = £4,011,582$

Gain/loss on futures:

As the exchange rate has moved adversely for the UK company a gain should be expected on the futures hedge.

\$/£1

\$/£1

Sell – on 10 July	1.71035
Buy back – on 26 August	<u>(1.65750)</u>
Gain	<u>0.05285</u>

This gain is in terms of \$ per £ hedged. Hence, the total gain is:

$$0.05285 \times 39 \text{ contracts} \times £100,000 = \$206,115$$

Alternatively, the contract specification for the futures states that the tick size is 0.00001\$ and that the tick value is \$1. Hence, the total gain could be calculated in the following way:

$$0.05285 / 0.00001 = 5,285 \text{ ticks}$$

$$5,285 \text{ ticks} \times \$1 \times 39 \text{ contracts} = \$206,115$$

This gain is converted at the spot rate to give a £ gain of:

$$\$206,115 / 1.65770 = £124,338$$

Total cost:

$$£4,011,582 - £124,338 = \mathbf{£3,887,244}$$

This total cost is the actual cost less the gain on the futures. It is close to the receipt of £3,886,389 that the company was originally expecting given the spot rate on 10 July when the hedge was set up. (\$6.65m/1.71110). This shows how the hedge has protected the company against an adverse exchange rate move.

Summary

All of the above is essential basic knowledge. As the exam is set at a particular point in time you are unlikely to be given the futures price and spot rate on the future transaction date. Hence, an effective rate would need to be calculated using basis. Alternatively, the future spot rate can be assumed to equal the forward rate and then an estimate of the futures price on the transaction date can be calculated using basis. The calculations can then be completed as above.

The ability to do this would normally earn four marks in an exam. Equally, another one or two marks could be earned for reasonable advice such as the fact that a futures hedge effectively fixes the amount to be paid and that margins will be payable

during the lifetime of the hedge. It is some of these areas that we will now explore further.

Foreign exchange futures – other issues

Initial margin

When a futures hedge is set up the market is concerned that the party opening a position by buying or selling futures will not be able to cover any losses that may arise. Hence, the market demands that a deposit is placed into a margin account with the broker being used – this deposit is called the '**initial margin**'.

These funds still belong to the party setting up the hedge but are controlled by the broker and can be used if a loss arises. Indeed, the party setting up the hedge will earn interest on the amount held in their account with their broker. The broker in turn keeps a margin account with the exchange so that the exchange is holding sufficient deposits for all the positions held by brokers' clients.

In the scenario above the CME contract specification for the £/\$ futures states that an initial margin of \$1,375 per contract is required.

Hence, when setting up the hedge on 10 July the company would have to pay an initial margin of $\$1,375 \times 39 \text{ contracts} = \$53,625$ into their margin account. At the current spot rate the £ cost of this would be $\$53,625 / 1.71110 = \text{£}31,339$.

Marking to market

In the scenario given above, the gain was worked out in total on the transaction date. In reality, the gain or loss is calculated on a daily basis and credited or debited to the margin account as appropriate. This process is called '**marking to market**'.

Hence, having set up the hedge on 10 July a gain or loss will be calculated based on the futures settlement price of \$1.70925/£1 on 11 July. This can be calculated in the same way as the total gain was calculated:

\$/£1

Sell – on 10 July

1.71035

Settlement price – 11 July

(1.70925)

\$/£1

Gain	<u>0.00110</u>
------	----------------

Gain in ticks – $0.00110/0.00001 = 110$

Total gain – $110 \text{ ticks} \times \$1 \times 39 \text{ contracts} = \$4,290$

This gain would be credited to the margin account taking the balance on this account to $\$53,625 + \$4,290 = \$57,915$.

At the end of the next trading day (Monday 14 July), a similar calculation would be performed:

\$/£1

Settlement price – 11 July	1.70925
----------------------------	---------

Settlement price – 14 July	<u>(1.70805)</u>
----------------------------	------------------

Gain	<u>0.00120</u>
------	----------------

Gain in ticks – $0.00120/0.00001 = 120$

Total gain – $120 \text{ ticks} \times \$1 \times 39 \text{ contracts} = \$4,680$.

This gain would also be credited to the margin account taking the balance on this account to $\$57,915 + \$4,680 = \$62,595$.

Similarly, at the end of the next trading day (15 July), the calculation would be performed again:

\$/£1

\$/£1

Settlement price – 14 July	1.70805
Settlement price – 15 July	<u>(1.71350)</u>
Loss	<u>0.00545</u>

Loss in ticks – $0.00545/0.00001 = 545$

Total loss – $545 \text{ ticks} \times \$1 \times 39 \text{ contracts} = \$21,255$.

This loss would be debited to the margin account, reducing the balance on this account to $\$62,595 - \$21,255 = \$41,340$.

This process would continue at the end of each trading day until the company chose to close out their position by buying back 39 September futures.

Maintenance margin, variation margin and margin calls

Having set up the hedge and paid the initial margin into their margin account with their broker, the company may be required to pay in extra amounts to maintain a suitably large deposit to protect the market from losses the company may incur. The balance on the margin account must not fall below what is called the '**maintenance margin**'. In our scenario, the CME contract specification for the £/\$ futures states that a maintenance margin of \$1,250 per contract is required. Given that the company is using 39 contracts, this means that the balance on the margin account must not fall below $39 \times \$1,250 = \$48,750$.

As you can see, this does not present a problem on 11 July or 14 July as gains have been made and the balance on the margin account has risen. However, on 15 July a significant loss is made and the balance on the margin account has been reduced to \$41,340, which is below the required minimum level of \$48,750.

Hence, the company must pay an extra \$7,410 ($\$48,750 - \$41,340$) into their margin account in order to maintain the hedge. This would have to be paid for at the spot rate prevailing at the time of payment unless the company has sufficient \$ available to fund it. When these extra funds are demanded it is called a '**margin call**'. The necessary payment is called a '**variation margin**'.

If the company fails to make this payment, then the company no longer has sufficient deposit to maintain the hedge and action will be taken to start closing down the hedge. In this scenario, if the company failed to pay the variation margin the balance on the margin account would remain at \$41,340, and given the maintenance margin of \$1,250 this is only sufficient to support a hedge of $\$41,340/\$1,250 \approx 33$ contracts. As 39 futures contracts were initially sold, six contracts would be automatically bought back so that the markets exposure to the losses the company could make is reduced to just 33 contracts. Equally, the company

will now only have a hedge based on 33 contracts and, given the underlying transaction's need for 39 contracts, will now be underhedged.

Conversely, a company can draw funds from their margin account so long as the balance on the account remains at, or above, the maintenance margin level, which, in this case, is the \$48,750 calculated.

Foreign exchange options – the basics

Scenario

Imagine that today is the 30 July. A UK company has a €4.4m receipt expected on 26 August. The current spot rate is £0.7915/€1. They are concerned that adverse exchange rate fluctuations could reduce the £ receipt but are keen to benefit if favourable exchange rate fluctuations were to increase the £ receipt. Hence, they have decided to use €/£ exchange traded options to hedge their position.

Research shows that €/£ options are available on the CME Europe exchange.

The contract size is €125,000 and the futures are quoted in £ per €1. The options are American options and, hence, can be exercised at any time up to their maturity date.

Setting up the hedge

1. Date? – September:

The available options mature at the end of March, June, September and December. The choice is made in the same way as relevant futures contracts are chosen.

2. Calls/Puts? – Puts:

As the contract size is denominated in € and the UK company will be selling € to buy £, they should take the options to sell € for £ – put options.

3. Which exercise price? – £ 0.79250/€1

An extract from the available exercise prices showed the following:

Exercise price £/€1	Put premiums £/€1
0.79000	0.00465
0.79250	0.00585

As the company is selling €, it wants the maximum net £ receipt for each € sold. The maximum net receipt is the exercise price minus the premium cost.

This is calculated below:

Exercise price £/€1	Put premiums £/€1	Net receipt £/€1
0.79000	0.00465	$0.7900 - 0.00465 = 0.78535$
0.79250	0.00585	$0.79250 - 0.00585 = 0.78665$

Hence, the company will choose the 0.79250 exercise price as it gives the maximum net receipt. Alternatively, the outcome for all available exercise prices could be calculated.

In the exam, either both rates could be fully evaluated to show which is the better outcome for the organisation or one exercise price could be evaluated, but with a justification for choosing that exercise price over the other.

4. How many? – 35

This is calculated in a similar way to the calculation of the number of futures. Hence, the number of options required is:
 $\text{€}4.4\text{m} / \text{€}0.125\text{m} \approx 35$

Summary

The company will buy 35 September put options with an exercise price of £0.79250 /€1

Premium to pay – $\text{£}/\text{€}0.00585 \times 35 \text{ contracts} \times \text{€}125,000 = \text{£}25,594$

Outcome on 26 August:

On 26 August the following was true:

Spot rate – £ 0.79650/€1

As there has been a favourable exchange rate move, the option will be allowed to lapse, the funds will be converted at the spot rate and the company will benefit from the favourable exchange rate move.

Hence, $€4.4m \times 0.79650 = £3,504,600$ will be received. The net receipt after deducting the premium paid of £25,594 will be £3,479,006.

Note:

Strictly a finance charge should be added to the premium cost as it is paid when the hedge is set up. However, the amount is rarely significant and, hence, it will be ignored in this article.

If we assume an adverse exchange rate move had occurred and the spot rate had moved to £ 0.78000/€1, then the options could be exercised and the receipt arising would have been:

Receipt	€4,400,000	
Exercise option:		
Pay – 35 x 125,000	(€4,375,000)	
Receive – 4.375m x 0.79250		£3,467,188
Underhedged amount	€25,000	
Buy £ at spot (£0.78/€1)	(€25,000)	£19,500
	0	£3,486,688
Deduct premium cost		(£25,594)
Net receipt – see Note 1		<u>£3,461,094</u>

Notes:

1. This net receipt is effectively the minimum receipt as if the spot rate on 26 August is anything less than the exercise price of £ 0.79250/€1, the options can be exercised and approximately £3,461,094 will be received. Small changes to this net receipt may

occur as the €25,000 underhedged will be converted at the spot rate prevailing on the 26 August transaction date. Alternatively, the underhedged amount could be hedged on the forward market. This has not been considered here as the underhedged amount is relatively small.

2. For simplicity it has been assumed that the options have been exercised. However, as the transaction date is prior to the maturity date of the options the company would in reality sell the options back to the market and thereby benefit from both the intrinsic and time value of the option. By exercising they only benefit from the intrinsic value. Hence, the fact that American options can be exercised at any time up to their maturity date gives them no real benefit over European options, which can only be exercised on the maturity date, so long as the options are tradable in active markets. The exception perhaps is traded equity options where exercising prior to maturity may give the rights to upcoming dividends.

Summary

Much of the above is also essential basic knowledge. You are unlikely to be given the spot rate on the transaction date. However, the future spot rate can be assumed to equal the forward rate which is likely to be given in the exam. The ability to do this may earn up to six marks in the exam. Equally, another one or two marks could be earned for reasonable advice.

Foreign exchange options – other terminology

This article will now focus on other terminology associated with foreign exchange options and options and risk management generally. All too often students neglect these as they focus their efforts on learning the basic computations required. However, knowledge of them would help students understand the computations better and is essential knowledge if entering into a discussion regarding options.

Long and short positions

A '**long position**' is one held if you believe the value of the underlying asset will rise. For instance, if you own shares in a company you have a long position as you presumably believe the shares will rise in value in the future. You are said to be long in that company.

A '**short position**' is one held if you believe the value of the underlying asset will fall. For instance, if you buy options to sell a company's shares, you have a short position as you would gain if the value of the shares fell. You are said to be short in that company.

Underlying position

In our example above where a UK company was expecting a receipt in €, the company will gain if the € gains in value – hence the company is long in €. Equally the company would gain if the £ falls in value – hence, the company is short in £. This is their '**underlying position**'.

To create an effective hedge, the company must create the opposite position. This has been achieved as, within the hedge, put options were purchased. Each of these options gives the company the right to sell €125,000 at the exercise price and buying these options means that the company will gain if the € falls in value. Hence, they are short in €.

Therefore, the position taken in the hedge is opposite to the underlying position and, in this way, the risk associated with the underlying position is largely eliminated. However, the premium payable can make this strategy expensive.

It is easy to become confused with option terminology. For instance, you may have learnt that the buyer of an option is in a long position and the seller of an option is in a short position. This seems at variance with what has been stated above, where buying the put options makes the company short in €. However, an option buyer is said to be long because they believe that the value of the option itself will rise. The value of put options for € will rise if the € falls in value. Hence, by buying the €/£ put options the company is taking a short position in €, but is long the option.

Hedge ratio

The hedge ratio is the ratio between the change in an option's theoretical value and the change in the price of the underlying asset. The hedge ratio equals $N(d_1)$, which is known as delta. Students should be familiar with $N(d_1)$ from their studies of the Black-Scholes option pricing model. What students may not be aware of is that a variant of the Black-Scholes model (the Grabbe variant – which is no longer examinable) can be used to value currency options and, hence, $N(d_1)$ or the hedge ratio can also be calculated for currency options.

Hence, if we were to assume that the hedge ratio or $N(d_1)$ for the €/£ exchange traded options used in the example was 0.95 this would mean that any change in the relative values of the underlying currencies would only cause a change in the option value equivalent to 95% of the change in the value of the underlying currencies. Hence, a €0.01 per £ change in the spot market would only cause a €0.0095 per £1 change in the option value.

This information can be used to provide a better estimate of the number of options the company should use to hedge their position, such that any loss in the spot market is more exactly matched by the gain on the options:

Number of options required = amount to hedge / (contract size x hedge ratio)

In our example above, the result would be:

$\text{€}4.4\text{m} / (\text{€}0.125\text{m} \times 0.95) \approx 37 \text{ options}$

Conclusion

This article has revisited some of the basic calculations required for foreign exchange futures and options questions using real market data, and has additionally considered some other key issues and terminology in order to further build knowledge and confidence in this area.

Business valuations

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Business valuations**

Business valuation is 'an art not a science'. These are the words used by many ACCA financial management tutors (including myself) when introducing this topic to students preparing for *Advanced Financial Management*. The words imply that when trying to value the equity capital of a business, there is range of possible correct answers, all of which can be justified as being the most appropriate. To a certain extent this is true but, as I like to put it, 'there are different degrees of correctness'.

Questions on Business Valuations are included in every *Financial Management* exam. The questions have typically tested the 'basic' equity valuation methods of:

- net assets
- dividend valuation model (or dividend growth model)
- earnings model using P/E ratio or earnings yield

The *Advanced Financial Management* syllabus builds on those methods tested at the lower level paper. The concept is the same – to find the value of equity. However, the techniques and methods are more sophisticated. As I stated above, 'there are different degrees of correctness'.

The primary purpose of this article is to demonstrate how to tackle an *Advanced Financial Management* business valuation question. The detailed understanding of this topic will be gained from your studies, whichever mode you choose to use. My aim is to show you how to successfully apply this knowledge under exam conditions.

Equity valuation – categorising the methods

As stated above, there are more methods and models that can be used to find an equity share price in *Advanced Financial Management*. The official textbooks explain these in detail and choose different ways of categorising them within their material. I prefer to take a simple view of equity valuation by allocating the methods into two main categories:

- Pre-acquisition
- Post-acquisition

Under the first category, the question will be asking the students to ascertain an equity value for a company. The entity may be a private company and, hence, no stock market price exists or that even if the company is listed, the market price may not be appropriate for the relevant situation. The valuation methods appropriate here are:

- net assets
- dividend valuation model (or dividend growth model)
- earnings model using P/E ratio or earnings yield
- net assets + calculated intangible value (CIV)
- free cash flows (FCF)

Past questions have, in my view, clearly indicated which method should be used to arrive at the share price. However, it is fair to say that the free cash flow model has been tested more than any other method.

Post-acquisition valuation requires a different mindset and series of methods. Here, students will need to ascertain the value of the combined companies after acquisition. More importantly, past exam requirements have requested students to ascertain the percentage gain or loss to both groups of shareholders – those of both the buying and selling companies.

The post-acquisition valuation methods are:

- bootstrapping – applying the price earnings ratio of the buyer to the combined expected earnings of the two entities
- combining the pre-acquisition values of the two companies and appending these with the fair value of the synergies
- free cash flows (FCF) – present value of the combined companies FCF using the relevant discount rate.

As I have stated above, your preparation should include ample time to study and understand the equity valuation methods above, allowing you to apply your knowledge successfully in the exam room.

Below is a worked sample question illustrating how I would tackle a 25-mark exam style question, based broadly on previous exam content.

Borgonni and Venitra

Borgonni Co is a very successful entity. The company has consistently followed a business strategy of aggressive acquisitions, looking to buy companies that it believes were poorly managed and hence undervalued. Borgonni can be described as a modern day conglomerate and its business interests stretch far and wide.

Its board of directors has chosen the takeover targets with care. Always looking for companies with potential, but which were poorly managed and having a below par market value, Borgonni has maintained its price earnings (P/E) ratio on the stock market at 12.2.

Borgonni's 20X3 figures show a profit after tax of \$886m and it has 375m shares in issue.

Venitra Pvt is a well-established owner-managed business. In financial terms it has a rather chequered history with its up and downs corresponding directly with the state of the global economy. Over the past five years, its profits have fallen each year with the 20X3 values standing at:

	\$m
Revenue	<u>1,500</u>
Operating profit	480
Interest	<u>(137)</u>
Profit before tax	343
Taxation @ 25%	<u>(86)</u>
Profit after tax	<u>257</u>
Number of shares in issue	150m

\$m

EPS

\$1.72

However, with economists predicting an upturn in the Western economies, Venitra's management team feel that revenue will increase by 6% per annum up to and including 20X7. The company's operating profit margin is not expected to change for the foreseeable future.

Operating profits are shown after deducting non-cash expenses (including tax allowable depreciation) of \$125m. This is expected to increase in line with sales. However, the company has recently spent \$210m on purchase of non-current assets. Venitra's management believes this value will have to increase by 10% per annum until 20X7 to enable the company to remain competitive. Venitra has estimated its overall cost of capital to be approximately 12%, but this assumes it will maintain its debt to equity ratio at 40:60.

Some of Venitra's major shareholders are not so confident about the future and would like to sell the business as a going concern. The minimum price they would consider would be the fair value of the shares, plus a 10% premium. Venitra's CFO believes the best way to find the fair value of the shares is to discount the forecasted free cash flows of the firm, assuming that beyond 20X7 these will grow at a rate of 3% per annum indefinitely.

Requirement

(a) As at 1 January 20X4, prepare a schedule of Venitra's forecast free cash flows for the firm. Ascertain the fair value of the Venitra's equity on a per share basis.

(10 marks)

(b) Borgonni intends to make an offer to Venitra based upon a share for share swap. Borgonni will exchange one of its shares for every two Venitra shares. Assuming that Borgonni can maintain its earnings rating at 12.2, calculate the percentage gain in equity value that will be earned by both groups of shareholders?

(8 marks)

(c) What factors should the Venitra shareholders consider before deciding whether to accept or reject the offer made by Borgonni?

(7 marks)

(25 marks)

Solution

At this stage, you can choose one of two ways to follow my approach to answering this question. My solution to each part along with the relevant explanation is shown below.

(a) Venitra – Forecast Free Cash Flows and Value of Equity

	A	B	C	D	E	F	G	H
1		20X3	20X4	20X5	20X6	20X7	20X8+	
2		\$m	\$m	\$m	\$m	\$m	\$m	
3		Reference only						
4	Revenue	1,500	1,590	1,685	1,787	1,894		
5								
6	Operating profit		509	539	572	606		
7	Taxation @25%		(127)	(135)	(143)	(152)		
8	Add: Non-cash expenses	125	133	140	149	158		
9	Less: Investment in NCA	(210)	(231)	(254)	(280)	(307)		
10	FCF		284	290	298	305	314	
11	Discount @12%		0.893	0.797	0.712	0.636	7.067	
12			254	231	212	194	2,219	
13								
14	Value of the entity (\$m)	3,110						
15	Equity value (60%) (\$m)	1,866						
16	Value per share (Po) (\$)	12.44						
17	P/E (\$12.44/\$1.72)	7.23						

Notes and explanations:

- As we are preparing a valuation as at 1 January 20X4, I have set up columns for each future period. I need to prepare a detailed forecast for the first four years only. After 20X7, the FCF of the firm will increase at the rate of 3% per annum. I have started with a 20X3 column just as a reference point.
- Revenue has been increased by 6% per annum using the 20X3 sales as a base point.
- The operating margin in 20X3 was 32% ($480/1,500$). This will be maintained for the foreseeable future.
- One key factor is to ignore the interest payment. FCF for the firm must EXCLUDE interest. This is because the cost of capital used to discount these flows is the company WACC. The WACC takes into account the interest element and its tax benefit.
- Income tax on company profits is charged at 25%. In this case, it is to be paid in the same year as the profits are earned.
- The operating profit is after deducting non-cash expenses, which are allowable for taxation. These include tax allowable depreciation. In this question, these expenses will increase in line with sales and they have to be added back after the tax charge has been computed.
- Venitra needs to set aside cash each year to maintain its non-current asset (NCA) base. The amount of capital expenditure will increase by 10% per annum for the next four years.
Please note that in some past questions, it has been assumed that the non-cash expenses equal the required investment in NCAs. Hence, the add back and deduction will cancel out.
- The forecast FCF for the firm is a simple totalling up process for the first four years. After 20X7, the FCF are expected to grow at a rate of 3% per annum indefinitely. Therefore, the 20X8 value is calculated as $\$305m \times 1.03$.
- As stated in point 4, the relevant discount rate to apply to the FCF of the firm is Venitra's WACC. This has been estimated as 12%. The first four discount factors have been copied from the discount tables provided at the end of the exam paper. The discount factor for 20X8 and beyond must take into account both a 3% per annum growth rate as well as a cost of capital of 12%. The financial mathematics for a delayed perpetuity with an annual growth rate is $(1/(0.12 - 0.03) \times 0.636)$.
- The value of the entity is the total of the present value of the forecast FCF. However, this amount represents a combination of the debt and equity together. Venitra's equity is equal to 60% of the value of the firm.
- The question requirement is to ascertain the equity value per share. Therefore, $\$1,866m / 150m = \12.44 . This is the fair value of one share of Venitra.
- Finally, I have computed the P/E ratio for Venitra. Although this was not specifically asked for, this value will be needed for part (b).

(b) Percentage gain in equity value – both groups of shareholders

The first stage is to compute the current market price per share for Borgonni:

	\$m
20X3 – Earnings	886
	x
P/E ratio	<u>12.2</u>
Value of equity	<u>\$10,809m</u>
No of issued shares	375m
Value per share (Po)	\$28.82

Borgonni expects to maintain its P/E ratio after acquiring Venitra. Therefore, the post-acquisition value of the two entities combined together can be ascertained by applying Borgonni's P/E ratio to the sum of the latest earnings of each company. As the P/E ratio of Borgonni (12.2) exceeds that of Venitra (7.23) this is known as 'bootstrapping'.

	\$m
Borgonni – 20X3 PAT	886

\$m

Venitra – 20X3 PAT

257

1,143

P/E ratio – Borgonni

12.2

Post-acquisition value

\$13,945m

The purchase is to be funded via a share for share exchange. Borgonni will issue one new share in its company in return for every two shares in Venitra.

Borgonni issued share capital

375m

Additional shares issued (150m/2)

75m

New total issued share capital

450m

The new equity value for a Borgonni share is now $\$13,945\text{m}/450\text{m} = \30.99 .

However, although many candidates may stop at this point (believing they have reached Utopia!) the requirement has not been addressed. The question asks candidates to ascertain the gain that will be made on the equity value to each group of shareholders. Looking at each in turn:

Borgonni shareholders' gain $((\$30.99 - \$28.82)/\$28.82) \times 100 = 7.53\%$

To compute the gain for the Venitra shareholders, the candidate must first compute the post-acquisition value of a Venitra share. Venitra shareholders gave up two shares in their company to receive one new Borgonni share. Therefore, the equivalent post-acquisition value of a Venitra share will be $\$30.99/2 = \15.50 .

The fair value of a Venitra share, per part (a), was \$12.44. Therefore, the Venitra shareholders gain **24.60%**.

(c) Factors to consider – for the Venitra shareholders

There is no one correct answer to this part. As long as the candidate produces a reasonable number of valid points, they will earn decent marks.

My answer would read as follows:

- Venitra shareholders wanted a gain of at least 10% on the fair value of the shares. Based upon the figures, they are gaining nearly 25%, which is likely to encourage them to accept the offer.
- The share for share exchange may be beneficial for tax planning. Any capital gain earned on the sale of the shares will be rolled over until the gain is realised in cash.
- Venitra may decide to reject this bid believing that Borgonni will make a more lucrative offer in the future.
- The fair value of the Venitra shares has been based upon forecasts and estimates. Some sensitivity analysis needs to be carried out to ensure the value is robust.
- There is no guarantee that Borgonni can maintain its P/E ratio at 12.2. There may well be an element of dilution given the much lower P/E of Venitra. Hence, the post-acquisition value is then uncertain.
- Not all Venitra shareholders want to sell the company. The constitution of the company may allow the takeover to be blocked unless a certain percentage majority of the shareholders agree.
- Venitra shareholders may also feel that as the economic conditions are improving, their business prospects and value will get better. They may reject Borgonni's approach and stay as an independent company.

As you can see, business valuation questions require you to have a disciplined approach and to demonstrate that you have studied and understand this key area of the syllabus. Although equity valuations are an 'art not a science', you have to produce an answer that is pleasing to the eyes of the examining and marking team.

Sunil Bhandari, freelance tutor

www.SunilBhandari.com



Related Links

- **[Student Accountant hub page](#)**

International project appraisal

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **International project appraisal**

International project appraisal is an integral part of the *Advanced Financial Management* syllabus.

The purpose of this series of articles is to assist your preparation for the exam by demonstrating how to attempt exam questions on this area of the syllabus. Coupled with a comprehensive mode of study and revision, you should be ready for whatever the exam may contain. International project appraisal will be a large component of your studies, and I will demonstrate a systematic method of answering a question on this topic for each section of the exam.

In this article, I will demonstrate how to answer a 'Section B' style 25-mark question.

Penn Co

Penn Co is successful company based in a European country where the local currency is the dollar and inflation has been stable at 5% pa. Income tax is charged on company profits at the rate of 25% and is payable in the same year as the profits are earned.

The company is listed on several major stock exchanges as it has operations all over the globe. Its market capitalisation is

\$655m. The company has bonds with varying maturities trading at \$145m.

The treasury department of the company regularly computes the company's nominal cost of capital and this has been fairly stable at 10%. However, when Penn Co have carried out projects in developing markets it has used a nominal risk-adjusted rate of 12%.

Penn Co's primary business is construction and laying of train tracks and tramlines. Their main consumers are governments due to Penn Co's position as the market leader. Penn Co has a record of completing long and complex contracts within schedule, as well as conducting its business in an ethical manner.

The CEO of Penn Co recently attended a trade delegation to Africa where he met the prime minister of the fast developing country called Zanadia. The prime minister and his political team provided Penn's CEO with an outline of a contract that the Zanadian government would like to award to Penn Co.

Tramline project

Zanadia is situated on the African West coast, with the local currency being the dinar. Although being relatively small when compared to its neighbours, its economy has grown at over 15% pa for the past five years, but this has led to an inflation rate currently running at 30% pa. The democratically elected government has taken full credit for this economic prosperity.

The prime minister is adamant that the performance of the country is a result of trade links he created with European-based multinational corporations (MNCs). He believes that by encouraging investment from these entities in his country, the MNCs will generate substantial returns with minimal risk, as many of the projects are government contracts.

There has been varied success when European MNCs have invested in Zanadia, with political interference, particularly from the prime minister, being blamed for below par returns. Rumours have been rife that the prime minister has been ordering his government to make ad hoc requests for payments to be made at various times during the contracted period. The government themselves have stated that, on a very rare basis, penalty charges have been levied when companies have not been keeping to schedule.

The Zanadian government's latest project is to create an environmentally friendly electric tramline network to connect all areas of Enat, which is Zanadian's capital city. The project will ultimately take 20 years to complete. However, the initial contract will be to lay the tramline to connect Enat to the national airport located 23 kilometres away. Providing this is completed to the satisfaction of the Zanadian government, they will extend the contract to allow initial supplier connect the rest of the city.

Following the recent meeting between the prime minister and the CEO of Penn Co, the prime minister has authorised his government officials to release financial projections to Penn Co to allow it to assess the financial viability of the contract.

Financial details

The government will pay a fixed price of dinar 5000m for the initial contract to lay the tramline between Enat and the national airport. This will be paid in stages:

Time period	%
6 months	10
12 months	20
18 months	40
24 months	30

Machinery needs to be purchased in Zanadia at a cost of dinar 1000m at the start of the project. The other costs will be locally incurred labour costs, which are estimated to be 30% of the revenue. All values are given in nominal terms. The government has already purchased sufficient materials from a low cost provider based in the US for the initial 23 kilometres of the tramline. They only require Penn Co to lay and test the tramline.

The government taxes company profits at 40% and this is to be paid in the year in which the profit is earned. The government has no provision to offset tax allowable depreciation (TAD).

The current spot rate Dinar/\$ 150 – 175 and this is expected to change in the future based upon the relative inflation rates.

After two years, the prime minister will personally review the work carried out by Penn Co and he expects to extend the contract to complete Enhat's city tramline. The exact terms of this contract extension will be subject to negotiation but the returns are expected to be substantial.

Requirement

- Prepare a financial assessment of the project covering the initial two-year period assuming Penn Co appraises projects by discounting nominal \$m cash flows at the appropriate cost of capital. State clearly any assumptions that you make. *(11 marks)*
- Explain the main risks and issues faced by Penn Co if it chooses to undertake this project. *(14 marks)*

(25 marks)

The key to a good *Advanced Financial Management* answer is to have a clear plan when answering the question. Knowing where to start and how to progress through in a smooth and efficient way is vital.

Time allocation

The 25-mark question needs to be completed in 45 minutes, allowing 1.8 minutes per mark. A split of 20 minutes for part (a) and 25 minutes for part (b) is a good starting point in terms of allocation, but with students often finding the numerical elements challenging, allowing a minimum of 20 minutes for part (b) will give flexibility.

Understand the requirements

What are you been asked to do? Read the requirements and understand the key words. Match them to your *Advanced Financial Management* knowledge.

In this case:

(a) Compute a net present value in \$m. 'Nominal' cash flows means adjusted for relevant inflation. 'Appropriate' cost of capital – my initial thoughts are either the company's weighted average cost of capital (WACC) or a risk adjusted WACC. I need to list out any assumptions I make.

(b) 'Risks and issues' – I assume I will be able to derive most of these from the main body of the question coupled with the relevant *Advanced Financial Management* knowledge areas.

Which part to attempt first?

From experience, most candidates attempt the question in the order it is set. There are no instructions that says you *have* to do this. Decide which order you feel most comfortable with, but ensure you make a valid attempt at both parts of the question.

As for me, *I would attempt Part (b) first*. I feel I am more likely to be in control of my time this way. However, I will show my answer in the order the question was set.

The read through

I understand the requirements. Now, I need to digest the details of the question. My approach is a simple one.

Part (a) – Answer

To address this requirement I need to show a disciplined approach. Let me break this down into stages and explain my thought process for each step.

(1) Dinar cash flows

Description	Now Dinar (m)	6 months Dinar (m)	12 months Dinar (m)	18 months Dinar (m)	24 months Dinar (m)
Revenue		500	1,000	2,000	1,500
Variable cost (30%)		(150)	(300)	(600)	(450)
Taxable cash flows		350	700	1,400	1,050
Taxation @ 40%		(140)	(280)	(560)	(420)
Initial investment	(1,000)				
Project cash flows	(1,000)	210	420	840	630

The points to notes here:

- Columnar layout corresponding with the timing of the cash flows specified in the question.
- Cash flows are in nominal dinars; they have already been adjusted for Zanadian inflation.
- Revenue is dinar 5000m allocated per the percentages specified in the question. I see no reason to show a working for this.
- Variable cost is just 30% of the revenue figures as indicated on the schedule.
- Sub-totalled to show the taxable 'profits'. As there is no TAD in this question, these are equal to the operating cash flows.
- Taxation is a relevant cash flow. On the read through stage, I noted that there was no time delay in the payment of the tax to the Zanadian government.
- Initial investment is a simply copy and paste.
- The total project cash flows show that Penn Co will need to initially buy dinars (sell \$s) to make the investment. Subsequently, Penn Co will be receiving dinars that it will convert into \$s. Care needs to be taken when choosing/calculating the spot rates.

(2) Spot rates and conversion to \$m

Project cash flows	(1,000)	210	420	840	630
Spot rate (W1) Dinar/\$	150	196	217	243	269
Free cash flows (\$m)	(6.67)	1.07	1.94	3.46	2.34

The question stated that the current spot rate was dinar 150 – 175/\$. As Penn Co needs to sell \$s initially, the bid rate of dinar 150/\$ will apply. As all other cash flows are receipts in dinars, a working is needed to compute the projected offer spot rates via the purchasing power parity theory (PPPT) formula.

		Dinar/\$
Now		175
6 months	$(175 + 217) / 2$	196
12 months	$175 \times 1.30 / 1.05$	217
18 months	$(217 + 268) / 2$	243
24 months	$217 \times 1.30 / 1.05$	269

The PPPT formula is used to calculate the annual spot rates. The intervening half-year rates are average values.

Finally, many students fail to convert the foreign cash flows into the home currency correctly. Based on my experience, I have seen many answers where confusion has reigned supreme, as students are not sure whether to divide or multiply. In this case the project cash flows are in dinars and our spot rates are dinar/\$. We divide the dinar cash flows to convert to \$m.

(3) Final answer

Description	Now Dinar (m)	6 months Dinar (m)	12 months Dinar (m)	18 months Dinar (m)	24 months Dinar (m)
Revenue		500	1,000	2,000	1,500
Variable cost		(150)	(300)	(600)	(450)
	_____	_____	_____	_____	_____
Taxable cash flows		350	700	1,400	1,050
Taxation @ 40%		(140)	(280)	(560)	(420)
Initial investment	(1,000) _____	_____	_____	_____	_____
Project cash flows	(1,000)	210	420	840	630
Spot rate (W1) Dinar/\$	150	196	217	243	269
Free cash flows (\$m)	(6.67)	1.07	1.94	3.46	2.34
Cost of capital (12%)	1.000 _____	0.945 _____	0.893 _____	0.844 _____	0.797 _____
Present values (\$m)	(6.67) _____	1.01 _____	1.73 _____	2.92 _____	1.87 _____
Net present			+\$0.86m		

value (\$m)

Discounting at the risk adjusted WACC of 12% (see assumptions below) should be the easy part. The discount factors for 12 and 24 months can be taken from the tables provided at the back of the exam. Those same tables provide the formula to use to calculate the six-month and 18-month discount factors remembering that $r = 0.12$ and $n = 0.5$ and 1.5 respectively.

(4) Assumptions

The requirement invites the students to state any assumptions. Here is a sample of some that could be made:

- The nominal risk adjusted cost of capital of 12% is the appropriate discount rate given that the project is based in a developing country.
- There is no additional taxation to pay in Penn Co's home country on the remitted dinar cash flows.
- Inflation rates in both countries will remain constant at their respective rates for the next two years.
- The PPPT formula provides a materially accurate assessment of the projected spot rates.
- There is no residual value for the machinery after two years, as it will continue in use after this initial period.

-

Part (b) – Answer

As stated above, students could choose to 'front load' the answer to this requirement. My thought process here is to derive as many relevant points from the information in the scenario that relate to 'risks and issues' and expand on these. The examiner is looking for students to apply their knowledge to the details given in the question. It is these points that will ensure you achieve a sound pass mark for this requirement.

In addition, I will include the relevant factual points that can be found in any of the *Advanced Financial Management* approved textbooks. They will certainly earn some marks.

Risks and issues facing Penn Co

- Sensitivity analysis – irrespective of the final NPV, the values used to arrive at this number are subject to estimation errors. The ones of particular concern are the cost of capital and predicted spot rates. Penn Co should carry out sensitivity analysis to identify how any changes to these variables affect the NPV.
- Recent history – although many MNCs have invested in Zanadia, this has not always led to success. Penn Co needs to investigate and ascertain a little more detail as to why this has been the case. Blame could lie with both parties to the contract.
- Ad hoc Payments – Penn Co needs to obtain some clarity on this matter. These charges would reduce the NPV of the project and may even change the decision that the company takes. The terms of the contract need to be carefully reviewed to ensure that Penn Co is aware of what the Zanadian government expects.
- Supplier – Penn Co has a reputation for manufacturing as well as laying tramlines. In this case, they will be installing lines purchased from another supplier. There may well be quality and specification issues.

- Prime minister – the prime minister has a lot of influence over this contract. He will ‘personally’ review the work carried out by Penn Co before deciding to extend the contract. The criteria on which his decision will be made are not specified and may well be highly judgmental.
- 20-year contract – this is a long project and the risks associated with time are very high. The returns are said to be substantial but again not quantified.
- Other issues – there are a number of other issues Penn Co should accrue for:
 - The company would need to be aware of local customs and work practices.
 - The legal and regulatory issues would need to be quantified.
 - If managers would be recruited in Zanadia or sent from Penn Co’s home country.
 - The project will not damage Penn Co’s business and ethical reputation.

International project appraisal questions are challenging, but far from impossible. Students need to follow a disciplined approach. The format is very similar to when preparing a standard ‘home country based’ NPV which candidates have practised many times as it is part of both *Financial Management* and *Advanced Financial Management*.

For projects based abroad it is about starting with the relevant nominal foreign currency cash flows. The key differences are the computation of the spot rates and conversion of foreign currency cash flows to domestic currency values. Discounting at risk-adjusted rates should not be unexpected given the change in risk levels when investing abroad.

There are certain skills that we have not seen tested above such as royalties, transfer pricing and double taxation. These will be part of my next article when we return to Penn Co and look at another international project it is considering as an investment opportunity.

Sunil Bhandari, freelance tutor

www.SunilBhandari.com



Related Links

- **[Student Accountant hub page](#)**



Think Ahead

International project appraisal - part 2

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **International project appraisal - part 2**

The first part of this article highlighted the importance of international project appraisal within the *Advanced Financial Management* syllabus. There was also a demonstration as to how to tackle a Section B, 25-mark question using material from your studies.

It is just as important to prepare for a Section A 50-mark test on international project appraisal. There will be a large section of text containing far more information than a Section B type question. However, this is compensated by a greater mark allocation and, hence, more time to produce an acceptable answer. As shown in the previous article, students need to adopt a methodical approach to ensure they are rewarded with high marks.

Let us now return to Penn Co and consider another international investment opportunity.

Penn Co

Penn Co is a successful company based in the European country, Ayjai. The local currency is the dollar (\$), inflation has been stable at 2.75% pa and income tax is charged on profits, in the year in which they are earned, at a rate of 25% pa. The company

is listed on several major stock exchanges as it has operations all over the globe. Its market capitalisation is \$655m. The company has bonds with varying maturities trading at \$145m.

Penn's nominal cost of capital has been stable at 10%. However, Penn Co uses a nominal risk-adjusted rate of 12% when carrying out projects in developing markets.

Penn Co's main operation is constructing and laying of train tracks and tramlines. Due to its position as the market leader, its primary consumers are governments. Penn Co is renowned for its ethical business style, and ability to complete long and complex contracts within schedule.

Penn Co sets up wholly owned subsidiary companies in each country where it has business interests, including in Nuruk.

Nuruk

Nuruk is a fully-fledged member of the euro zone and shares a border with Ayjai. Its currency is the euro (€). Nuruk is a well-developed country and, unlike most of the euro zone, its economy is growing at a healthy rate.

The primary reason for Nuruk's current economic state is its low level of taxation. Income tax is charged at 20% pa and can be paid up to one year after profits are earned. In addition, the Nurukian government reacted to the global recession with a substantial fiscal expenditure plan, leading to the enhancement of the national railway network.

Since 2009, the government have invested in replacing and upgrading the state-owned national railway network to allow the lines to run the new 'SuperFast2 (SPF2)' trains. The government committed to a 10-year plan to ensure SPF2 trains could operate on lines nationwide.

Penn Co, via its Nurukian subsidiary, has benefited from the government investment in the railway network. The subsidiary was granted preferred supplier status by the government in 2009. It has been the primary, but not the exclusive, business partner to the government. To date, Penn Co have supplied the entire specialist train track required to run the SPF2 and have consulted and advised the various construction companies, contracted by the government, on the laying and testing process. Currently, all stakeholders are content with the progress made.

Final Phase

The final phase of the project will take five years to complete. The track is to be laid on a national heritage site, the Linus mountain range, by which there are many small villages.

The government has been scrutinised by both the villagers and environmental protest groups, concerned that the new line would cause substantial ecological damage. In 2010, the government pushed back the start date to 1 January 2014 in order to hold a public enquiry and hear the concerns of the stakeholders. They decided that environmental considerations should be prioritised

when laying the SPF2 rail line and it should be considered a 'special case'. The government accepted these findings and decided that Penn Co would be the most suited company to carry out the upgrades due to its ethical approach.

Penn Co is required to supply, fit and test the line via its subsidiary. The government will closely monitor the project due to the outcome of the enquiry and, in addition, has allocated extra resources to this phase, as it understands the task of laying the new rail-line will be onerous.

Penn Co wishes to consider the financial and other implications of the project before making a final decision. The subsidiary will need to buy specialist machinery at the commencement of the project for €1,000m. The company can claim tax allowable depreciation (TAD) on only €250m of this investment, claimed on a straight-line basis over the life of the project.

Penn Co's treasury department believes at this financial investment will not alter the company's gearing level nor will the project affect its business risk profile. However, the necessary amount of funds to purchase specialist machinery will have to be raised in Ajjain dollars via the financial markets.

One key stipulation of the public enquiry was to specify how many metres of line could be laid in each calendar year:

Year ending 31 December	Metres
2014	5,700
2015	6,500
2016	10,900
2017	8,100
2018	6,300

The government will pay Penn Co, €55,000 per metre at the end of 2014, increasing by 3% pa. Material and local labour costs are expected to be €23,000 per metre at the end of 2014, with expected increases at a rate of 5% pa thereafter. Fixed operating costs will increase by €40m at the end of 2014 and this amount will rise by 6% pa.

Penn Co has a standard policy that all its foreign subsidiaries must make a fixed annual royalty payment of \$15,000 per metre back to the holding company at the end of each respective year. This is a fair arms length value to cover the investment made by Penn Co to develop the train track technology.

Working capital funds will be needed from 1 January 2014. The initial amount can be estimated to be 10% of the revenue earned at the end of year 2014. Each year, this will need to be adjusted by €10 for each €100 change in annual sales revenue. Working capital will be recovered in full on 31 December 2018. On the same day, the Nurukian government has guaranteed to purchase from Penn Co the specialist machinery for a nominal value of €500m.

Economic forecasters believe that the mid-point spot exchange rate on 1 January 2014 will be €0.7810/\$. The Ayjain Central Bank expects the dollar to devalue at a rate of 5% pa. The current risk free rate is 4.5% pa. The estimated standard deviation of the future free cash flows is 30%.

A bilateral tax treaty exists between the countries of Ayjai and Nuruk – hence, taxable profits earned in Nuruk will be liable to the differential income tax rate on company profits that applies between the two countries. The Ayjain government expects this to be paid in the same year as the taxable profits are earned.

Offer from Elders Inc

Elders Inc is the largest construction company based in Nuruk. Since 2009, it has laid and tested a substantial amount of the new SPF2 train line in Nuruk. It has worked closely with Penn Co as it supplied this train track.

The board of directors (BoD) were bemused that the Nurukian government did not offer them the SPF2 contract for the final phase. They believe that they have gone through the learning curve and could do the work on an efficient basis.

The BoD decided to approach Penn Co with an offer of \$1,200m to purchase the contract from them in two years time (31 December 2015). Penn Co's lawyers have advised them that the Nurukian government has not expressly precluded Penn Co from exiting the contract early, but advise Penn Co to consider their ethical stance should they decide to do so.

Alternative Sources of Finance

The chief financial officer (CFO) of Penn Co has concerns about the substantial initial investment required to start the project, relative to Penn Co's market value. The company's financial advisers agree with the CFO and are suggesting two alternative methods of raising the funds.

- €1,000m five-year 6.25% syndicated bank loan – Penn Co's advisers believe that a number of Nurukian banks would be willing to participate in such a transaction. They also believe that they may be able to persuade the Nuruk government to provide a subsidised interest rate of 4% pa on an element of this loan.
- To raise the required funds using Islamic finance in the form of sukuk bonds. The advisers feel that the project's characteristics are within the Sharia law regulations and this would give Penn Co access to low cost finance.

Requirement

Prepare a report to the Board of Directors (BoD) of Penn Co that:

a) Provides a financial assessment of the final phase of the Nuruk train line project as at 1 January 2014. All cash flows are to be presented in nominal terms and the project's dollar free cash flows are to be discounted at the appropriate nominal cost of capital. **Ignore** the offer from Elders Inc and the alternative finance options. (22 marks)

b) A discussion of the assumptions made in arriving at the financial assessment.
(5 marks)

c) An assessment of the offer made by Elders Inc to purchase the contract from Penn Co in two years time. This should include an estimate of the financial value of the real option. (9 marks)

d) A discussion of the two alternative finance options specifically addressing:

(i) If Penn Co raised the funds from the banks based in Nuruk, how this would affect the financial assessment of the project.

No further calculations are required.

(4 marks)

(ii) The key differences that Penn Co should be aware of between raising money via the Islamic finance option as opposed to traditional forms of debt capital. (6 marks)

Professional marks awarded for format, structure and presentation of the report.

(4 marks)

(50 Marks)

Students will not be surprised to see a scenario-based question 1 containing a vast amount of information. Several areas of the syllabus will be tested, including international project appraisal. Before focusing on the primary topic, I wish to demonstrate my step-by-step approach to answering question 1.

Understand The Requirements and Allocate Your Time

This question represents 50% of the exam – therefore, the answer should be completed in 90 minutes. However, the requirements of the question should be understood. 'Topic recognition' as I call it entails identifying which part of the syllabus is being targeted by each requirement. Simultaneously, I will allocate my time based upon the standard approach of 1.8 minutes per mark.

a) Keywords 'financial assessment', 'project's dollar nominal cash flows' and 'discounted' would trigger my thoughts. I have to prepare a schedule of free cash flows and compute the net present value (NPV). 22 marks would indicate a time allotment of 40 minutes. However, there are four professional marks for structure and presentation, which I can spread across the requirements. Revised time allocation – 45 minutes.

b) 'Assumptions' relating to the financial assessment – lots of scope to score marks here within nine minutes.

c) 'Real option' takes my thought process directly to the Black-Scholes Option Pricing model (BSOP). I have to compute the value of this PUT option and add the relevant discussion points. Allocate 17 minutes.

d) The topic under scrutiny here appears to be two different forms of debt finance. However, the requirements need to be interpreted very carefully.

(i) How raising loan finance will affect the project appraisal. My initial thoughts are to explain the Adjusted Present Value (APV) appraisal method. (8 minutes)

(ii) Islamic finance – I need to apply my knowledge of Islamic finance (sukuk bonds) to answer this final requirement. (11 minutes)

Answer Format

The question has clearly stated that the answer should be presented in a report format. The best way to do this is have appendices showing the computational elements, followed by the discussion parts in the main body of the headed report. In this case, I would layout my answer:

- Appendix 1 – NPV and relevant workings
- Appendix 2 – Real option valuation using BSOP model
- Headed report – With four subheadings matching the requirements.

From reading the examiner's report published after each exam, there appear to be a worrying number of candidates who don't format their answer as requested, and then missing out on the 'easy-to-earn' marks.

The Read Through

I understand the requirements. Now, I need to digest the details of the question. My approach is a simple one.

Let me now return and concentrate on preparing an answer on the international project appraisal aspects of this question.

Appendix 1 – NPV and Workings

Description	Time 0 €m	Time 1 €m	Time 2 €m	Time 3 €m	Time 4 €m	Time 5 €m	Time 6 €m
Revenue (W1)	_____	313.50	368.23	636.01	486.81	389.99	_____
Variable cost (W2)		131.10	156.98	276.40	215.67	176.13	
Incremental fixed costs		40.00	42.40	44.94	47.64	50.50	
Royalty (W3)		63.44	68.72	109.48	77.29	57.11	
TAD (250/5)		50.00	50.00	50.00	50.00	50.00	
Total costs	_____	284.54	318.10	480.82	390.60	333.73	_____
Taxable cash flows		28.96	50.13	155.19	96.21	56.25	
Taxation @ 20%			(5.79)	(10.03)	(31.04)	(19.24)	(11.25)

Add: TAD		50.00	50.00	50.00	50.00	50.00	
Initial investment	(1000.00)						
Scrap proceeds						500.00	
Working capital	(31.35)	(5.47)	(26.78)	14.92	9.68	39.00	
€m	(1031.35)	73.49	67.56	210.08	124.86	626.01	(11.25)
Spot rate (W4) €/ \$	0.7810	0.7420	0.7049	0.6696	0.6361	0.6043	0.5741
	\$m	\$m	\$m	\$m	\$m	\$m	\$m
Remitted amounts	(1320.55)	99.05	95.84	313.74	196.28	1035.89	(19.60)
Royalty income (W3)		85.50	97.50	163.50	121.50	94.50	
Taxation on royalty Income @ 25%		(21.38)	(24.38)	(40.88)	(30.38)	(23.63)	

Additional tax on € Taxable profits (W5)		(1.95)	(3.56)	(11.59)	(7.56)	(4.65)	
Free cash flows	(1320.55)	161.22	165.41	424.78	279.84	1102.11	(19.60)
Cost of capital (10%)	1.000	0.909	0.826	0.751	0.683	0.621	0.564
Present values (\$m)	(1320.55)	146.55	136.63	319.01	191.13	684.41	(11.05)
Net present value (\$m)			+146.13				

My explanations and workings are as follows:

- Columnar layout corresponding with the timing of the cash flows specified in the question. Even though the project will finish in Year 5, there will be some taxation to pay one year later.
- Revenue needs to be supported with a working:

(W1) Revenue	Time 1	Time 2	Time 3	Time 4	Time 5
--------------	--------	--------	--------	--------	--------

Metres	5,700	6,500	10,900	8,100	6,300
Price (€)	55,000	56,650	58,350	60,100	61,903
	_____	_____	_____	_____	_____
€m	313.50	368.23	636.01	486.81	389.99
	_____	_____	_____	_____	_____

I have noted that €55,000 is the nominal price at the end of Year 1 and then it increases by 3% pa. So many past questions have asked students to adopt this approach for converting real cash flows into nominal values.

- When computing the variable cost, my working incorporates the 5% pa increase in the unit cost from Year 2 onwards.

(W2) Variable cost	Time 1	Time 2	Time 3	Time 4	Time 5
Unit cost	23,000	24,150	25,358	26,625	27,957
	_____	_____	_____	_____	_____
Metres x unit cost (€m)	131.10	156.98	276.40	215.67	176.13
	_____	_____	_____	_____	_____

- Incremental fixed costs can be directly entered on to the schedule of cash flows accruing for the 6% pa increase from Year 2.
- Royalty – this needs some care. I have incorporated this twice on the schedule of cash flows. The royalty will be income earned in dollars for Penn Co in Ayjai. However, it will be an operational cost for the Nurukian subsidiary and the dollar values need to be converted, at the predicted spot rate, into euros.

(W3) Royalty	Time 1	Time 2	Time 3	Time 4	Time 5
--------------	--------	--------	--------	--------	--------

Metres x \$15,000 (\$m)	\$85.50	\$97.50	\$163.50	\$121.50	\$94.50
Spot rate (€/ \$) – (W4)	€0.7420 _____	€0.7049 _____	€0.6696 _____	€0.6361 _____	€0.6043 _____
€m	63.44 _____	68.72 _____	109.48 _____	77.29 _____	57.11 _____

The predicted spot rate is a mid-point number (an average of the bid and offer rates) and is the value of one dollar in euros. The dollar is predicted to devalue by 5% pa.

(W4) Spot rates	Time 0	Time 1	Time 2	Time 3	Time 4	Time 5	Time 6
Spot rate (€/ \$)	€0.7810	€0.7420	€0.7049	€0.6696	€0.6361	€0.6043	€0.5741

- TAD is **not** a cash flow. It is an allowable expense so I can compute the taxable profit and the relevant taxation cash flow. The TAD is then added back.
- Taxation is computed at the rate of 20% of the taxable profit. However, it will be paid one year after the profit was earned.
- Initial investment and scrap proceeds are just a copy and paste.
- Working capital – my ‘thought process’ is to assume the project needs to have a unique bank account. It needs a cash investment on 1 January 2014 of €31.35m (10% x €313.50m). At end of the first year an incremental adjustment is needed that can be computed as 10% x (€368.23m – €313.50m). Similar adjustments are made at the end of T2, T3 and T4. At the end of the project, I assume this bank account is closed and whatever is left is withdrawn.
- The euro cash flows are converted into dollars at the predicted spot rates. As I stated in my last article, some students make errors at this point. In this case, I have to divide the euro cash flows by the spot rate (euro per dollar) to find the dollar amounts.
- As mentioned above, the dollar value of the royalty appears twice on the schedule. Penn Co will receive the royalties and these will be subject to taxation in Ayjai.
- The bi-lateral tax agreement mentioned in the question leads to an additional cash flow. The working clarifies the position.

(W5) Additional taxation	Time 1	Time 2	Time 3	Time 4	Time 5
	€m	€m	€m	€m	€m
Taxable profit	28.96	50.13	155.19	96.21	56.25
(25–20)% x Taxable Profit	(1.45)	(2.51)	(7.76)	(4.81)	(2.81)
Spot Rate (€/£) – W4	€0.7420	€0.7049	€0.6696	€0.6361	€0.6043
	_____	_____	_____	_____	_____
\$m	(1.95)	(3.56)	(11.59)	(7.56)	(4.65)
	_____	_____	_____	_____	_____

- As the project will not alter Penn Co's business and financial risk, the appropriate cost of capital is 10%, the company's WACC.

Extract from the Report

Let me turn my attention to the report. I will show you an extract from this so as you can get a feel as to what you need to produce in the exam.

To: BoD of Penn Co

From: xxxxxx

Subject: Nurukian Train Line Project

Date: xx-xx-xx

Financial assessment

I have prepared a forecast of the nominal free cash flows for the Nurukian train line project in Appendix (1). After discounting these at the Penn Co's current cost of capital (10%), the project increases shareholder wealth by just under \$150m. Based on this value, the company should accept this project.

All forecasts are subject to estimation errors. This should be taken into account when the BoD arrives at its final decision.

Assumptions

There are a number of assumptions that have been made when computing the NPV. Some of these are considered below:

- Inflation – specific inflation rates have been incorporated into the appraisal and are expected to remain constant for the five-year period.
- Taxation – the current tax rates and allowances used to arrive at the taxation cash flows may vary over the life of the project.
- Scrap proceeds – the Nuruk government have guaranteed to purchase the machinery for a value of €500m. This may be subject to the condition of the machine as there will be wear and tear.
- Exchange rates – future spot rates are affected by many factors and, hence, the values used in the assessment may be incorrect.
- Finance – the project requires €1,000m (\$1280m) initial finance. It has been assumed that this will be raised in the Ajjain financial markets. This is a large value relative to the company's current entity value. The project may be too big for Penn Co to undertake.

Sensitivity analysis should be carried out to identify how changes in key variables affect the NPV.

Appendix (2) and Remaining Part of the Report

The primary objective of this second article was to show how to deal with international project appraisal within Section A of *Advanced Financial Management*. This has now been achieved. Appendix (2) and the remaining elements of this report are on other key areas of the *Advanced Financial Management* syllabus. Although, these are just as important, they are not the focus of these series of articles. My intention was to provide you with a logical way of attacking questions on international project appraisal only.

International project appraisal is an important element within the *Advanced Financial Management* syllabus. Students preparing to attempt this exam should ensure they study this topic very carefully. Know your subject well.

As I have demonstrated in this series of articles, a question on this area can appear in either section of the exam. As long as you take a disciplined approach and apply the knowledge you have gained from your studies, you can earn a mark worthy of a pass.

Sunil Bhandari, freelance tutor

www.SunilBhandari.com



Related Links

Aspects of Islamic finance

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Aspects of Islamic finance**

This article aims to continue to develop the understanding of Islamic finance and follows the first article on this subject, 'Islamic finance – theory and practical use of sukuk bonds'. The first article focused on Sukuk finance and case studies where Sukuk finance was utilised. This article is more general and considers different aspects, issues and developments of Islamic finance. It also extends the explanations provided in the appendix to the first article.

Questions are likely to focus on the application of Islamic finance as an additional source of finance, and assessing its benefits and drawbacks, in different business scenarios and situations.

Overview

Islamic virtues and tenets specify the need for ethical behaviour and fair treatment. Within a business context, this means that organisations should maintain high ethical standards in all business dealings. Specifically, business and enterprise should be conducted with honesty and integrity, maintaining truthfulness and morality in all dealings. In particular, such business and enterprise should not capitalise on the misfortune of others or take unfair advantage. For example, higher prices should not be

charged to an individual because they lack knowledge and information about the fair price of a product that they are purchasing. Profit creation should be the result of business activity that benefits society at large.

Within this context, the Islamic finance framework is based on certain prohibitions. In particular, money (and money substitute products such as gold and silver) should not be viewed as commodities, but rather as means of exchange. Therefore, interest (or *riba*) cannot be paid or received on loans. Furthermore, although it is fully acceptable to engage in profitable business activities, such business should be ethical. In particular dealing in alcohol, pork-related products, armaments, gambling and other socially detrimental activities is not acceptable. Engaging in activities involving speculation is also not allowed, limiting the use of derivative instruments and money markets, which are based on interest.

Operationalising Islamic finance

Organisations need access to short-term and long-term sources of finance. The basic, fundamental function of banks is to provide a channel that enables the flow of financial resources from investors to borrowers, and thereby provides a source of finance for organisations. Investors invest their excess funds to earn interest, and borrowers use the funds in business activity to generate profits, some of which are then used to pay interest on the borrowings. Among other sources of finance that involve the payment and receipt of interest are corporate and government bonds.

The question that could be asked is how could finance flow between investors and borrowers without involving interest. The answer provided by Islamic finance, in its basic form, is through profit-sharing arrangements or partnerships. The article on Islamic finance published in February 2013 explained it as follows:

In an Islamic bank, the money provided in the form of deposits is not loaned, but is instead channelled into an underlying investment activity, which will earn profit. The depositor is rewarded by a share in that profit, after a management fee is deducted by the bank.

Islamic finance institutions (IFIs), including banks, could raise finance via *Mudaraba* and *Musharaka* equity-type contracts through multi-partnership contracts (see below). Here, investors (known as *rub-ul-mal*) would invest funds with the IFI (known as the *mudareb* or investment manager). The funds are then pooled and used in profit-making projects while also keeping within Sharia rules. Therefore, the IFI would effectively become the *rub-ul-mal* and the corporation that uses the funds for investment purposes becomes the *mudareb*. In each case, the emphasis is on partnerships, and the profits earned are shared between the corporation, the bank and the investors. It is possible that all three parties share the losses as well, if the business venture is not successful.

However, with corporations requiring different modes of finance and IFIs keen on providing these, different types of Islamic financial products have been developed. The challenge for IFIs is to ensure that the products comply with Sharia rulings, as well as normal financial regulations and law.

Common Islamic financial products

IFIs offer two broad categories of financial products: equity-based and fixed income-based. The appendix to the February 2013 article on Islamic finance explained many of these financial products and it is recommended that this article is read in conjunction with the first article for further detail.

Equity-based financial products

Equity-based financial products consist of Mudaraba and Musharaka contracts. With these contracts, the investor or IFI (rub-ul-mal) and the investment manager or corporation (mudareb) share the profits from the business venture, in which the funds are invested, in a pre-arranged agreement. The key differences between the two contracts are two-fold.

With a Mudaraba contract:

- all losses are borne solely by the investor (IFI), although provisions can be set up to carry forward these losses against future profits, and
- the mudareb, as the expert in the business venture takes the sole responsibility for running the business.

With a Musharaka contract:

- losses are shared between the two parties in proportion to their monetary investment or investment-in-kind, and
- both parties would participate in managing and running the venture jointly.

Diminishing Musharaka contracts are a recent innovation where not only are the profits shared between rub-ul-mal and the mudareb, but the mudareb would pay greater amounts to the rub-ul-mal. In this way the mudareb owns greater and greater proportion of the asset, until eventually the ownership of the asset is passed to the mudareb entirely.

Fixed-income based financial products

With Murabaha contracts, the IFI purchases the asset and then sells it to the business or individual at cost plus a fair profit. The business or individual pays for the asset in pre-agreed instalments and over a pre-agreed time period.

Ijara contracts are similar to short-term leases where the IFI purchases an asset for the business or individual to use. The lease payments, the lease period and payment terms are agreed at the start of the contract. The lessor is responsible for the maintenance and insurance of the asset. Provisions can be made to allow the lessee to purchase the asset for a nominal fee at the end of the contract.

Sukuk bonds were covered in some detail in the February 2013 article on Islamic finance and it is recommended that you study that article in detail. Sukuk bonds have been based on underlying securitised Islamic contracts such as Ijara and Mudaraba, as well on individual or groups of physical assets. Some Sukuk bonds have been based on securitised Murabaha contracts, but there is some debate on whether these comply with Sharia rulings, as they may be viewed as debt on debt and therefore attracting riba. Some Sharia rulings have allowed minor proportions of Murabaha and Istisna contracts within the securitised asset portfolio, used as the underlying asset portfolio.

Salam contracts are similar to forward contracts, where a commodity (or service) is sold today for future delivery. Cash is received immediately from the IFI and the quantity, quality, and the future date and time of delivery are determined immediately. The sale will probably be at a discount so that the IFI can make a profit. In turn, the IFI would probably sell the contract to

another buyer for immediate cash and profit, in a parallel Salam arrangement. Salam contracts are prohibited for commodities such as gold, silver and other money-type assets.

Istisna contracts are often used for long-term, large construction projects of property and machinery. Here, the IFI funds the construction project for a client that is delivered on completion to the IFI's client. The client pays an initial deposit, followed by instalments, to the IFI, the amount and frequency of which are determined at the start of the contract.

Sharia boards

Sharia Boards (SBs) ensure that all products and services offered by IFIs are compliant with the principles of Sharia rules. They review and oversee all new product offerings made by the IFI and make judgments on an individual case-by-case basis, regarding their acceptability with Sharia rulings. Additionally, SBs often oversee Sharia compliant training programmes for an IFI's employees and participate in the preparation and approval of the IFI's annual reports.

SBs are normally made up of a mixture of Islamic scholars and finance experts to ensure that fair and reasonable judgments are made. Where necessary, the finance experts can explain the products to the Islamic scholars. The Islamic scholars often sit on several SBs of a number of different IFIs. SBs are in-turn supervised by the International Association of Islamic Bankers (IAIB).

SBs face several challenges when making judgments. Sharia law can be open to different interpretations, leading to different outcomes on the acceptability of the same products by different SBs and Islamic scholars. Furthermore, precedents set by SBs are not binding, and changes in SB's personnel over time may shift the balance of the SB's collective opinions and judgments on the acceptability of existing and new products.

SBs need considerable resources to operate effectively, especially where Sukuk finance is concerned. IFIs need to ensure that their SB members are well informed about the developments and trends in global financial markets.

Benefits, drawbacks and challenges

Benefits

Corporations, individuals and IFIs engaged in raising and issuing funding based on Islamic finance virtues may be viewed as belonging in stakeholder-type partnerships that are engaged in deriving benefits from ethical, fair business activity. The result of these partnerships is one of mutual interest, trust and co-operation. The ethical stance and fair dealing of Islamic finance virtues means that partnerships, business activity and profit creation comes from benefiting the community as a whole.

Since the virtues of Islamic finance and enterprise prohibit speculation and short-term opportunism, it encourages all parties to take a longer term view of success from the partnership. It focuses all the parties' attention on creating a successful outcome to the venture. This should result in a more stable financial environment. Indeed, literature in this area suggests that had banks and other financial institutions conducted their business activity based on Islamic finance principles, the negative impact of the banking and sovereign debt financial crises would have been much reduced.

IFIs or conventional financial institutions with products based on Islamic finance principles gain access to Muslim funds across the world and provide finance for organisations and individuals who need them. As the February 2013 article stated, it is estimated that Islamic financial assets have exceeded \$1,600bn worldwide. As the world emerges from the global financial crisis and business activity increases, this should increase. Furthermore, access to Islamic finance is not restricted to Muslim communities only. The wider business community could have access to new sources of finance. This may be particularly attractive to corporations focused on ethical investments that Islamic finance virtues stipulate.

Drawbacks and challenges

Because of the prohibitions of *riba* and on speculation, IFIs may be slower to react to market demand and changes. They may lack sufficient flexibility in their product offering when compared to conventional financial institutions and may be less able to take advantage of short-term opportunities.

Moral hazard and principal-agent issues may be more pronounced between IFIs and organisations and individuals to whom they lend funds. This is because Islamic finance virtues stipulate close relationships from partnership-like arrangements. However, information asymmetry between the IFI and the borrower of funds will always exist. Therefore, costs related to increased level of due diligence and negotiating are probably higher for IFIs.

Costs related to developing new financial products may also be higher for IFIs because not only will the products have to comply with normal financial laws and regulations but also with Sharia rules. As stated above, the resources required by SBs can be considerable.

Added to this, because these financial products need to go through stages of compliance and layers of complications before they are approved, the approval process can take time. The pace of innovation of new Islamic financial products may be considerably slower than that of conventional products. This may make the IFI less able to compete with conventional financial institutions and may make it restrict its activities to smaller, niche markets.

Some Islamic financial products may not be compatible with international financial regulation – for example, a diminishing Musharaka contract may not be an acceptable mortgage instrument in law, although it could be constructed as such. The need to ensure that such products comply with regulations may increase legal and insurance costs.

The interpretation of Sharia rulings may allow certain Islamic finance products to be acceptable in some markets, but not in others. This has led to some Islamic scholars, who are experts in Sharia and finance, to criticise a number of product offerings. For example, some Murabaha contracts have been criticised because their repayments have been based on prevailing interest rates rather than on economic or profit conditions within which the asset will be used. Some Sukuk bonds have faced similar criticisms in that their repayments have been based on prevailing interest rates, they have been credit-rated and their redemption value is based on a nominal value rather than on a market value, and thereby, perhaps, making them too close to conventional bonds and their repayments too similar to *riba*. On the other hand, the opposite argument could be that in order to make Islamic financial products competitive in all markets, their valuations need to be comparable. Therefore, benchmarking them using conventional means is necessary.

So far the discussion on drawbacks and challenges has focused on IFIs, as providers of Islamic finance. It is also important to consider the drawbacks and challenges that corporations may face when using Islamic finance.

From the above discussion, the costs related to developing and gaining approval for Islamic financial products is likely to be passed down to customers and possibly make these products more expensive. In addition to this, access to new products and flexibility within existing products may be limited, due to the more complicated approval process that is necessary. These more expensive and less flexible sources of finance may make the corporation using them less competitive when compared to rivals who have access to cheaper, more flexible sources of finance.

The partnership nature of Islamic finance contracts may also cause agency type issues within corporations. These may be more prevalent in joint venture type situations or where the diverse range of stakeholders may make it more difficult for corporations to determine and act upon the importance of various stakeholder groups. For example, in the case of a Musharaka contract, where the IFI and the organisation are both involved in the management of a project, dealing with other stakeholder groups may be more challenging.

Before the financial crisis, trading in asset backed and securitised Sukuk products, issued by corporations, has been limited (a notable exception was Sukuk products denominated in Malaysian ringgits). Furthermore, since the financial crisis, issuance in new Sukuk products has reduced somewhat.

Using Islamic finance may also increase the cost of capital for a corporation. For example, it may be more difficult to demonstrate that repayments for Mudaraba, Musharaka and Sukuk contracts are like debt, and therefore they may not attract a tax-shield. However, an equivalent organisation which raises the same finance using conventional debt finance may be able to lower its cost of capital due to tax-shields and therefore increase the value of its investment.

Conclusion

The increasing global interest in and use of Islamic finance means that this is an important source of finance which organisations need to consider. Its many attributes that are common to ethical investment and finance would make it an attractive source of finance particularly to organisations that place importance on ethical issues. The innovations in Islamic financial products by IFIs, such as Sukuk and diminishing Musharaka, have meant new and innovative Islamic finance products are being developed and are coming into the market. This is likely to continue as the impact of the financial crisis recedes.

However, IFIs face a number of challenges such as agency-related issues, increased costs, lack of flexibility, difficulties with complying with Sharia rulings, and also normal regulations and law. The IFIs (and the wider Islamic finance regulatory bodies) need to put into place mechanisms and strategies that will help to overcome these challenges, and thereby ensure that Islamic finance-based products compete and compare with conventional riba-based financial products.

Written by a member of the *Advanced Financial Management* examining team



Related Links

Islamic finance

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Islamic finance**

This article looks at Islamic finance as a growing and important source of finance, including the success and failure of the use.

The growth and popularity of the use of Islamic finance has been exceptional since the Central Bank of Bahrain issued the first sovereign sukuk bonds in 2001. It is estimated that by the end of 2012 Islamic financial assets will have exceeded \$1,600bn, which is around 1–2% of global financial assets worldwide.

A successful *Advanced Financial Management* student must: 'Demonstrate an understanding of the role of, and developments in, Islamic financing as a growing source of finance for organisations; explaining the rationale for its use, and identifying its benefits and deficiencies.'

This article is focused on Islamic finance as a growing and important source of finance. In particular, it looks at the success and failure of the use of sukuk bonds to finance the purchase of assets. However, for the purpose of reference, the attached appendix explains the basic principles of Islamic finance.

Sukuk finance

What is sukuk finance? The official definition provided by the Accounting and Auditing Organization for Islamic Financial Institutions (AAOIFI), the Bahrain-based Islamic financial standard setter, is 'certificates of equal value representing undivided shares in the ownership of tangible assets, usufructs and services or (in the ownership of) the assets of particular projects or special investment activity.'

Sukuk is about the finance provider having ownership of real assets and earning a return sourced from those assets. This contrasts with conventional bonds where the investor has a debt instrument earning the return predominately via the payment of interest (riba). Riba or excess is not allowed under Sharia law.

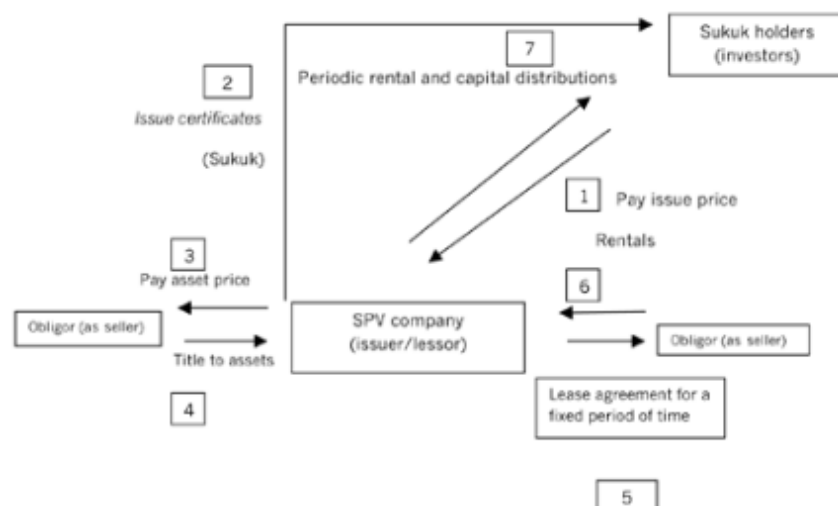
There has been considerable debate as to whether sukuk instruments are akin to conventional debt or equity finance. This is because there are two types of sukuk:

- Asset based – raising finance where the principal is covered by the capital value of the asset but the returns and repayments to sukuk holders are not directly financed by these assets.
- Asset backed – raising finance where the principal is covered by the capital value of the asset but the returns and repayments to sukuk holders are directly financed by these assets.

There are fundamental differences between these. The diagrams set out below explain the mechanics of how each sukuk operates.

Asset-based sukuk

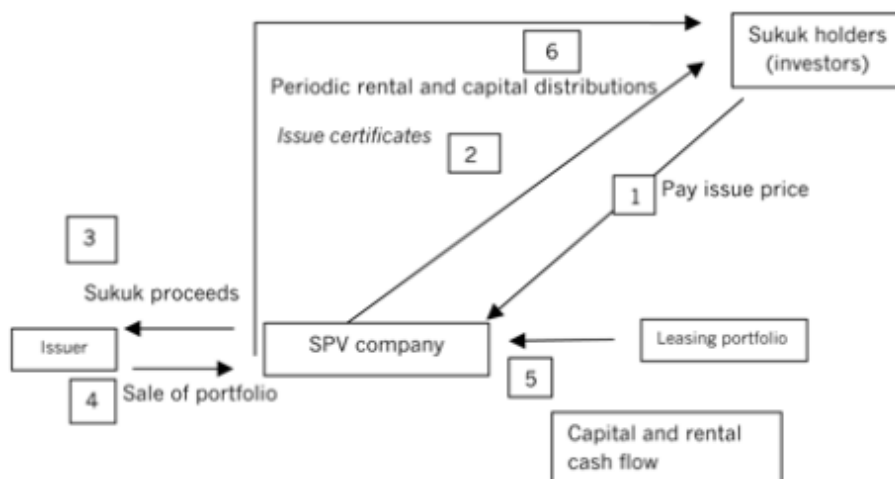
Sukuk Al-ijarah: financing acquisition of asserts or raising capital through sale and lease back.



1. Sukuk holders subscribe by paying an issue price to a special purpose vehicle (SPV) company.
2. In return, the SPV issues certificates indicating the percentage they own in the SPV.
3. The SPV uses the funds raised and purchases the asset from the obligor (seller).
4. In return, legal ownership is passed to the SPV.
5. The SPV then, acting as a lessor, leases the asset back to the obligor under an Ijarah agreement.
6. The obligor or lessee pays rentals to the SPV, as the SPV is the owner and lessor of the asset.
7. The SPV then make periodic distributions (rental and capital) to the sukuk holders.

Asset-backed sukuk

Sukuk: Securitisation of Leasing Portfolio



1. Sukuk holders subscribe by paying an issue price to a SPV company.
2. In return, the SPV issues certificates indicating the percentage they own in the SPV.
3. The SPV will then purchase a portfolio of assets, which are already generating an income stream.
4. In return, the SPV obtains the title deeds to the leasing portfolio.
5. The leased assets will be earning positive returns, which are now paid to the SPV company.
6. The SPV then makes periodic distributions (rental and capital) to the sukuk holders.
7. With an asset-based sukuk, ownership of the asset lies with the sukuk holders via the SPV. Hence, they would have to maintain and insure the asset. The payment of rentals provides the return and the final redemption of the sukuk is at a pre-agreed value. As the obligor is the lessee, the sukuk holders have recourse to him if default occurs. This makes this type of sukuk more akin to debt or bonds.

Asset-backed sukuk certainly have the attributes of equity finance – the asset is owned by the SPV. All of the risks and rewards of ownership passes to the SPV. Hence, should the returns fail to arise the sukuk holders suffer the losses. In addition, redemption for the sukuk holders is at open market value, which could be nil.

Sukuk finance case studies

For Emirates airline, the use of sukuk finance has been a huge success. The company issued its first sukuk (Islamic Bond), with a seven-year term, in 2005, which was listed on the Luxembourg Stock Exchange. The \$550m was repaid in full in June 2012.

‘The repayment of our first ever sukuk bond is part of Emirates’ varied financing strategy and reflects our robust financial position,’ said Sheikh Ahmed bin Saeed Al Maktoum, chairman of the Emirates Group and chief executive of Emirates airline.

Emirates’ initial injection of equity finance at the time of its creation 24 years ago has been supplemented by a variety of financing options, including leasing, EU/US export credits, commercial asset-backed debt, Islamic financing, conventional bonds, as well as sukuk.

Tim Clark, Emirates’ president, recently stated that the airline had traditionally used European debt to finance purchase of its fast-growing Boeing and Airbus fleet. The French banks were particularly forthcoming with finance solutions.

However, since the global debt crisis in 2008, the traditional debt markets have taken a risk averse position – even with a business like Emirates, which has an unbroken profit-making record.

Clark outlined that obtaining funding for new planes using sukuk could be tricky because Islamic finance, in addition to forbidding payment of interest, prohibits pure monetary speculation and requires deals to involve concrete assets. It would be harder to win a seal of approval from Islamic finance scholars for a sukuk that was based on assets, which the airline did not yet own.

For new aircraft, it is not impossible but it is much more complicated as the cash would have to go from investors through a special purpose vehicle to the manufacturer before a lease-back arrangement is put in place. Hence, using existing assets to obtain sukuk finance is far easier.

Emirates currently has two aircraft-based sukuk instruments that have been issued globally, and is backed by existing aircraft: a \$500m issue from GE Capital in November 2009, and a \$100m deal for Nomura in July 2010.

Emirates is not the only success story when it comes to the use of sukuk finance. Dubai shopping mall developer Majid Al Futtaim decided against issuing a conventional bond because of pricing concerns. It mandated its banks to set up a separate sukuk programme. Turkish Airlines has followed suit and will finance the purchase of its expanding fleet with Islamic bonds.

The sukuk market has been relatively resilient during the instability in global financial markets, which has made it more difficult for even highly rated companies around the world to issue conventional bonds. That is partly because Islamic investors in the Gulf remain cash-rich, partly due to the limited supply of sukuk, and partly since sukuk investors tend to hold the bonds until maturity. If these bonds are not being sold on to other investors, there is little or no chance of the bond value fluctuating.

Recent events have shown the same is not true for conventional bonds. The influence of the credit rating agencies with their regular reassessment of government and corporate credit ratings has caused downward movement in prices. As one commentator recently stated: ‘Equities are the only game in town – bonds carry more risk.’

However, the story of Dubai World, the sovereign investment fund of the Dubai royal family, gives the other side of the story when it comes to the use of Islamic finance. On 25 November 2009, the financial world was shocked when Dubai World requested a restructuring of \$26bn in debts. The main concern was the delay in the repayment of the \$4bn sukuk, or Islamic bond, of Dubai World's developer Nakheel, which was especially known for construction of the Dubai Palm Islands.

The Nakheel sukuk was quite a complicated instrument. It was broadly based on the aforementioned Ijarah structure. In theory, the SPV has legal ownership over the asset in this sale and leaseback arrangement. However, in this case the SPV only had a long leasehold interest for a period of 50 years. The issue is that leasehold right is not seen as a real right or property right under UAE law as applicable in Dubai. What may have seemed secure was not.

Nevertheless, the Nakheel sukuk was backed by a few additional guarantees that should have provided sukuk investors with some recourse. As such, these guarantees gave investors the confidence to invest in the sukuk. A guarantee from the state-owned parent company, which implicitly provides a government guarantee for the sukuk (despite the fact that the prospectus clearly stated otherwise), had reassured investors. This misplaced assumption misled investors in their risk–return decision on the investment.

The issue, however, did not end there; the complications worsened when the parent company that acted as guarantor found itself in a situation that made it no better placed than Nakheel to repay the sukuk. Dubai World is also just a holding company for a number of other companies beside Nakheel. However, all of Dubai World's subsidiaries have their own creditors and their own debts to service, and the important thing for Nakheel sukuk-holders is that the creditors of Dubai World, through the guarantee, are subordinated to the creditors of the subsidiaries of Dubai World.

A public statement on 30 November 2009 by the Dubai Finance Department director-general, that the Dubai World debts are 'not guaranteed by the government', appears to correctly reflect the legal position, as the Dubai government was not required by the lenders, and nor did it provide, any contractual guarantees in respect of the Dubai World debt.

As history tells us, Nakheel did not default on its Islamic bond. The well reported \$10bn bailout, including providing \$4.1bn to assist Nakheel directly from Dubai's rich neighbours Abu Dhabi, calmed the markets. But this was only part of the solution. Nakheel also issued new sukuk bonds to some of its creditors in lieu of amounts due to them. This was a key part of the company's restructuring.

In a prospectus attached to the new sukuk, Nakheel revealed that it wrote down the value of its property and project portfolio by almost Dh74bn (US\$20.14bn) in 2009 as its fortunes flagged. The company also said it changed tactics in response to the financial crisis, forging ahead with a selection of its projects and putting others on hold.

Conclusion

The global debt crisis sent shockwaves through the financial markets and, at the time of writing this article, the western banks remain reluctant to loan cash to the business community. Islamic finance, and in particular sukuk, has to some extent filled the gap left by the traditional debt markets.

The Sharia principle on which it is based is fundamentally important and should ensure it is a safe and sensible finance option for both the company needing the finance as well as the sukuk holder. Clearly, companies like Emirates have shown the way on how to make sukuk one part of their finance portfolio.

However, the Nakheel story paints a different picture. A complicated Ijarah structure and a lack of legal clarity as to ownership of the underlying asset have clouded the water. If the Abu Dhabi bailout failed to materialize, then the story may have been significantly different.

Sunil Bhandari is a freelance tutor (www.SunilBhandari.com)

Appendix – The basic principles of Islamic finance

The Islamic economic model has developed over time based on the rulings of Sharia on commercial and financial transactions.

The Islamic finance framework is based on:

- equity, such that all parties involved in a transaction can make informed decisions without being misled or cheated
- pursuing personal economic gain but without entering into those transactions that are forbidden (for example, transactions involving alcohol, pork-related products, armaments, gambling and other socially detrimental activities). Also, speculation is also prohibited (so options and futures are ruled out)
- the strict prohibition of interest (riba = excess).

As stated above, earning interest (riba) is not allowed.

In an Islamic bank, the money provided in the form of deposits is not loaned, but is instead channelled into an underlying investment activity, which will earn profit. The depositor is rewarded by a share in that profit, after a management fee is deducted by the bank.

A typical illustration would be how an Islamic bank may purchase a property from a seller and resell it to a buyer at a profit. The buyer will be allowed to pay in instalments. Compare this to a typical mortgage where the bank lends money to the buyer and charges interest.

Hence, returns are made from cash returns from a productive source – for example, profits from selling assets or allowing the use of an asset (rent).

In Islamic banking there are broadly two categories of financing techniques:

- ‘fixed Income’ modes of finance – murabaha, ijara, sukuk
- equity modes of finance – mudaraba, musharaka.

Fixed income modes

(a) Murabaha

Murabaha is a form of trade credit or loan. The key distinction between a murabaha and a loan is that, with a murabaha, the bank will take actual constructive or physical ownership of the asset. The asset is then sold to the 'borrower' or 'buyer' for a profit but they are allowed to pay the bank over a set number of instalments.

The period of the repayments could be extended, but no penalties or additional mark-up may be added by the bank. Early payment discounts are not within the contract.

(b) Ijara

Ijara is the equivalent of lease finance. It is defined as when the use of the underlying asset or service is transferred for consideration. Under this concept, the bank makes available to the customer the use of assets or equipment such as plant or motor vehicles for a fixed period and price. Some of the specifications of an Ijara contract include:

- the use of the leased asset must be specified in the contract
- the lessor (the bank) is responsible for the major maintenance of the underlying assets (ownership costs)
- the lessee is held for maintaining the asset in proper order.

(c) Sukuk

Companies often issue bonds to enable them to raise debt finance. The bond holder receives interest and this is paid before dividends.

This is prohibited under Islamic law. Instead, Islamic bonds (or sukuk) are linked to an underlying asset, such that a sukuk holder is a partial owner in the underlying assets and profit is linked to the performance of the underlying asset. So, for example, a sukuk holder will participate in the ownership of the company issuing the sukuk and has a right to profits (but will equally bear their share of any losses).

Equity modes

(a) Mudaraba

Mudaraba is a special kind of partnership where one partner gives money to another for investing it in a commercial enterprise. The investment comes from the first partner (who is called 'rab ul mal'), while the management and work is an exclusive responsibility of the other (who is called 'mudarib').

The Mudaraba (profit sharing) is a contract, with one party providing 100% of the capital and the other party providing its specialist knowledge to invest the capital and manage the investment project. Profits generated are shared between the parties according to a pre-agreed ratio. In a Mudaraba only the lender of the money has to take losses.

This arrangement is therefore most closely aligned with equity finance.

(b) Musharaka

Musharaka is a relationship between two or more parties that contribute capital to a business, and divide the net profit and loss pro rata. It is most closely aligned with the concept of venture capital. All providers of capital are entitled to participate in management, but are not required to do so. The profit is distributed among the partners in pre-agreed ratios, while the loss is borne by each partner strictly in proportion to their respective capital.



Related Links

- [Student Accountant hub](#)



Think Ahead

Our sites

[myACCA](#)

[ACCA Mail](#)

[ACCA Careers](#)

[ACCA Learning Community](#)

[Your Future](#)

Useful links

[Make a payment](#)

[ACCA-X online courses](#)

Determining interest rate forwards and their application to swap valuation

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Determining interest rate forwards and their application to swap valuation**

The previous article on bonds (see 'Related links') considered the relationship between bond prices, the yield curve and the yield to maturity. It demonstrated how bonds can be valued and how a yield curve may be derived using bonds of the same risk class but of different maturities. It also showed how individual company yield curves may be estimated.

This article follows on from that, and will show how interest rate forwards may be determined from the spot yield curve. It will then briefly discuss what they mean, before proceeding to show how they may be used in determining the value of an interest rate swap. The second article addresses the learning required in the E1 and E3 areas of the *Advanced Financial Management Syllabus and Study Guide*.

Determination of interest rate forwards

Supposing that a bank assesses and quotes the following rates to a company, based on the annual spot yield curve for that company's risk class:

One-year: 3.50%
 Two-year: 4.60%
 Three-year: 5.40%
 Four-year: 6.10%
 Five-year: 6.30%

This indicates that the company would have to: pay interest at 3.50% if it wants to borrow a sum of money for one year; pay interest at 4.60% per year if it wants to borrow a sum of money for two years; pay interest at 5.40% per year if it wants borrow a sum of money for three years; and so on.

Alternatively, for a two-year loan, the company could opt to borrow a sum of money for only one year, at an interest rate of 3.50%, and then again for another year, commencing in one year's time, instead of borrowing the money for a total of two years.

Although the company would be uncertain about the interest rate in one year's time, it could request a forward rate from the bank that is fixed today – for example, through a 12v24 forward rate agreement (FRA). The question then arises: how may the value of the 12v24 FRA be determined?

A forward rate commencing in one year for a borrowed sum lasting a year can be calculated as follows:

The compound factor for the total interest rate for two years = 1.046^2
 The compound factor for the one-year interest rate = 1.035

The one-year forward interest rate can be calculated as follows:
 $1.046^2 / 1.035 = 1.0571$ or 5.71%

The interest rate forward commencing in one year for a borrowed sum lasting a year would be 5.71%. The 12v24 FRA = 5.71%.

Similarly, annual interest rate forward rates for future years can be calculated as follows:

The interest rate forward starting in year 3, for a year (24v36 FRA) =
 $1.0540^3 / 1.0460^2 = 1.0702$ or 7.02%
 [or: $1.0540^3 / (1.0350 \times 1.0571)$]

The interest rate forward starting in year 4, for a year (36v48 FRA) =
 $1.0610^4 / 1.0540^3 = 1.0823$ or 8.23%
 [or: $1.0610^4 / (1.0350 \times 1.0571 \times 1.0702)$]

The interest rate forward starting in year 5, for a year (48v60 FRA) =
 $1.0630^5 / 1.0610^4 = 1.0710$ or 7.10%
 [or: $1.0630^5 / (1.0350 \times 1.0571 \times 1.0702 \times 1.0823)$]

In summary:

Year	Annual spot yield curve	Annual forward rates
1	3.50%	-
2	4.60%	5.71%
3	5.40%	7.02%
4	6.10%	8.23%
5	6.30%	7.10%

Supposing the company wants to borrow a sum of money for three years on the basis of the above rates:

- i. it could pay annual interest at a rate of 5.40% in each of the three years, or
- ii. it could pay interest at a rate 3.50% in the first year, 5.71% in the second year and 7.02% in the third year, or
- iii. it could pay annual interest at a rate of 4.60% in each of the first two years and 7.02% in the third year.

Using interest rate forwards to value a simple interest rate swap contract

Supposing the above company has \$100m borrowings in the form of variable interest rate loans repayable in five years and pays interest based on the above yield curve. It expects interest rates to increase in the future and is therefore keen to fix its interest rate payments.

The bank offers to swap the variable interest rate payments for a fixed rate, such that the company pays a fixed rate of interest to the bank in exchange for receiving a variable rate of return from the bank based on the above yield rates less 50 basis points. The variable rate receipts from the bank will then be used to pay the interest on the loan.

The fixed equivalent rate of interest the company will pay the bank for the swap can be calculated as follows:

The current expected amounts of interest the company expects to receive from the bank, based on year 1 spot rate and years 2, 3, 4 and 5 forward rates are:

Year 1	$0.0300 \times \$100\text{m} = \3.00m
Year 2	$0.0521 \times \$100\text{m} = \5.21m
Year 3	$0.0652 \times \$100\text{m} = \6.52m
Year 4	$0.0773 \times \$100\text{m} = \7.73m
Year 5	$0.0660 \times \$100\text{m} = \6.60m

Note: The rates used to calculate the annual amounts are reduced by 50 basis points or 0.5%.

At the start of the swap, the net present value of the swap receipts based on the variable rates from the bank will be the same as the costs based on the fixed amount paid to the bank.

Let's say R is the fixed amount of interest the company will pay the bank, then

$$\begin{aligned} &(\$3.00\text{m} - R) \times 1.035^{-1} + \\ &(\$5.21\text{m} - R) \times 1.046^{-2} + \\ &(\$6.52\text{m} - R) \times 1.054^{-3} + \\ &(\$7.73\text{m} - R) \times 1.061^{-4} + \\ &(\$6.60\text{m} - R) \times 1.063^{-5} \\ &= 0 \end{aligned}$$

Removing the brackets, the above expands to:

$$2.90m - 0.966R + 4.76m - 0.914R + 5.57m - 0.854R + \\ 6.10m - 0.789R + 4.86m - 0.737R = 0$$

Simplifying this, adding all the \$ flows together and R-flows together, gives:

$$24.19m - 4.26R = 0$$

$$\$5.68m = R$$

$$\text{In percentage} = \$5.68m / \$100m = 5.68\%$$

In practice the receipts and payments of the swap would be netted off such that the company will expect to pay \$2.68m (\$5.68m – \$3.00m) to the bank in year one, and expect to receive \$0.84m (\$6.52m – \$5.68m) from the bank in year three, and so on for the other years. The present values of these net annual flows, discounted at the yield curve rates, will be zero. The fixed rate of 5.68% is lower than the five-year spot rate of 6.30% because some of the receipts and payments related to the swap contract occur in earlier years when the spot yield curve rate is lower.

Although at the commencement of the contract, the present value of the swap is zero, as interest rates fluctuate, the value of the swap will change. For example, if interest rates increase and the company pays interest at a fixed rate, then the swap's value to the company will increase. The value of the swap contract will also change as the swap approaches maturity, and the number of receipts and payments reduce.

Conclusion

The previous article and this article considered the relationship between bonds, interest rates, spot and forward yield curves, culminating with how the forwards rates can be used to determine the equivalent fixed rate of a simple swap contract. The examples used were simplified into annual cash flows and rates, and students undertaking *Advanced Financial Management* should be able to demonstrate their knowledge and understanding of these areas to this extent.

In practice, valuation of bonds and related products is more complicated because of factors such as: cash flows occurring more frequently than once a year, early redemption of products, change in product values, and so on. However, these aspects are beyond the scope of the *Advanced Financial Management* syllabus.

Written by a member of the *Advanced Financial Management* examining team



Related Links

- [Student Accountant hub page](#)

Bond valuation and bond yields

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Bond valuation and bond yields**

Bonds and their variants such as loan notes, debentures and loan stock, are IOUs issued by governments and corporations as a means of raising finance. They are often referred to as fixed income or fixed interest securities, to distinguish them from equities, in that they often (but not always) make known returns for the investors (the bond holders) at regular intervals. These interest payments, paid as bond coupons, are fixed, unlike dividends paid on equities, which can be variable. Most corporate bonds are redeemable after a specified period of time. Thus, a 'plain vanilla' bond will make regular interest payments to the investors and pay the capital to buy back the bond on the redemption date when it reaches maturity.

This article, the first of two related articles, will consider how bonds are valued and the relationship between the bond value or price, the yield to maturity and the spot yield curve. It addresses, in part, the learning required in Sections B3a and B3e of the the *Advanced Financial Management Syllabus and Study Guide*.

Bond value or price

Example 1

How much would an investor pay to purchase a bond today, which is redeemable in four years for its nominal value or face value

of \$100 and pays an annual coupon of 5% on the nominal value? The required rate of return (or yield) for a bond in this risk class is 4%.

As with any asset valuation, the investor would be willing to pay, at the most, the present value of the future income stream discounted at the required rate of return (or yield). Thus, the value of the bond can be determined as follows:

Year	Year 1	Year 2	Year 3	Year 4
Cash flows	\$5	\$5	\$5	\$105
PV (4%)	$5 \times 1.04^{-1} = 4.81$	$5 \times 1.04^{-2} = 4.62$	$5 \times 1.04^{-3} = 4.44$	$105 \times 1.04^{-4} = 89.75$

Value/price (sum of the PV of flows) = \$103.62

Note: Mathematically, for year one, $5 / 1.04^1$ is the same as 5×1.04^{-1} , and similarly for all the years

If the required rate of return (or yield) was 6%, then using the same calculation method, the price of the bond would be \$96.53. And where the required rate of return (or yield) is equal to the coupon – 5% in this case – the current price of the bond will be equal to the nominal value of \$100.

Thus, there is an inverse relationship between the yield of a bond and its price or value. The higher rate of return (or yield) required, the lower the price of the bond, and vice versa. However, it should be noted that this relationship is not linear, but convex to the origin.

Bond price



The plain vanilla bond with annual coupon payments in the above example is the simpler type of bond. In addition to the plain vanilla bond, candidates – as part of their *Advanced Financial Management* studies and exam – are required to have knowledge of, and be able to deal with, more complicated bonds such as: bonds with coupon payments occurring more frequently than once a year; convertible bonds and bonds with warrants which contain option features; and more complicated payment features such as repayment mortgage or annuity type payment structures.

Yield to maturity (YTM) (also known as the [Gross] Redemption Yield (GRY))

If the current price of a bond is given, together with details of coupons and redemption date, then this information can be used to compute the required rate of return or yield to maturity of the bond.

Example 2

A bond paying a coupon of 7% is redeemable in five years at nominal value (\$100) and is currently trading at \$106.62. Estimate its yield (required rate of return).

$$\$106.62 = \$7 \times (1+r)^{-1} + \$7 \times (1+r)^{-2} + \dots + \$107 \times (1+r)^{-5}$$

The internal rate of return approach can be used to obtain r . Since the current price is higher than \$100, r must be lower than 7%.

Initially, try 5% as r:

$$\$7 \times 4.3295 [5\%, \text{ five - year annuity}] + \$100 \times 0.7835 [\text{PV } 5\%, \text{ five - year}] = \$30.31 + \$78.35 = \$108.66$$

Try 6% as r:

$$\begin{aligned} & \$7 \times 4.2124 [6\%, \text{ five - year annuity}] + \$100 \times 0.7473 [\text{PV } 6\%, \text{ five - year}] = \\ & \$29.49 + \$74.73 = \$104.22 \end{aligned}$$

$$\text{Yield} = 5\% + (108.66 - 106.62 / 108.66 - 104.22) \times 1\% = 5.46\%$$

The 5.46% is the yield to maturity (YTM) (or redemption yield) of the bond. The YTM is the rate of return at which the sum of the present values of all future income streams of the bond (interest coupons and redemption amount) is equal to the current bond price. It is the average annual rate of return the bond investors expect to receive from the bond till its redemption. YTMs for bonds are normally quoted in the financial press, based on the closing price of the bond. For example, a yield often quoted in the financial press is the bid yield. The bid yield is the YTM for the current bid price (the price at which bonds can be purchased) of a bond.

Term structure of interest rates and the yield curve

The yield to maturity is calculated implicitly based on the current market price, the term to maturity of the bond and amount (and frequency) of coupon payments. However, if a corporate bond is being issued for the first time, its price and/or coupon payments need to be determined based on the required yield. The required yield is based on the term structure of interest rates and this needs to be discussed before considering how the price of a bond may be determined.

It is incorrect to assume that bonds of the same risk class, which are redeemed on different dates, would have the same required rate of return or yield. In fact, it is evident that the markets demand different annual returns or yields on bonds with differing lengths of time before their redemption (or maturity), even where the bonds are of the same risk class. This is known as the term structure of interest rates and is represented by the spot yield curve or simply the yield curve.

For example, a company may find that if it wants to issue a one - year bond, it may need to pay interest at 3% for the year, if it wants to issue a two - year bond, the markets may demand an annual interest rate of 3.5%, and for a three-year bond the annual yield required may be 4.2%. Hence, the company would need to pay interest at 3% for one year; 3.5% each year, for two years, if it wants to borrow funds for two years; and 4.2% each year, for three years, if it wants to borrow funds for three years. In this case, the term structure of interest rates is represented by an upward sloping yield curve.

The normal expectation would be of an upward sloping yield curve on the basis that bonds with a longer period of maturity would require a higher interest rate as compensation for risk. Note here that the bonds considered may be of the same risk class but the longer time period to maturity still adds to higher uncertainty.

However, it is entirely normal for yield curves to be of many different shapes dependent on the perceptions of the markets on how interest rates may change in the future. Three main theories have been advanced to explain the term structure of interest rates or the yield curve: expectations hypothesis, liquidity-preference hypothesis and market-segmentation hypothesis. Although it is beyond the remit of this article to explain these theories, many textbooks on investments and financial management cover these in detail.

Valuing bonds based on the yield curve

Annual spot yield curves are often published by the financial press or by central banks (for example, the Bank of England regularly publishes UK government bond yield curves on its website). The spot yield curve can be used to estimate the price or value of a bond.

Example 3

A company wants to issue a bond that is redeemable in four years for its nominal value or face value of \$100, and wants to pay an annual coupon of 5% on the nominal value. Estimate the price at which the bond should be issued.

The annual spot yield curve for a bond of this risk class is as follows:

One-year	3.5%
Two-year	4.0%
Three-year	4.7%
Four-year	5.5%

The four-year bond pays the following stream of income:

Year	1	2	3	4
Payments	\$5	\$5	\$5	\$105

This can be simplified into four separate bonds with the following payment structure:

Year	1	2	3	4
Bond 1	\$5			
Bond 2		\$5		
Bond 3			\$5	
Bond 4				\$105

Each annual payment is a single payment in that particular year, much like a zero-coupon bond, and its present value can be determined by discounting each cash flow by the relevant yield curve rate, as follows:

Bond 1	$\$5 \times 1.035^{-1} = \4.83
Bond 2	$\$5 \times 1.04^{-2} = \4.62
Bond 3	$\$5 \times 1.047^{-3} = \4.36
Bond 4	$\$105 \times 1.055^{-4} = \84.76

The sum of these flows is the price at which the bond can be issued, \$98.57.

The yield to maturity of the bond is estimated at 5.41% using the same methodology as example 2.

Some important points can be noted from the above calculation; firstly, the 5.41% is lower than 5.5% because some of the returns from the bond come in earlier years, when the interest rates on the yield curve are lower, but the largest proportion comes in Year 4. Secondly, the yield to maturity is a weighted average of the term structure of interest rates. Thirdly, the yield to maturity is calculated after the price of the bond has been calculated or observed in the markets, but theoretically it is term structure of

interest rates that determines the price or value of the bond.

Mathematically:

$$\text{Bond price} = \text{Coupon} \times (1+r_1)^{-1} + \text{coupon} \times (1+r_2)^{-2} + \dots + \text{coupon} \times (1+r_n)^{-n} + \text{redemption value} \times (1+r_n)^{-n}$$

Where r_1, r_2 , etc are spot interest rates based on the yield curve and n is the number of time periods in which an amount of the coupon is paid and, finally, the value when the bond is redeemed.

In this article it is assumed that coupons are paid annually, but it is common practice to pay coupons more frequently than once a year. In these circumstances, the coupon payments need to be reduced and the time period frequency needs to be increased.

Estimating the yield curve

There are different methods used to estimate a spot yield curve, and the iterative process based on bootstrapping coupon paying bonds is perhaps the simplest to understand. The following example demonstrates how the process works.

Example 4

A government has three bonds in issue that all have a face or nominal value of \$100 and are redeemable in one year, two years and three years respectively. Since the bonds are all government bonds, let's assume that they are of the same risk class. Let's also assume that coupons are payable on an annual basis. Bond A, which is redeemable in a year's time, has a coupon rate of 7% and is trading at \$103. Bond B, which is redeemable in two years, has a coupon rate of 6% and is trading at \$102. Bond C, which is redeemable in three years, has a coupon rate of 5% and is trading at \$98.

To determine the yield curve, each bond's cash flows are discounted in turn to determine the annual spot rates for the three years, as follows:

$$\begin{aligned} \text{Bond A: } \$103 &= \$107 \times (1+r_1)^{-1} \\ r_1 &= 107/103 - 1 = 0.0388 \text{ or } 3.88\% \end{aligned}$$

$$\begin{aligned} \text{Bond B: } \$102 &= \$6 \times 1.0388^{-1} + 106 \times (1+r_2)^{-2} \\ r_2 &= [106 / (102 - 5.78)]^{1/2} - 1 = 0.0496 \text{ or } 4.96\% \end{aligned}$$

$$\begin{aligned} \text{Bond C: } \$98 &= \$5 \times 1.0388^{-1} + \$5 \times 1.0496^{-2} + 105 \times (1+r_3)^{-3} \\ r_3 &= [105 / (98 - 4.81 - 4.54)]^{1/3} - 1 = 0.0580 \text{ or } 5.80\% \end{aligned}$$

The annual spot yield curve is therefore:

Year

- | | |
|---|-------|
| 1 | 3.88% |
| 2 | 4.96% |
| 3 | 5.80% |

Discussion of other methods of estimating the spot yield curve, such as using multiple regression techniques and observation of spot rates of zero coupon bonds, is beyond the scope of the *Advanced Financial Management* syllabus.

As stated in the previous section, often the financial press and central banks will publish estimated spot yield curves based on government issued bonds. Yield curves for individual corporate bonds can be estimated from these by adding the relevant spread to the bonds. For example, the following table of spreads (in basis points) is given for the retail sector.

Rating	1 year	2 year	3 year
AAA	14	25	38
AA	29	41	55
A	46	60	76

Example 5

Mason Retail Co has a credit rating of AA, then its individual yield curve – based on the government bond yield curve and the spread table above – may be estimated as:

Year 1	Year 2	Year 3
4.17%	5.37%	6.35%

These would be the rates of return an investor buying bonds issued by Mason Retail Co would expect, and therefore Mason Retail Co would use these rates as discount rates to estimate the price or value of coupons when it issues new bonds. And Mason Retail Co's existing bonds' market price would reflect its individual yield curve.

Conclusion

This article considered the relationship between bond prices, the yield curve and the yield to maturity. It demonstrated how bonds can be valued and how a yield curve may be derived using bonds of the same risk class but of different maturities. Finally it showed how individual company yield curves may be estimated.

A following article will discuss how forward interest rates are determined from the spot yield curve and how they may be useful in determining the value of an interest rate swap. It will address the learning required in Sections E1 and E3 of the *Advanced Financial Management Syllabus and Study Guide*.

Written by the *Advanced Financial Management* examining team



Related Links

- [**Student Accountant hub page**](#)

Risk management

[Home](#) / [Students](#) / [Study resources](#) / [Advanced Financial Management \(AFM\)](#) / [Technical articles](#)
/ **Risk management**

The management of risk is a key area within a number of ACCA exams, and exam questions related to this area are common. It is vital that students are able to apply risk management techniques, such as using derivative instruments to hedge against risk, and offer advice and recommendations as required by the scenario in the question. It is also equally important that students understand why corporations manage risk in theory and in practice, because risk management costs money but does it actually add more value to a corporation? This article explores the circumstances where the management of risk may lead to an increase in the value of a corporation.

Risk, in this context, refers to the volatility of returns (both positive and negative) that can be quantified through statistical measures such as probabilities, standard deviations and correlations between different returns. Its management is about decisions made to change the volatility of returns a corporation is exposed to, for example changing a company's exposure to floating interest rates by swapping them to fixed rates for a fee. Since business is about generating higher returns by undertaking risky projects, important management decisions revolve around which projects to undertake, how they should be financed and whether the volatility of a project's returns (its risk) should be managed.

The volatility of returns of a project should be managed if it results in increasing the value to a corporation. Given that the market value of a corporation is the net present value (NPV) of its future cash flows discounted by the return required by its investors,

then higher market value can either be generated by increasing the future cash flows or by reducing investors' required rate of return (or both). A risk management strategy that increases the NPV at a lower comparative cost would benefit the corporation.

The return required by investors is the sum of the risk free rate and a premium for the risk they undertake. If investors hold well-diversified portfolios of investments then they are only exposed to systematic risk as their exposure to firm-specific risk has been diversified away. Therefore, the risk premium of their required return is based on the capital asset pricing model (CAPM).

Research suggests companies with diverse equity holdings do not increase value by diversifying company specific risk, as their equity holders have already achieved this level of risk diversification. Moreover, risk management activity designed to transfer systematic risk would not provide additional benefits to a corporation because, in perfect markets, the benefits achieved from risk management activity would at least equal the costs of undertaking such activity. Therefore, in a situation of perfect markets, it may be argued that risk management activity is at best neutral or at worst detrimental because costs would either equal or be more than the benefits accrued.

Such an argument would not apply to smaller companies which have concentrated, non-diversified equity holdings. In this case the equity holders, because they are exposed to both specific and systematic risk, would benefit from risk diversification by the company. Therefore, whereas larger companies may not create value from risk management activity, smaller companies can and should undertake risk management. However, empirical research studies have found that risk management is undertaken mostly by larger companies with diverse equity holdings and not by the smaller companies. The accepted reason for this is that the costs related to risk management are large and mostly fixed. Small companies simply cannot afford these costs nor can they benefit from the economies of scale that large companies can.

In addition to the ability of larger companies to undertake risk management, market imperfections may provide the motivation for them to do so. Market imperfections that exist in the real world, as opposed to the perfect world conditions assumed by finance or economic theory, may provide opportunities to reduce volatility in cash flows and thereby reduce the costs imposed on a corporation. The following discussion considers the circumstances which may result in providing such opportunities.

Taxation

Risk management may help in reducing the amount of tax that a corporation pays by reducing the volatility of the corporation's earnings. Where a corporation faces taxation schedules that are progressive (that is the corporation pays proportionally higher amounts of tax as its profits increase), by reducing the variability of that corporation's earnings and thereby staying in the same low tax bracket will reduce the tax payable.

According to academics, corporations could often find themselves in situations where they face progressive tax functions, for example, when they have previous losses which are not written off or, in the case of multinational corporations, due to the taxation treaties which exist between different countries. The amount of taxation that can be saved depends upon the corporation's individual circumstances.

Insolvency and financial distress

A corporation may find itself in a situation of being insolvent when it cannot meet its financial obligations as they fall due.

Financial distress is a situation that is less severe than insolvency in that a corporation can operate on a day-to-day basis, but it finds that these operations are difficult to conduct because the parties dealing with it are concerned that it may become insolvent in the future. When facing financial distress a corporation will incur additional costs, both direct and indirect, due to the situation it is facing.

The main indirect costs of financial distress relate to the higher costs of contracting with the corporation's stakeholders, such as customers, employees and suppliers. For example, customers may demand better warranty schemes or may be reluctant to buy a product due to concerns about the corporation's ability to fulfil its warranty; employees may demand higher salaries; senior management may ask for golden hellos before agreeing to work for the corporation; and suppliers may be unwilling to offer favourable credit terms.

Academics exploring this area postulate that because stakeholders are subject to the corporation's full risk, as opposed to only systematic risk, which is faced by the corporation's equity holders, the stakeholders would demand greater compensation for their participation. Where an organisation actively manages its risk and prevents (or reduces the possibility of) situations of financial distress, it will find it easier to contract with its stakeholders and at a lower cost. Hence, the more volatile the cash flows of a corporation, the more likely the need to manage its risk in order to reduce the costs related to financial distress.

External funding and agency costs

Another consequence of financial distress is the impact this may have on the corporation's ability to undertake profitable future investment. Financial distress may make the cost of external debt and equity funding so expensive that a corporation and its management may be forced reject profitable projects. Academics refer to this as the under investment problem.

Equity holders in effect hold a call option on a corporation's assets and debt holders can be considered to have written the option. In cases of low financial distress the company may be considered to be similar to an at-the-money option for its equity holders, and, therefore, they would be more willing to undertake risky projects as they would benefit from any increase in profitability, but the impact of any loss is limited. In the case of substantial financial distress, the option could be considered to be well out-of-the-money. In this situation there is little (or no) benefit to equity holders of undertaking new projects, as the benefits of these will pass to the debt holders initially. However, debt holders would be reluctant to lend to a severely distressed company in any case.

Therefore, when raising debt capital, a corporation that is subject to low levels of financial distress would face higher agency costs, with lenders imposing higher borrowing costs and more restrictive covenants. Whereas debt holders get a fixed return on their investment, any additional benefit due to higher profits would go to the equity holders. This would make the debt holders reluctant to allow the corporation to undertake risky projects or to lend more finance to the corporation because they would not gain any benefit from the risky projects.

A corporation that faces high levels of financial distress would find it difficult to raise equity capital in order to undertake new investments. If corporations try to raise equity finance for relatively less risky projects then the profits earned from such projects would initially go to the debt holders and the equity holders will gain only residual profits. Therefore equity holders would put

pressure on the corporation and its management to reject good, low risk projects, which may have been acceptable to the bondholders.

Therefore, risk management in reducing financial distress by reducing the volatility of the corporation's cash inflows may help the management to obtain an optimal mix of debt and equity, and to undertake profitable projects.

Capital structure and information asymmetry

Risk management can help a corporation obtain an optimal capital structure of debt and equity to maximise its value. Since risk management stabilises the variability of cash inflows, this would enable a corporation to take more debt finance in its capital structure. Stable cash flows indicate less risk and therefore debt holders would become more willing to lend to the corporation. Since debt is cheaper to finance than equity because of lower required rates of return and the tax shield, taking on more debt should increase the value of the corporation. Risk management can help achieve this.

Academics have observed that managers would prefer to use internally generated funds rather than going to the external markets for funds because it is cheaper and less intrusive on the corporation. They suggest that borrowing money from the external markets, whether equity or debt, would involve parties who do not have the complete information about the corporation. This information asymmetry would make the external sources of funds more expensive. If risk management stabilises the cash flows that the corporation receives from year to year, then this would enable managers to plan when the necessary internal funds will become available for future investments with greater accuracy. They will then be able to align their investment policies with the availability of funding.

Manager behaviour towards risk management

In his seminal paper, Rene Stulz suggests that managers, whose performance reward structure includes large equity stakes in a corporation, are more likely to reduce the corporation's risk, as opposed to managers whose performance reward structure is based primarily on equity options. Managers who hold concentrated equity stakes in a corporation face increased levels of risk when compared to other equity holders. As discussed previously, investors hold well-diversified portfolios and face exposure to systematic risk only. But managers with concentrated equity stakes would face both systematic and unsystematic risk. Therefore, they have a greater propensity to reduce the unsystematic risk.

However, if investors do not reward corporations that are reducing unsystematic risk, because they have diversified this risk away themselves. And if a corporation's managers use the corporation's resources to reduce unsystematic risk, thereby reducing the corporation's value. Then it is worth exploring under what circumstances would equity investors allow managers to act to reduce unsystematic risk and whether such actions could actually result in the value of the corporation increasing.

Stulz argues that encouraging managers to hold concentrated equity positions but allowing them to reduce unsystematic risk at the same time, may enable them to act in the best interests of the corporation and the result may be an increase in the corporate value. He explains that managers, who do not have to worry about risks that are not under their control (because they have hedged them away), would be able to focus their time, expertise and experience on the strategies and operations that they can

control. This focus may result in the increase in the value of the corporation, although the impact of this increase in value is not easily measurable or directly attributable to risk management activity.

As an aside, one could pose the question, why don't managers, who are rewarded by equity, diversify the risk of concentrated equity investments themselves? They could sell equity in their own corporation and replace it by buying equity in other corporations. In this way they do not have to hold concentrated equity positions and then would be like the normal equity holders facing only systematic risk. A research study on wealth management, which looked at concentrated equity positions and risk management, found that senior managers are reluctant to reduce their concentrated equity positions because any attempt to sell the equity would send negative signals to the markets, and cause their corporation's value to decrease unnecessarily.

Contrary to the behaviour of managers who hold concentrated equity stakes, managers who own equity options, which will be converted into equity at a future date, will actively seek to increase the risk of a corporation rather than reduce it. Managers who hold equity options are interested in maximising the future price of the equity. Therefore in order to maximise future profits and the price of the equity, they will be more inclined to undertake risky projects (and less inclined to manage risk). Equity options, as a form of reward, have been often criticised because they do not necessarily make managers behave in the best interests of the corporation or its equity investors, but encourage them to act in an overly risky manner.

A number of empirical studies looking at manager behaviour support the above discussion (see for example Tufano's study published in 1996 in the *Journal of Finance*).

Testing the impact of risk management

In addition to the above, empirical research studies have looked at the risk management policies and actions pursued by corporations and their impact on corporate value. Although the studies have provided varying results when studying each area of market imperfections and their impact, the overarching conclusion from these studies is that: corporations manage their risks in the belief that this would create or increase corporate value, although a direct link between risk management and a corresponding increase in corporate value has not been established.

Hence the belief held among managers is that the management of risk does create value, and certainly corporations and their senior managers seem to believe and act in a manner that it does. However, the jury is still out on whether risk management actually does lead to increased corporate value. There seem to be strong theoretical reasons for managing risk, but empirical research has not proven the impact of risk management activity on corporate value.

Written by a member of the examining team for *Advanced Financial Management*

Further reading

I would suggest that students read the following academic books and papers to supplement their knowledge and understanding.

- Arnold, G, 2008. *Corporate Financial Management*. 4th ed. Harlow: Pearson.
- Culp, C, 2002. *The Revolution in Corporate Risk Management: A Decade of Innovations in Process and Products*. *Journal of Applied Corporate Finance*, 14(4), 8–26.

- Jensen, M, and Meckling, W, 1976. *Theory of the Firm: Managerial Behaviour, Agency Costs and Ownership Structure*. Journal of Financial Economics, 3, 305–360.
- Myers, S, and Majluf, N, 1984. *Corporate Financing and Investment Decisions when Firms have Information that Investors do not have*. Journal of Financial Economics, 13, 187–221.
- Smithson, C, 1998. *Managing Financial Risk: A Guide to Derivative Products, Financial Engineering and Value Maximization*. New York: McGraw-Hill, pp492–517.
- Stulz, R, 1996. *Rethinking Risk Management*. Journal of Applied Corporate Finance, 9, 8–24.
- Tufano, P, 1996. *Who Manages Risk? An Empirical Examination of Risk Management Practices in the Gold Mining Industry*. Journal of Finance, 51, 1097–1137.



Related Links

- [Student Accountant hub page](#)



Think Ahead

Our sites

[myACCA](#)

[ACCA Mail](#)

[ACCA Careers](#)

[ACCA Learning Community](#)

[Your Future](#)

Useful links

[Make a payment](#)